

Global Financial Crisis

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DEFINITION: 4, 12, 13, 15, 25, 27–30, 32, 33, 36

IDEA: 4, 6, 7, 11, 13, 15, 16, 19, 20, 28, 33, 40, 43, 46, 49,
53, 57, 59, 66, 67, 70, 78–80, 83

EXAMPLE: 4, 11

FACT: 4–6, 9–11, 35, 50, 53, 59, 72, 78, 79

FACTS: 11, 12, 19, 20, 27–29

Beliefs, 6

Global Financial Crisis, 1

1 Introduction to the Global Financial Crisis

1.1 What is a Financial Crisis?

IDEA: Wealth is embodied in long-lived assets that pay off slowly over time. We don't have things that create massive value rapidly.

EXAMPLE: (A house provides housing services over a long period of time, a piece of machinery produces some good over a long period of time, human capital produces services and other economic value over a long period of time.)

IDEA: Some fraction of wealth needs to back short-term safe assets used in transactions (money).

EXAMPLE: (When you buy something, nobody is going to accept one-thousandth of my house.)

IDEA: Collectively, we cannot all convert our long-term assets to money at the same time. A "panic" happens when enough people get nervous and try to convert.

DEFINITION: A "financial crisis" occurs when a panic (or fear of panic) affects the functioning of the financial system.

IDEA: The panic itself doesn't mean people can't go to work or people can't get investment in order to expand their business, but if the panic happens and drains banks and other financial institution's ability to do any lending, then all of a sudden this can have an effect on the real economy.

1.2 Why Study the Global Financial Crisis?

1.3 Course Themes

Course Theme 01: Asset Bubbles

FACT: Prior to all financial crises, there is a large increase in the price of at least one asset class. When this price later falls, we retroactively label the original increase to be a "bubble".

FACT: In the Global Financial Crisis, the bubble was in housing.

FACT: Housing bubbles have been associated with every major financial crisis since WWII.

Course Theme 02: New Kinds of Money:

FACT: Through most of the history of civilization, our monetary needs were met primarily by metal-based currencies.

FACT: The rise of the modern state has allowed for sovereign-backed "fiat" money. But even stable governments have limits and the demand for money can exceed the supply of metal and sustainable fiat money. Additional monies are then "manufactured" by the financial system, using private debt backed by collateral.

Course Theme 03: The Anatomy of a Modern Panic

FACT: For hundreds of years, panics were easy to spot: depositors would run to banks to exchange their banknotes for gold, or to remove their deposits.

FACT: The Global Financial Crisis was different, with the panics occurring out of public sight, in the non-bank part of the financial sector. This area of the financial system is often called the "shadow banking" system.

Course Theme 04: Fighting the Panic

1.4 Course Organization

1.5 Bonus Content

2 The Common Causes of Financial Crises

2.1 Why does this matter

2.2 What Causes Financial Crises? Part 1

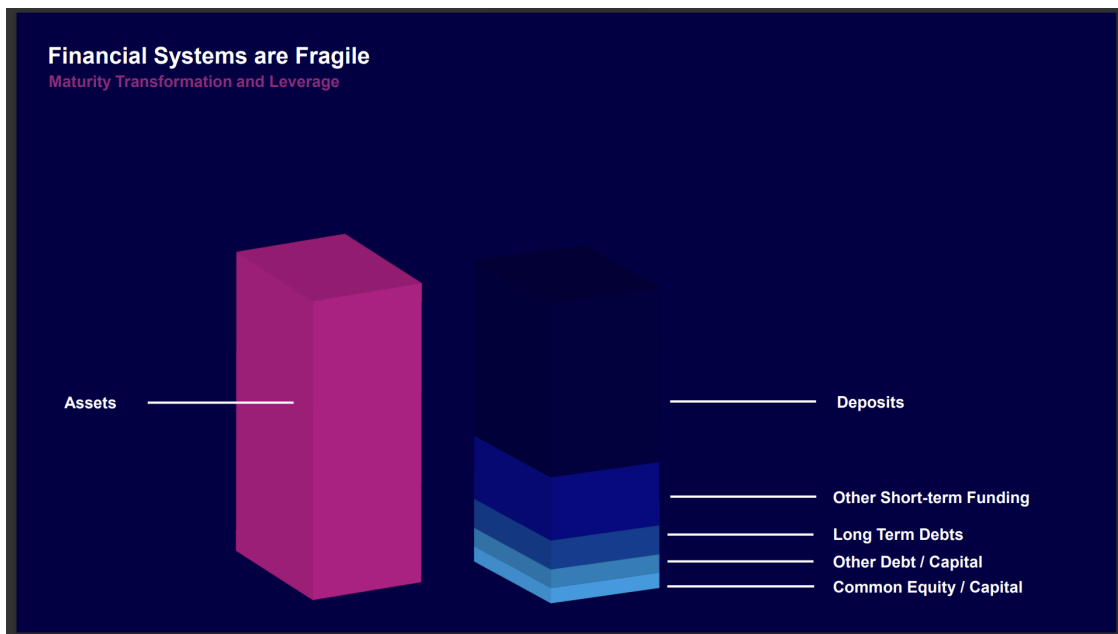


Figure 1. Lecture: What Causes Financial Crises? Part 1 1:26.

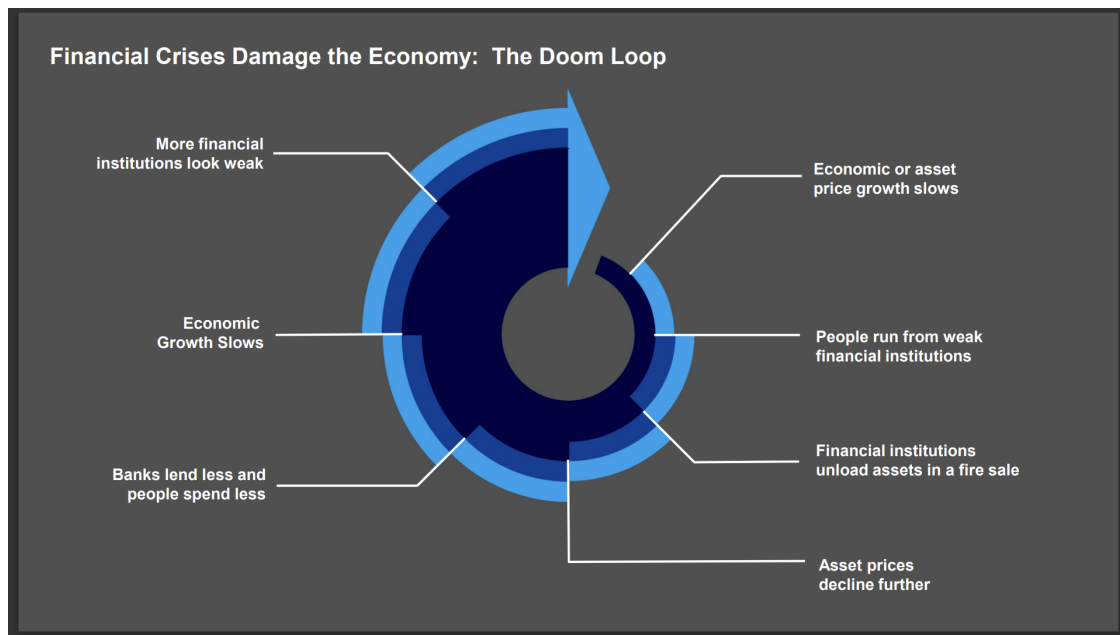


Figure 2. Lecture: What Causes Financial Crises? Part 1 4:18.

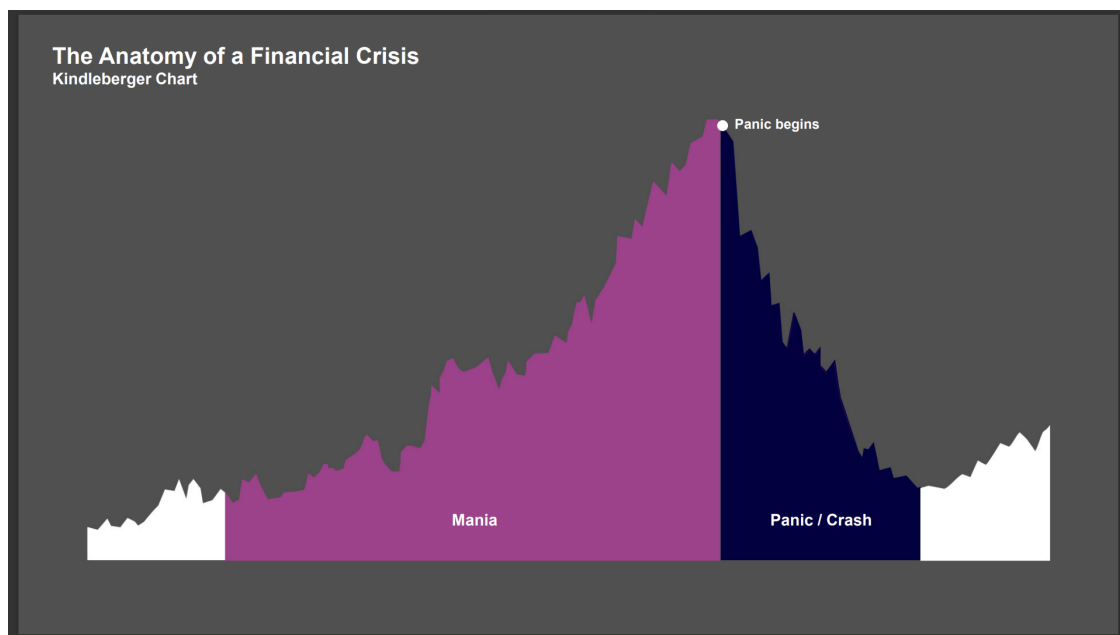


Figure 3. Lecture: What Causes Financial Crises? Part 2 1:47.

2.3 What Causes Financial Crises? Part 2

FACT: The causes of Financial crises:

- Monetary Policy and Global Savings Glut
- Fraud and Predation (increase the basic preconditions)
- Incentive Problems
- Moral Hazard
- Balkanized regulations and Supervisors
- **Beliefs**

IDEA: Elements of systemic financial crisis:

- Increased debt relative to income
- Financed by runnable sources
- Risk of contagion and panic

- Real economy suffers more due to distress in the financial sector
- Monetary policy is less effective due to indebtedness and leverage

IDEA: There's a set of things that help distinguish when a boom or mania tips into a panic or a crash. Three of them are:

- size and extent of the buildup of vulnerabilities.
- tools and financial capacity available to policymakers, governments, and central banks to respond
- what do they do with that authority? What do they do with that financial capacity? What are the choices they make? How wise and deft are those choices?

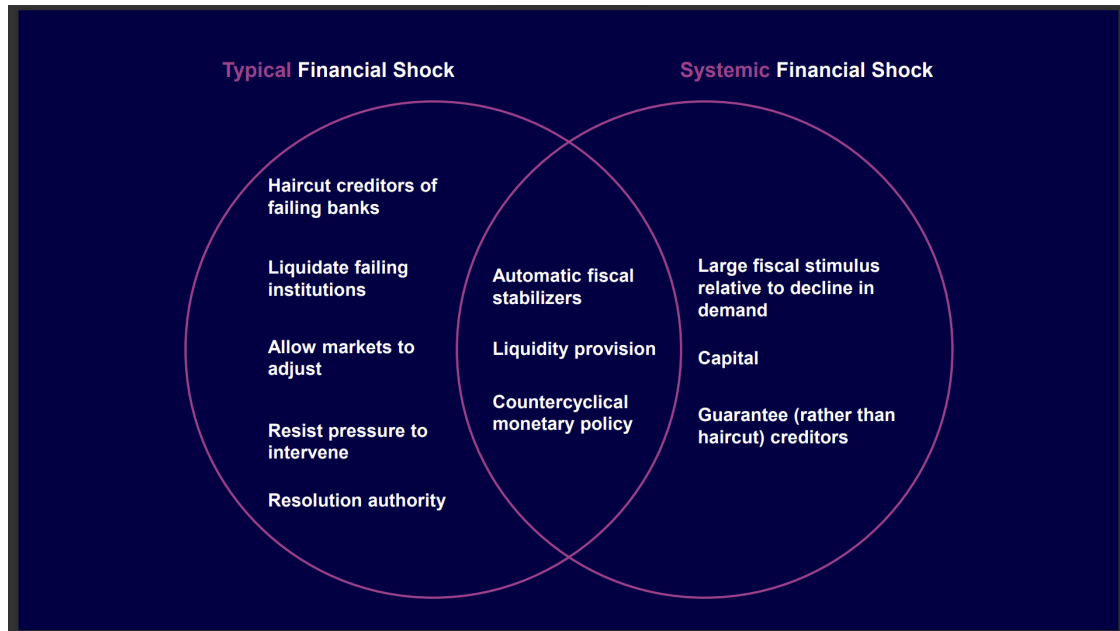


Figure 4. Lecture: What Causes Financial Crises? Part 2 5:20.

IDEA: In a panic,

- contagion risk is acute, even the solvent become illiquid
- the entire financial system is vulnerable to collapse
- asset prices fall
- contraction in credit threaten much greater overall economic damage

In this context, policy has to be aggressive in

- reducing the incentive to run
- repairing the lending capacity of the financial system
- offsetting the collapse in private demand

2.4 Causes of the 2008-2009 Financial Crisis

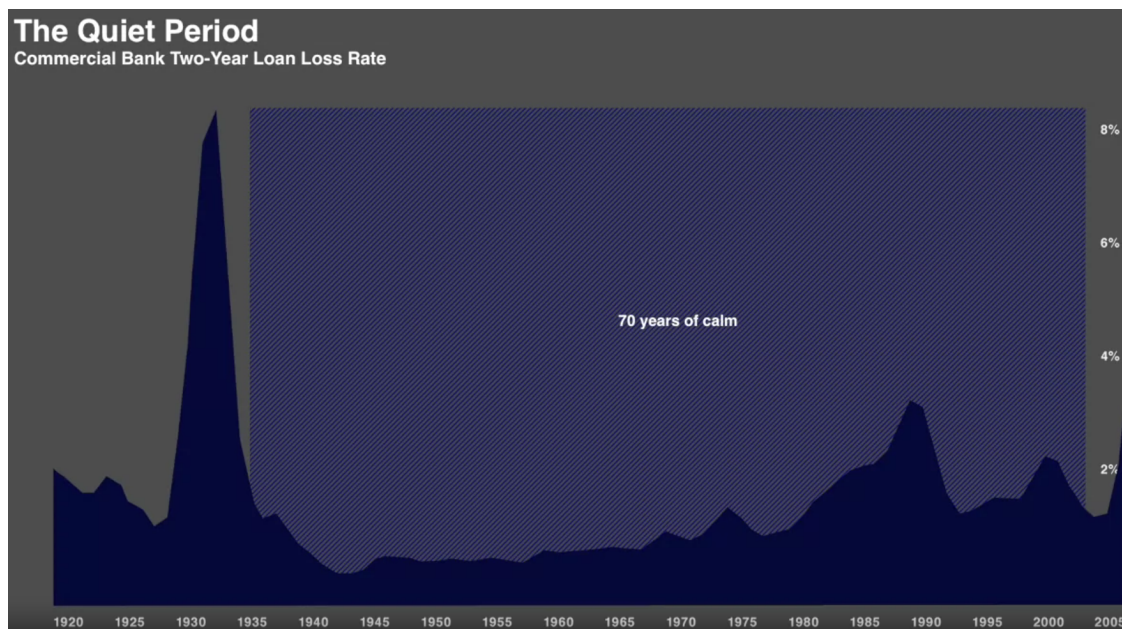


Figure 5. Lecture: Causes of the 2008-2009 Financial Crisis 0:38.

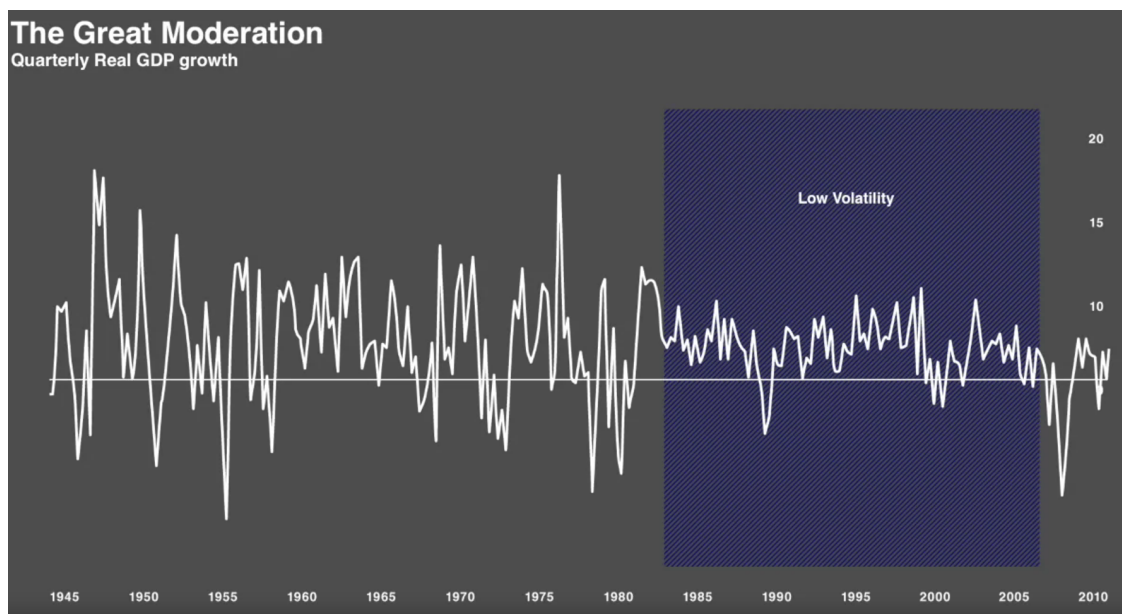


Figure 6. Lecture: Causes of the 2008-2009 Financial Crisis 1:37.

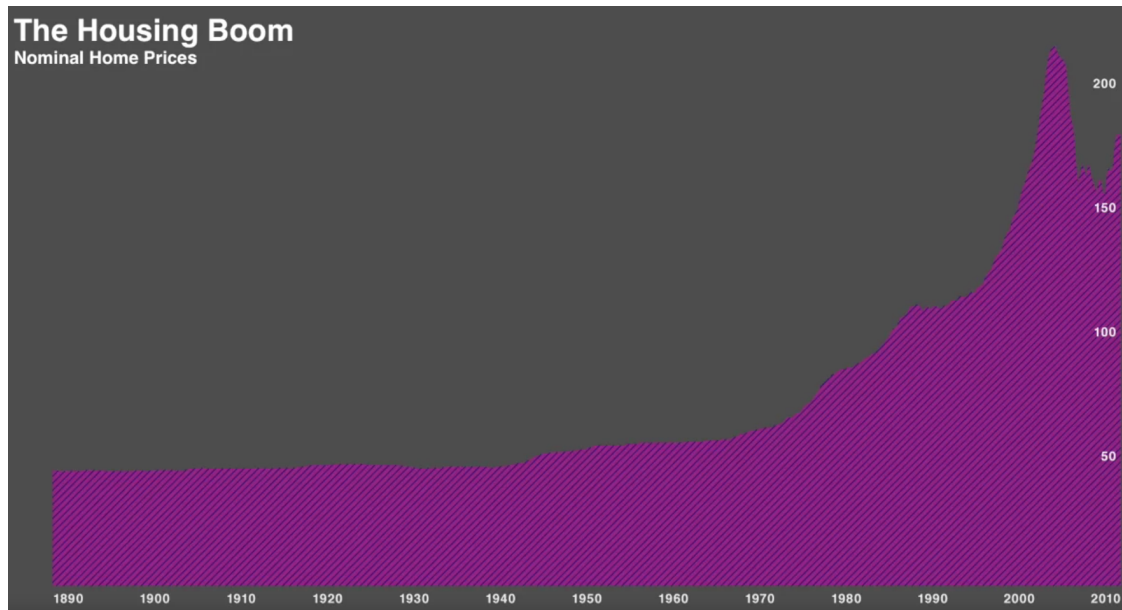


Figure 7. Lecture: Causes of the 2008-2009 Financial Crisis 1:57.

FACT: People were confident the value of homes would rise. They borrow a large amount relative to their income and lenders lend a large portion of someone's income because they expected to be a rising housing value.

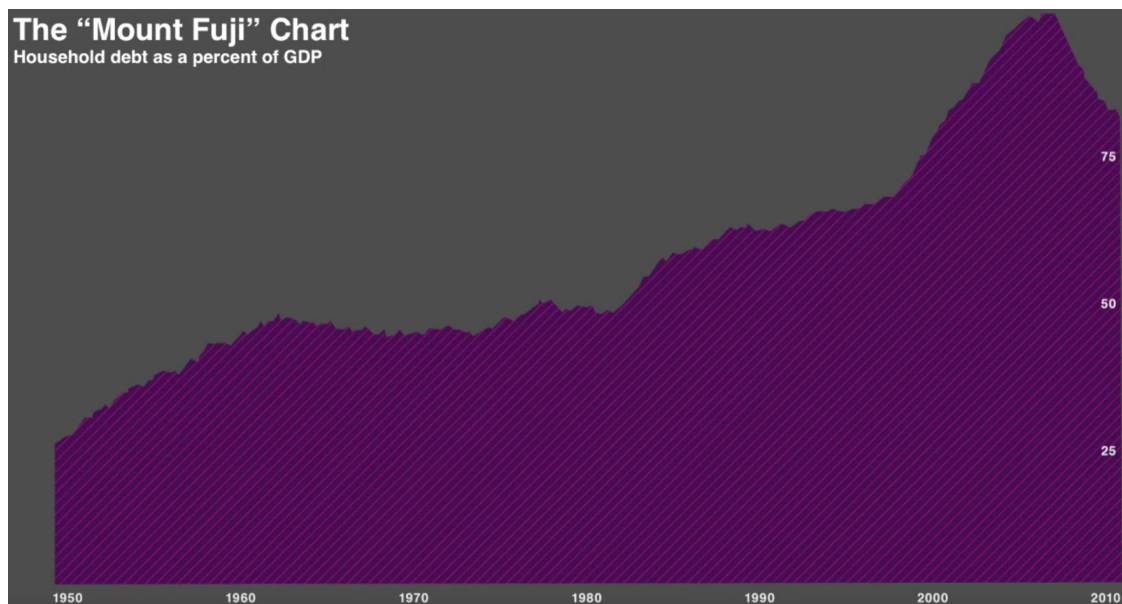


Figure 8. Lecture: Causes of the 2008-2009 Financial Crisis 2:21.

Dark blue shaded area at the bottom: the share risen in borrowing finance by banks.

FACT: Most of the increase in that borrowing was from individuals and companies, who were financed by other parts of the national system. Those institutions existed outside the safety net.

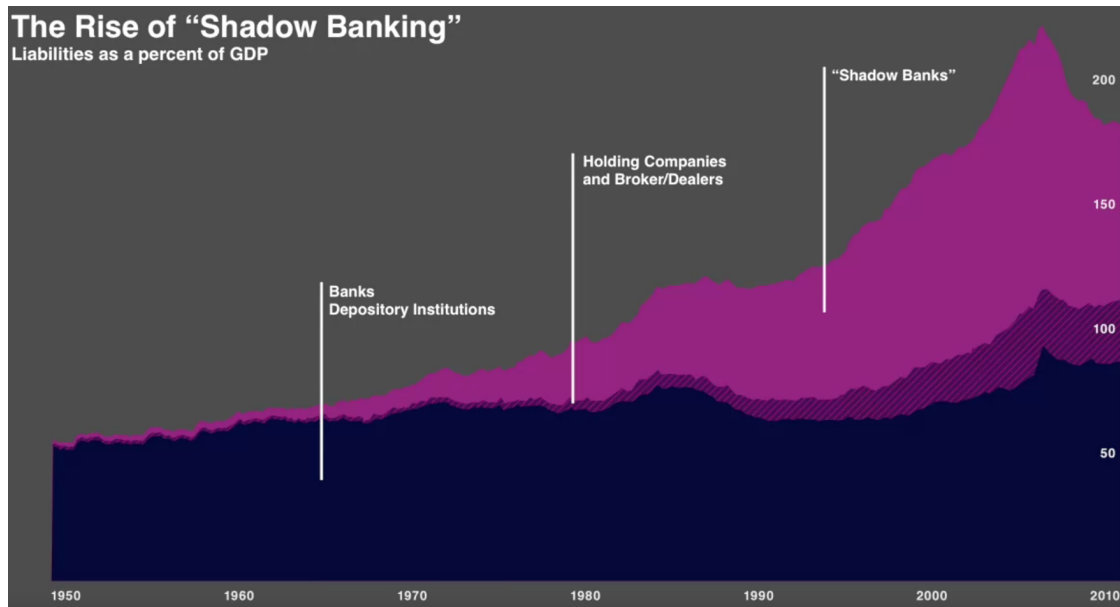


Figure 9. Lecture: Causes of the 2008-2009 Financial Crisis 2:36.

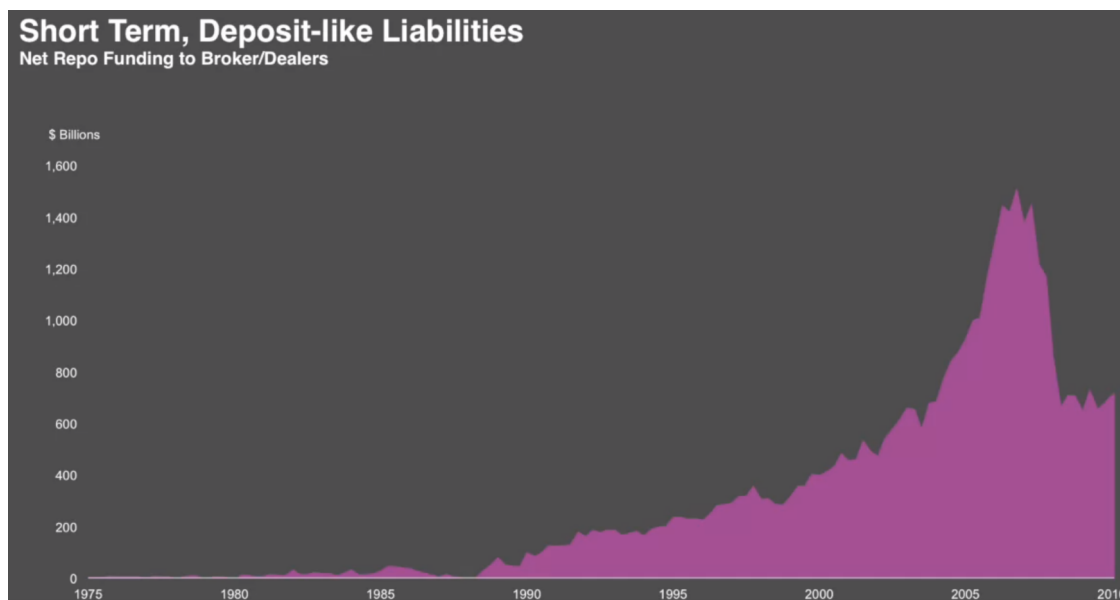


Figure 10. Lecture: Causes of the 2008-2009 Financial Crisis 3:40.

FACT: In the United States on the eve of the crisis, there was a long period where belief in a more stable future produced a huge boom in lending and in financial assets. Huge increase in debt relative to income, financed in very dangerous, runnable forms, with most of the increment financed outside the core of the banking system, and that created the conditions for panic and collapse.

FACT: At the same time, the US had a long trade where monetary policy was very accommodative, where real interest rates in the US and around the world were very low.

FACT: There was also a huge amount of fraud and predation in the US financial system. There were pockets of moral hazard most compellingly in the mortgage guarantors Fannie and Freddie. There was a mess of other incentive problems throughout the financial system, and we had a very Balkanized segmented regulatory structure.

2.5 What did We Miss and Why Did We Miss it?

FACT: Prior to the crisis, the U.S. financial system was designed with limited tools to constrain the build-up and risk.

FACT: The Federal Reserve and the bank supervisors had tools and authority to apply capital and liquidity requirements to banks and to bank holding companies, but those institutions represented only a small share of the American financial system.

FACTS: What we failed to do:

- We didn't limit leverage in investment banks or government-sponsored financial enterprises like Fannie and Freddie, or investment banks and non-bank financial institutions in some of the big insurance companies.
- couldn't limit the funding risk in those institutions.
- weren't able to limit the risks that were building up in money market funds.
- couldn't set margin requirements in derivatives.
- weren't able to limit the amount individuals could borrow relative to the value of their homes.
- had a set of constraints on authority that made it harder to reduce the risk. The build-up in vulnerabilities had happened in the decades before the crisis.

IDEA: Risk tends to find its way around the constraints during the period of bloom.

EXAMPLE: The tighter you set the limits on part of the system, you'll make that part of the system safer. You could make banks more stable but you'll shrink the market share of banks. You'll shrink the effective scope of those restrictions over time because over time, risk will migrate around them, and the relative size of those two different parts of the system will change in relative importance.

FACTS: What we did:

- pushed for risk management practices in banks and investment banks.
- introduced stress testing to think about the risks of a severe crisis.
- reduced risk in derivatives.
- limited the provision of leverage by banks to institutions like hedge funds, concerned by the risks we saw in the LTCM crisis in the late 90s.

IDEA: But supervision was focused too narrowly on compliance issues, like in money laundering and less focused on things about the stability of the entire system on systemic risk.

FACTS: What we didn't do:

- didn't raise capital requirements on banks and bank holding companies, despite these growing risks.
- didn't look for authority from the Congress to set limits on capital and leverage on funding.
- didn't raise down payment requirements on people who were borrowing to finance the purchase of a house.
- didn't limit risk taking in repo markets, securities, finance markets, and other markets that were critical to the way the system functioned.

IDEA: Our biggest mistake was failure in imagination. We failed to contemplate the possibility that we could face a classic panic and run like what made the Great Depression so terrible. We thought what happened in the great depression was inconceivable, unforeseeable, and, therefore, we didn't build the system to be strong enough to reduce the risk of that type of classic panic.

3 Housing and Mortgages

3.1 Introduction

3.2 Housing Bubbles and Financial Crises

FACTS: Reinhart and Rogoff (2008) identify the “big five” crises since WWII – Spain (1977), Norway (1987), Finland (1991), Sweden (1991), and Japan (1992) – plus 13 other bank-centered crises in developed countries.

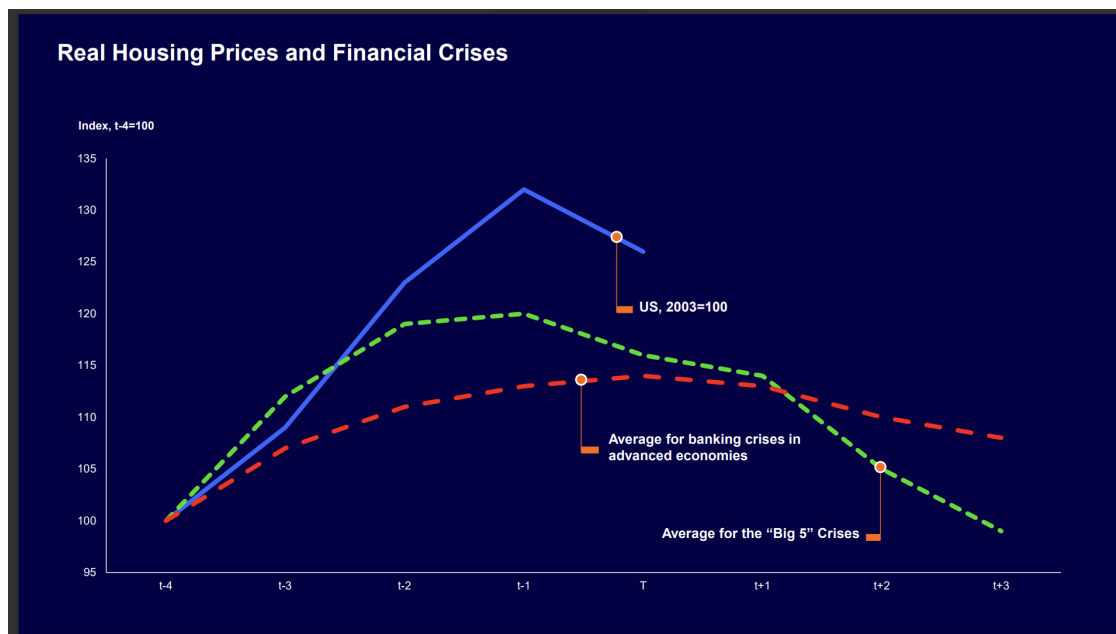


Figure 11. Lecture: Housing Bubbles and Financial Crises 2:59.

3.3 Mortgages: Fixed and Adjustable

FACTS: The standard mortgage in the United States has a 30-year term, with fixed interest rates. Alternative structures have adjustable interest rates for all or part of the term (“ARMs”, 2/28, 5/25 or 5/1, 7/23 or 7/1 etc.), with the adjustable rate set to some spread above a reference rate.

FACTS: Initial fixed rate will typically be lower than for 30-year fixed mortgage, sometimes called a “teaser”.

3.4 Prime Mortgages

FACTS: Fannie Mae and Freddie Mack loans rules:

- can't be bigger than a certain level
- have a certain ratio of loan to value (standard 20% down kind of mortgage)
- credit score for the borrower
- rules about occupancy

If you meet all of those criteria, Frannie or Freddie are allowed to buy the loans from the bank. The government agency guarantees the loan if the person with the prime loan doesn't make their payments.

3.5 Nonprime Mortgages

DEFINITION: Subprime loans target borrowers with tarnished credit histories and little savings for down payments. Government guarantee is absent.

DEFINITION: Near-prime mortgages (Alt-A) target borrowers with minor credit quality issuers or those who are unable or unwilling to provide full documentation of assets or income.

DEFINITION: Together, subprime and Alt-A are a category called nonprime mortgages.

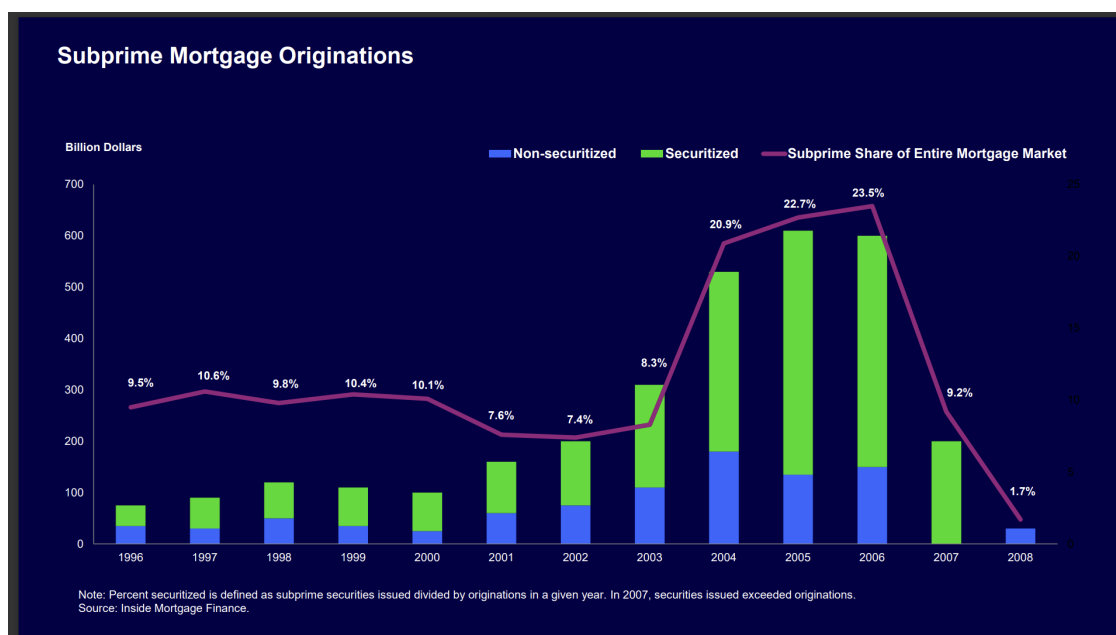


Figure 12. Lecture: Nonprime Mortgages 4:07.

3.6 Why Subprime Mortgages?

IDEA: It can work under the conditions that everybody expects housing prices to rise.

3.7 Some Foreclosure Facts

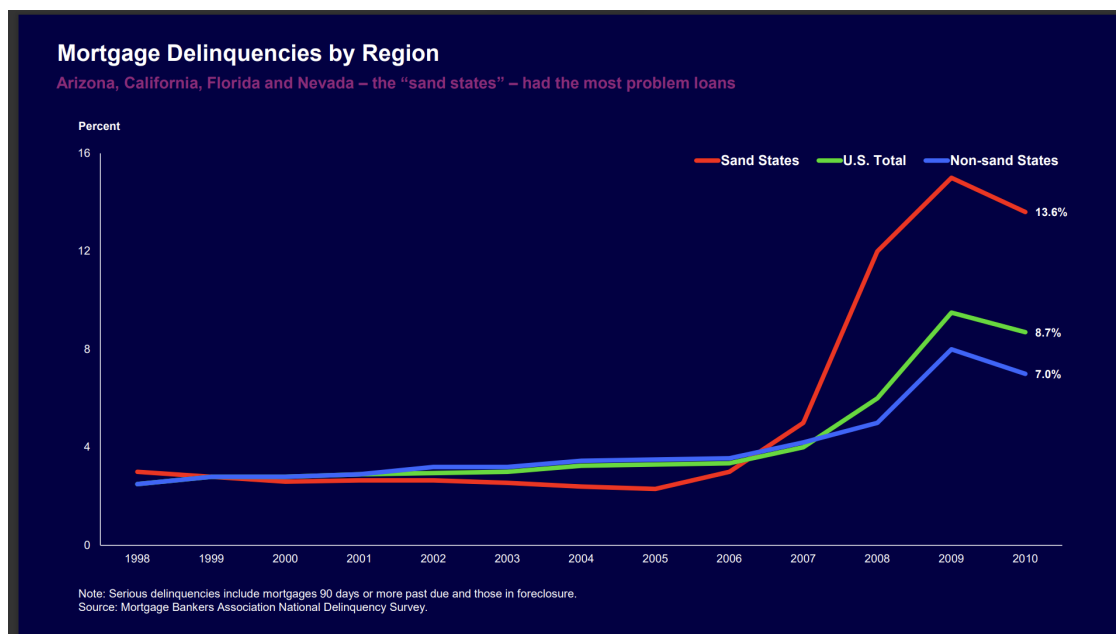


Figure 13. Lecture: Some Foreclosure Facts 1:42.

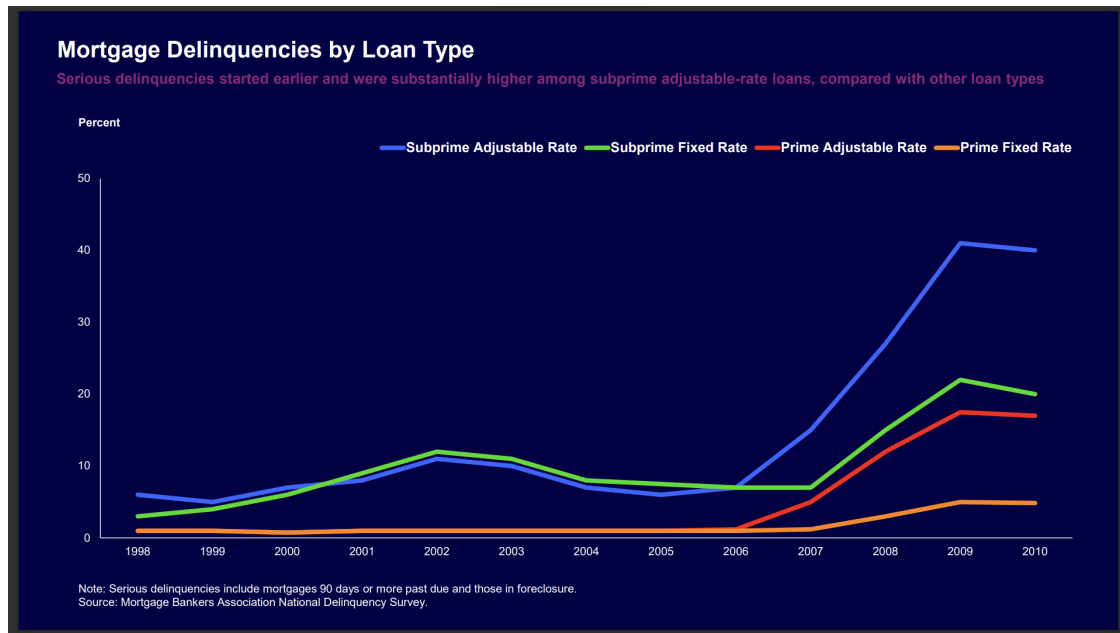


Figure 14. Lecture: Some Foreclosure Facts 3:47.

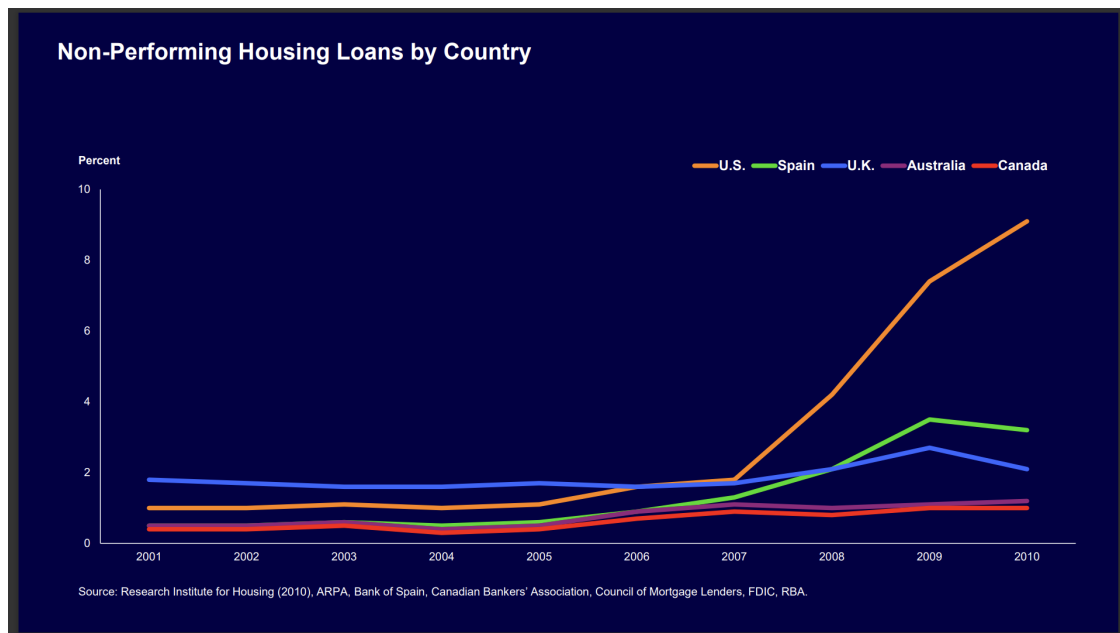


Figure 15. Lecture: Some Foreclosure Facts 5:58.

3.8 Crisis Terminology

4 Safe Assets and the Global Savings Glut

4.1 Shadow Banking

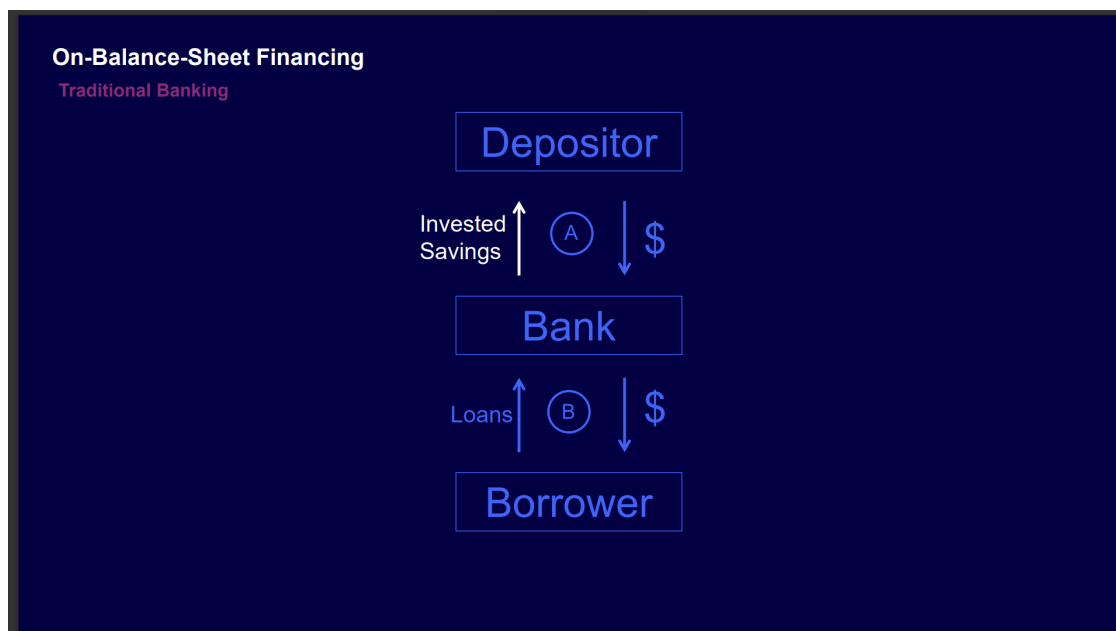


Figure 16. Lecture: Shadow Banking 2:42.

IDEA: Traditional banking system is limited. In the United States at this time, the protection is limited to \$100,000.

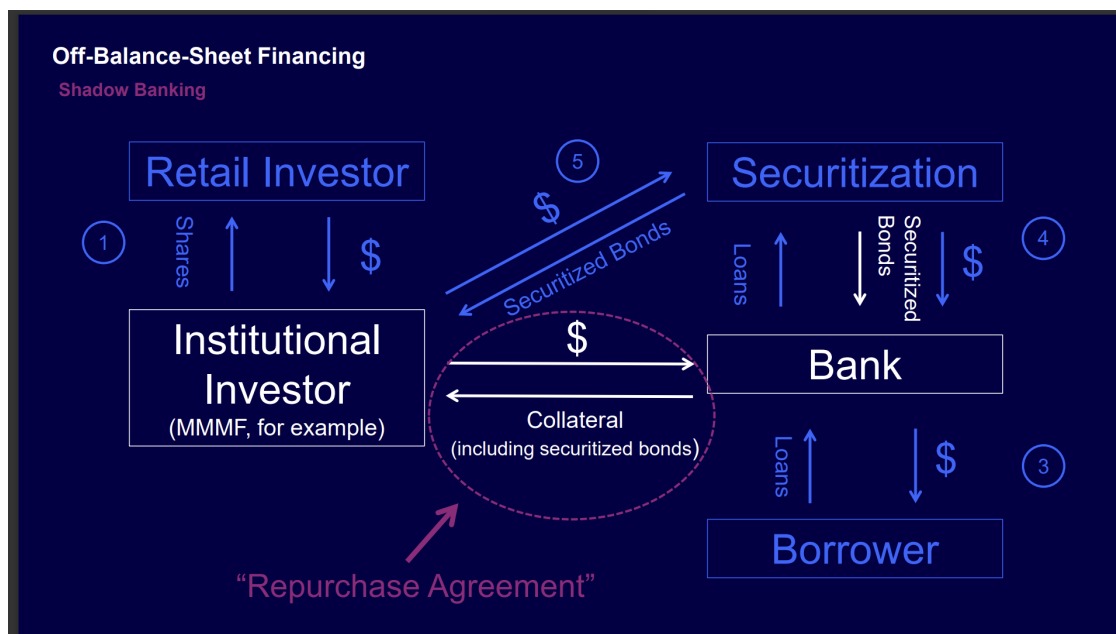


Figure 17. Lecture: Shadow Banking 8:40.

4.2 Safe Assets

DEFINITION: Information insensitivity, or "no adverse selection": the actions or decisions of economic agents are not significantly influenced by additional information

4.3 The Global Savings Glut Hypothesis

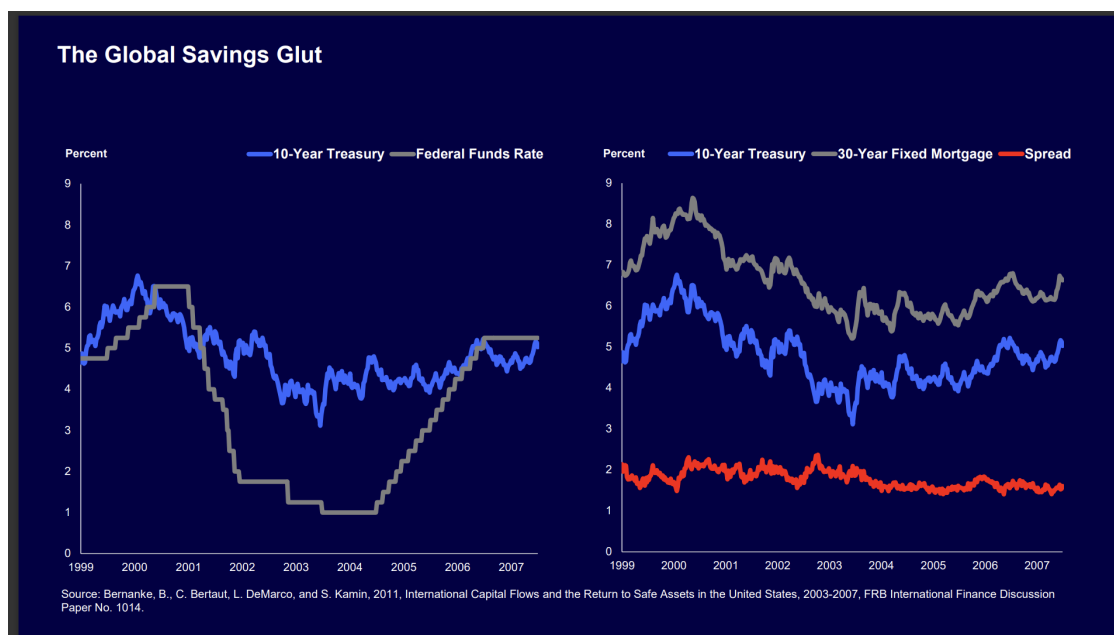


Figure 18. Lecture: The Global Savings Glut Hypothesis 5:20.

IDEA: 10-year treasury rate and 30-year fixed mortgage rate have similar trends. When the ten-year treasury bond does not move up as much as expected, mortgage rates stay low. And when mortgage rates stay low, that has a boomy effect on the housing market.

IDEA: Emerging market and commodity rich countries have large current account surpluses. They purchase safe government assets in the United States. The demand for those assets was very high. That kept effectively the price of those assets high. When the price of a bond is high, the rate on the bond is low.

IDEA: The link between short-term and long-term interest rates is as strong as it had been historically.

4.4 Global Savings Glut Data, Part I

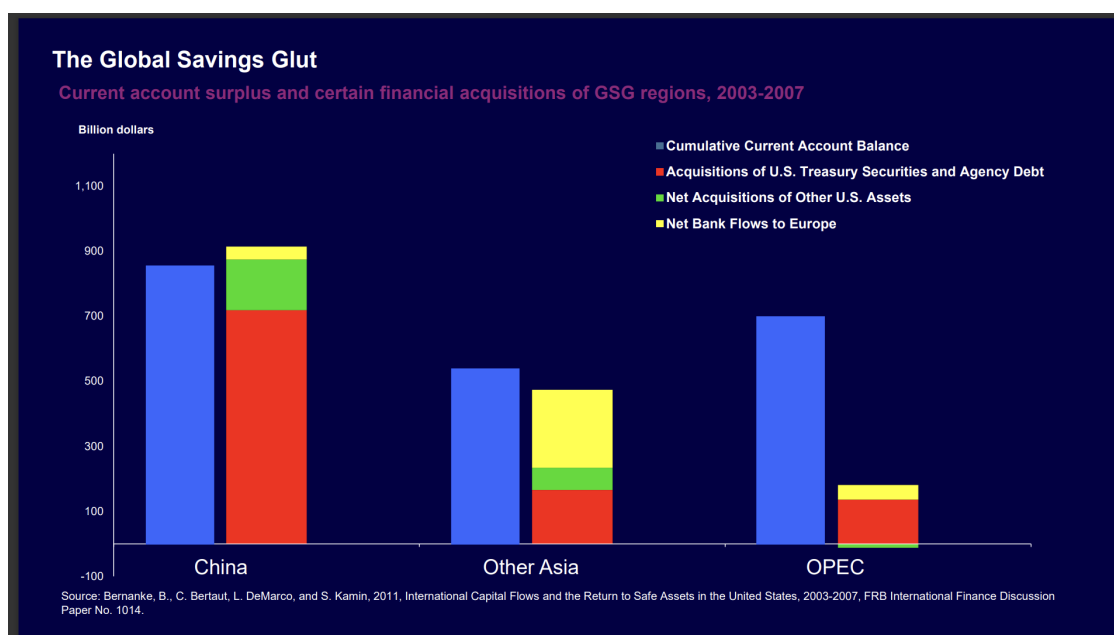


Figure 19. Lecture: Global Savings Glut Data, Part I 2:22.

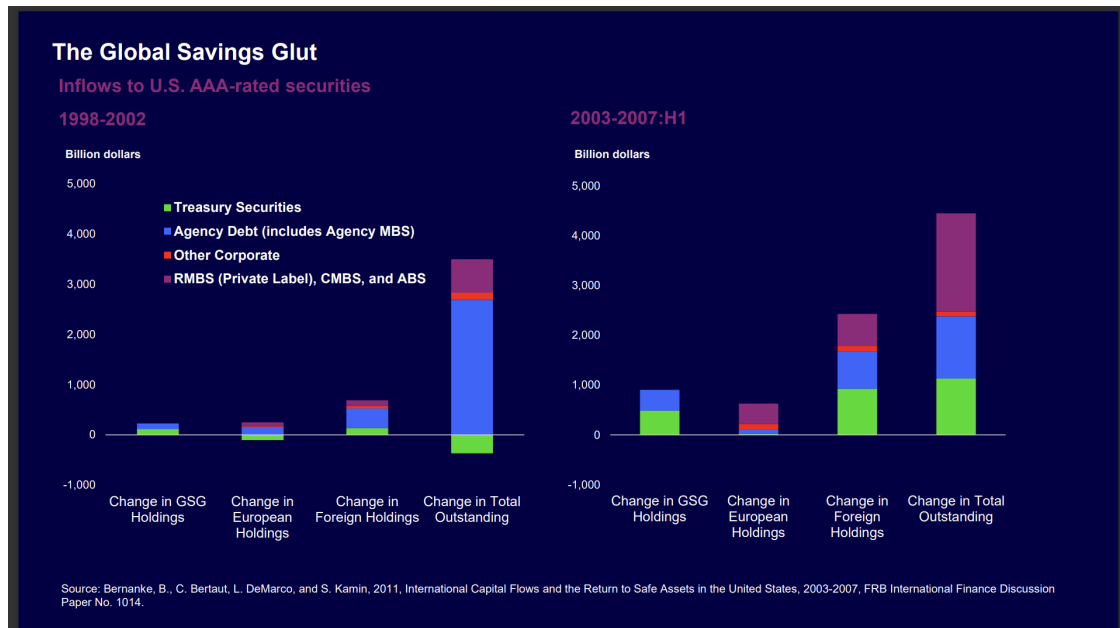


Figure 20. Lecture: Global Savings Glut Data, Part I 8:24.

4.5 Global Savings Glut Data, Part II

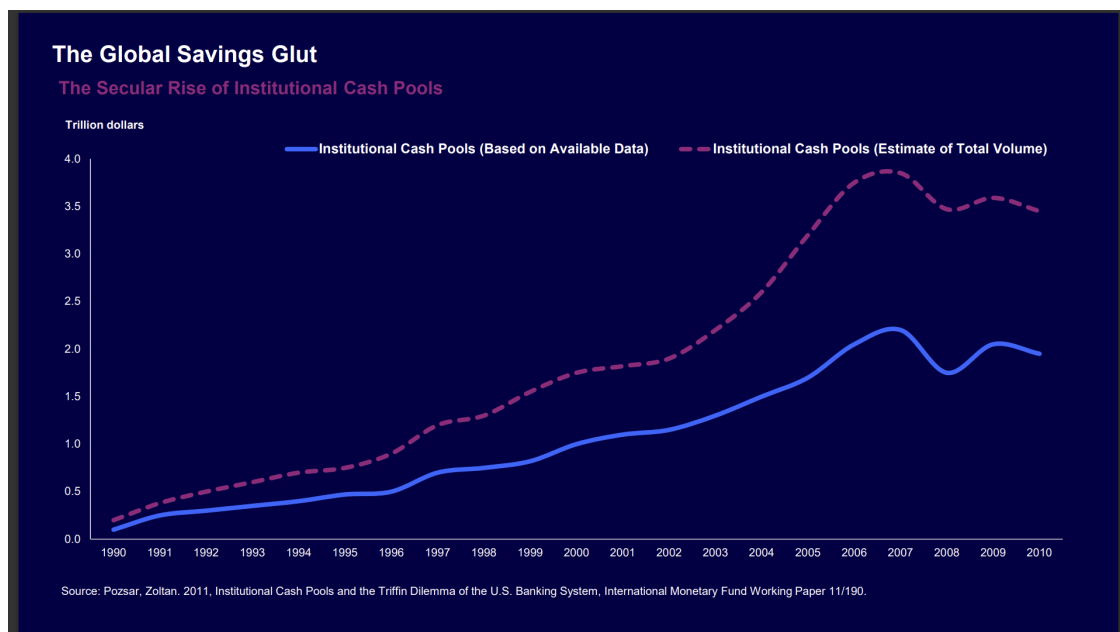


Figure 21. Lecture: Global Savings Glut Data, Part II 2:22.

4.6 Safe Asset Supply

Supply of Safe Assets						
Sources of Institutional Demand for Treasury Bills and Agency Debt						
Billion dollars	2005	2006	2007	2008	2009	2010
Outstanding Amounts						
Short-term Treasury securities	1,146	1,173	1,192	1,909	2,558	2,487
Short-term agency securities	568	489	560	903	844	618
Total	1,714	1,662	1,752	2,812	3,402	3,105
(-) Foreign Official Holdings						
Short-term Treasury securities	216	193	181	273	562	na
Short-term agency securities	112	110	80	130	34	na
Total	328	303	261	403	596	na
(-) Demand from Institutional Cash Pools						
Institutional cash pools (based on available data)	1,771	2,120	2,216	1,834	2,041	1,911
Institutional cash pools (estimate of total volume)	3,120	3,735	3,852	3,467	3,596	3,432
Average	2,446	2,928	3,034	2,651	2,819	2,672
= Deficit of safe, liquid, short-term products	-1,060	-1,569	-1,543	-242	-13	na

Source: Pozsar, Zoltan. 2011, Institutional Cash Pools and the Triffin Dilemma of the U.S. Banking System, International Monetary Fund Working Paper 11/190.

Figure 22. Lecture: Safe Asset Supply 2:22.

4.7 Manufacturing Safe Assets

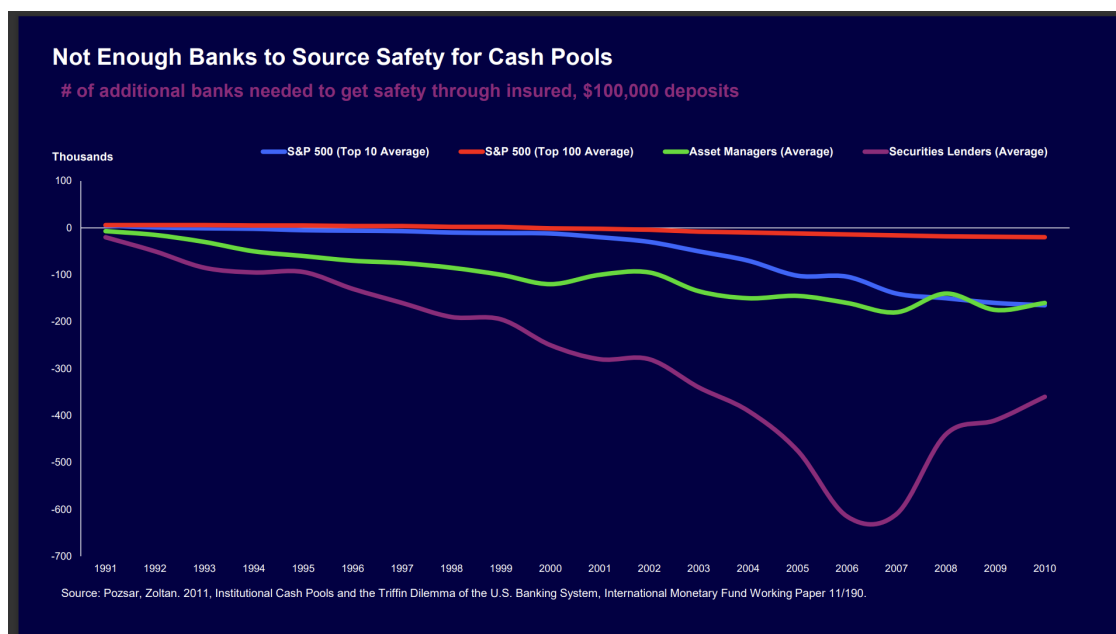


Figure 23. Manufacturing Safe Assets.

4.8 Securitization: Overview

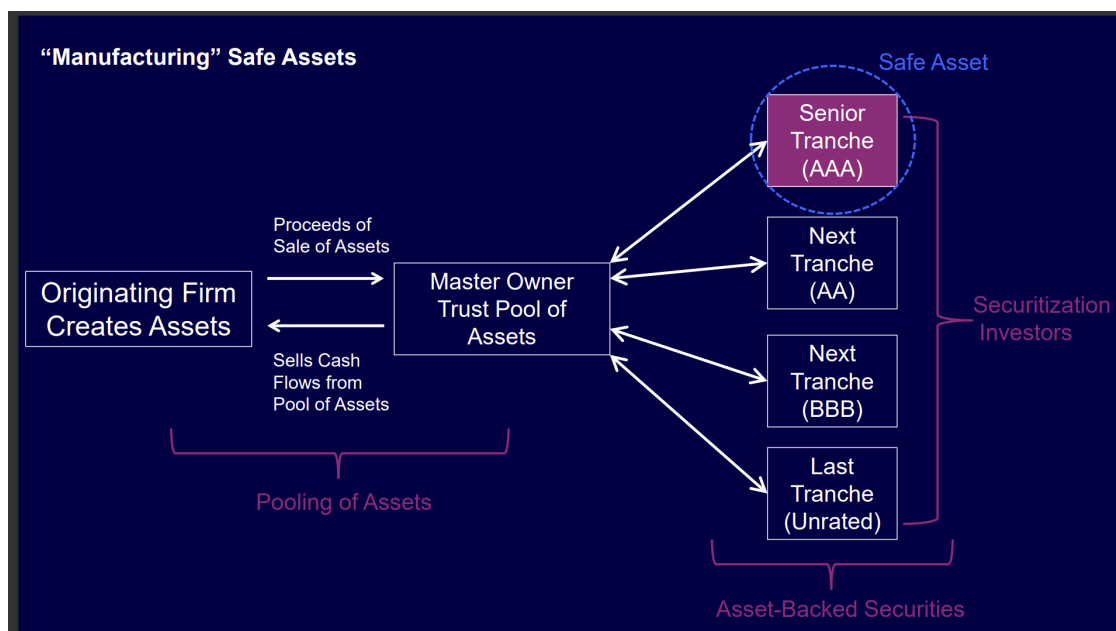


Figure 24. Lecture: Securitization: Overview 4:40.

FACTS: Each layer has to wait until the more senior layer has been paid before they get paid.

IDEA: The senior AAA tranche is considered safe as long as there is enough below it in the capital structure.

4.9 Securitization: Data, Part I

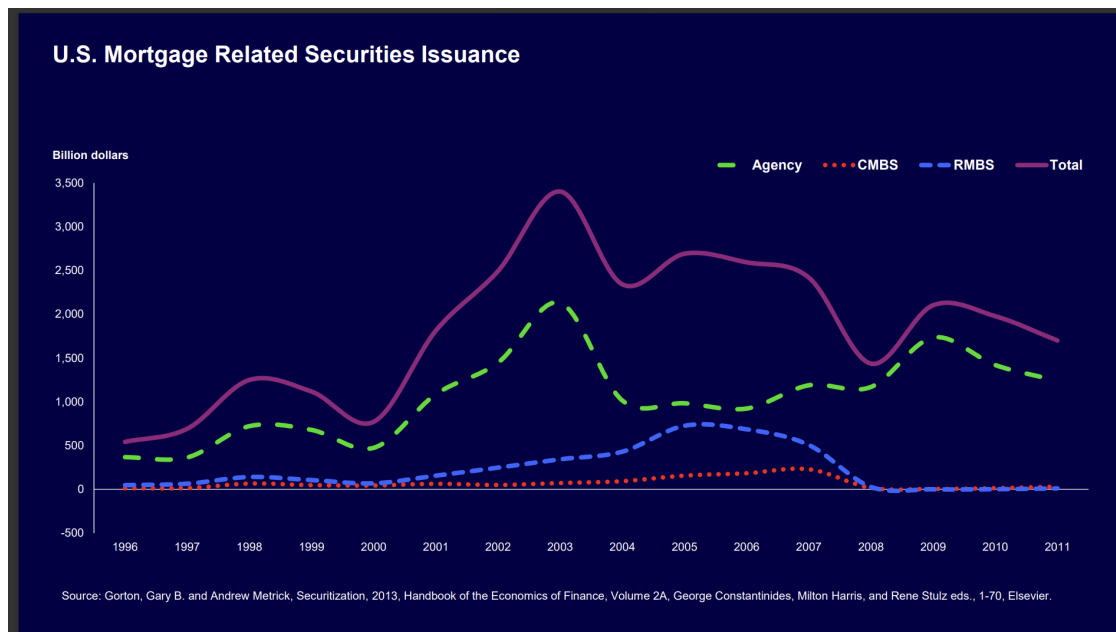


Figure 25. Securitization: Data, Part I 2:22.

4.10 Securitization: Data, Part II

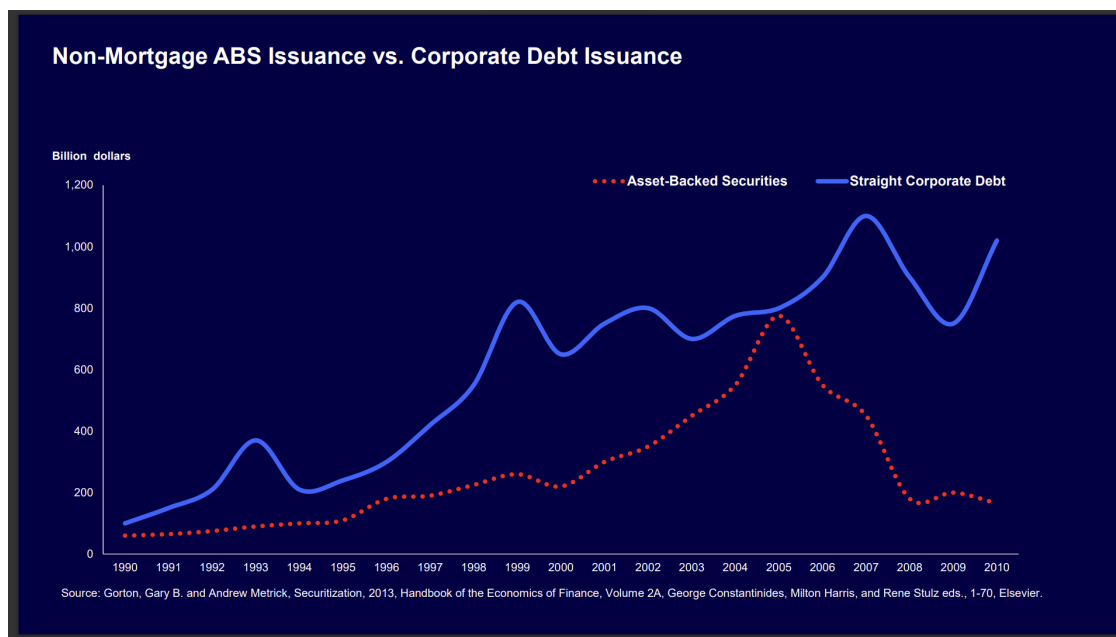


Figure 26. Securitization: Data, Part II 2:22.

FACTS: $\frac{2}{3}$ to $\frac{3}{4}$ of the securitization will receive the AAA highest rating.

5 The Housing Crisis

5.1 What Caused the Housing Crisis

IDEA:

1. bad behaviour: the mortgage demand comes from confused or misled buyers, mortgages that eventually self-destructed. Borrowers were moved into loans that weren't appropriate for them.
2. government failure
3. bubble thinking

FACTS: If you had a mortgage in the United States and you are itemizing your deductions on your tax return, then any interest you pay on your mortgage, you can deduct from your income before you pay taxes.

5.2 Bad Behavior, Part I

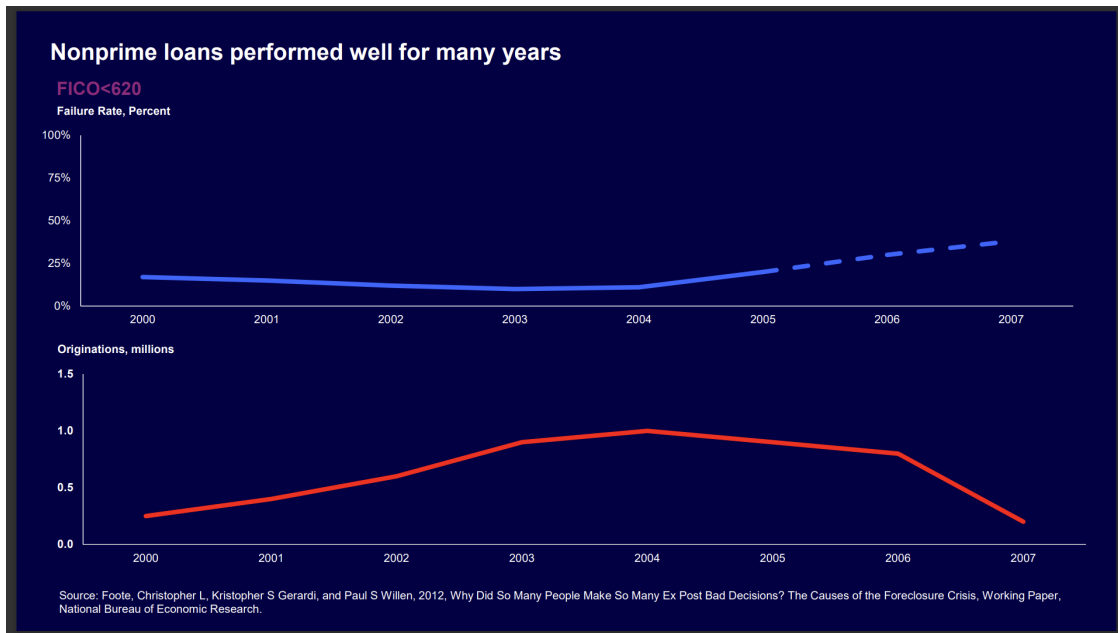


Figure 27. Lecture: Bad Behavior, Part I 1:05.

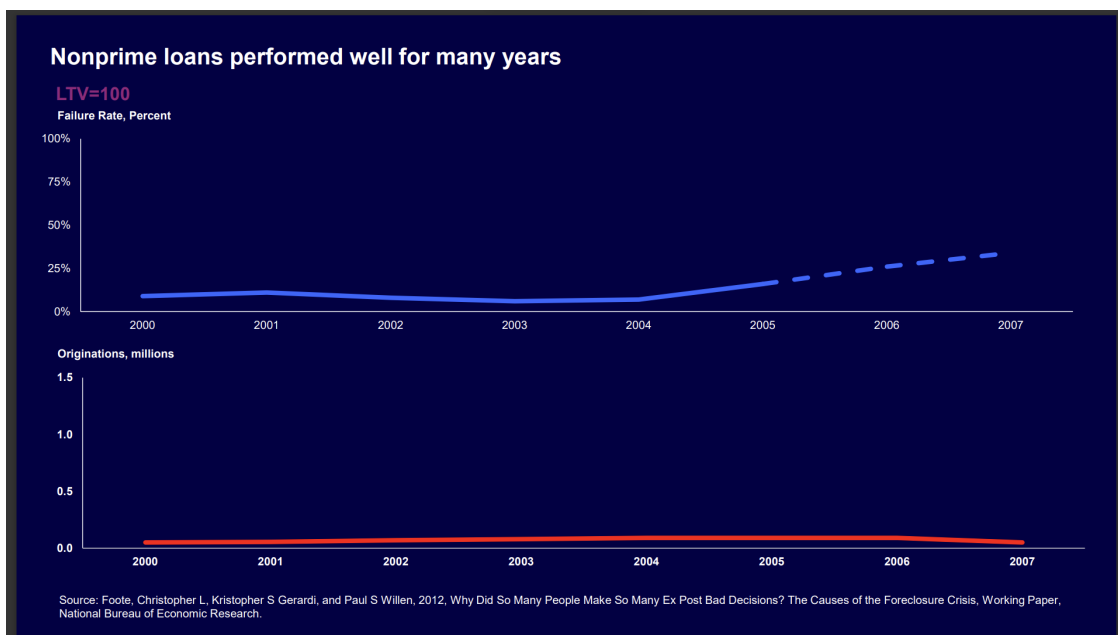


Figure 28. Lecture: Bad Behavior, Part I 3:07.

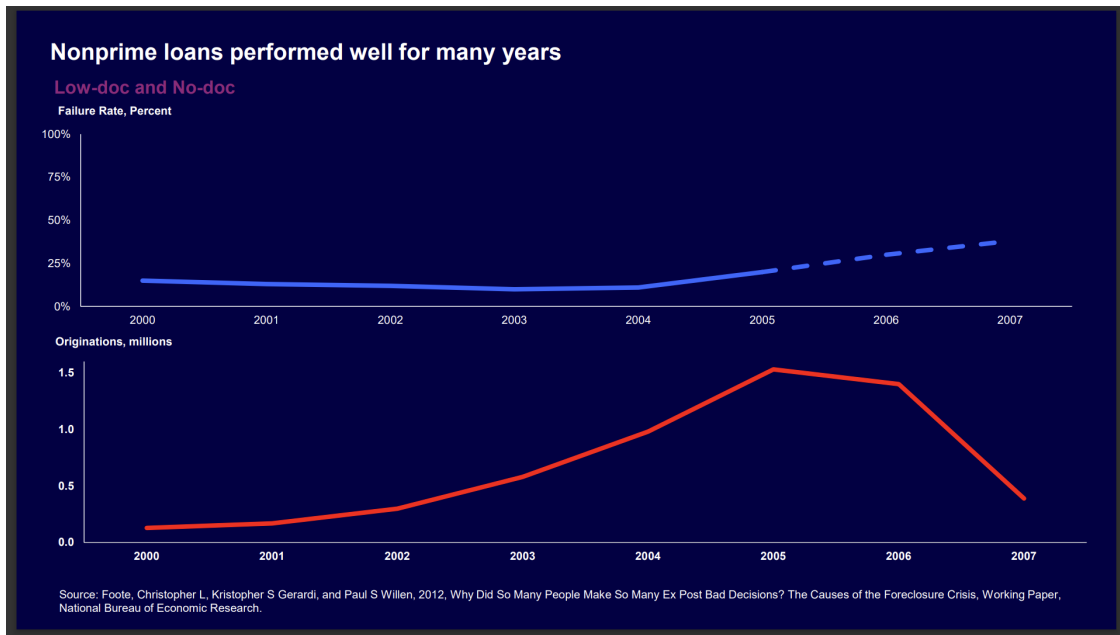


Figure 29. Lecture: Bad Behavior, Part I 3:48.

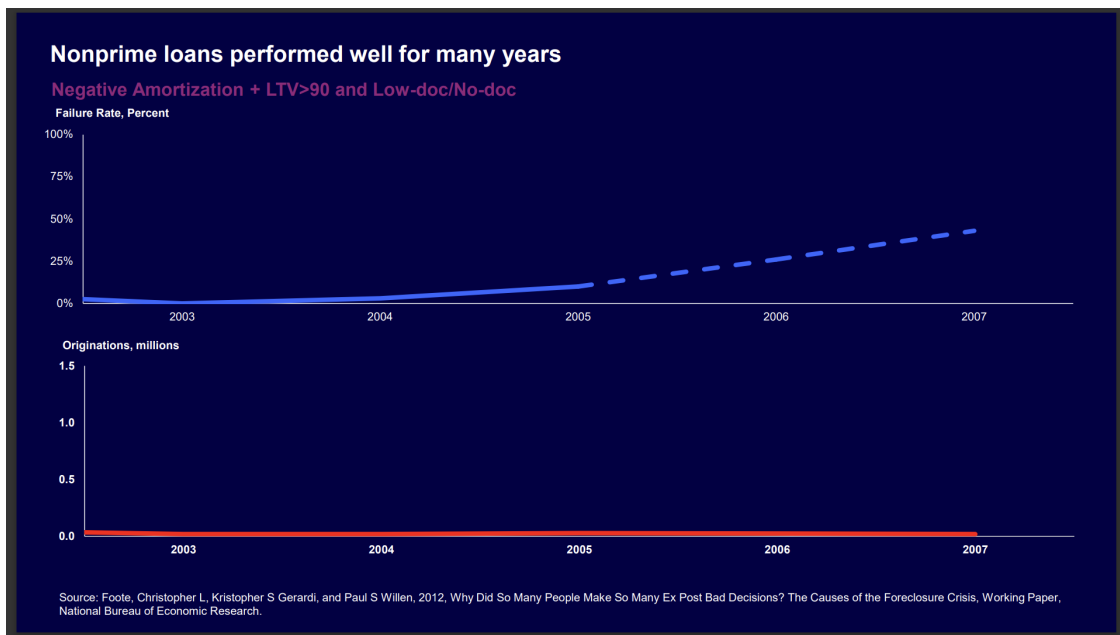


Figure 30. Lecture: Bad Behavior, Part I 5:05.

5.3 Bad Behavior, Part II

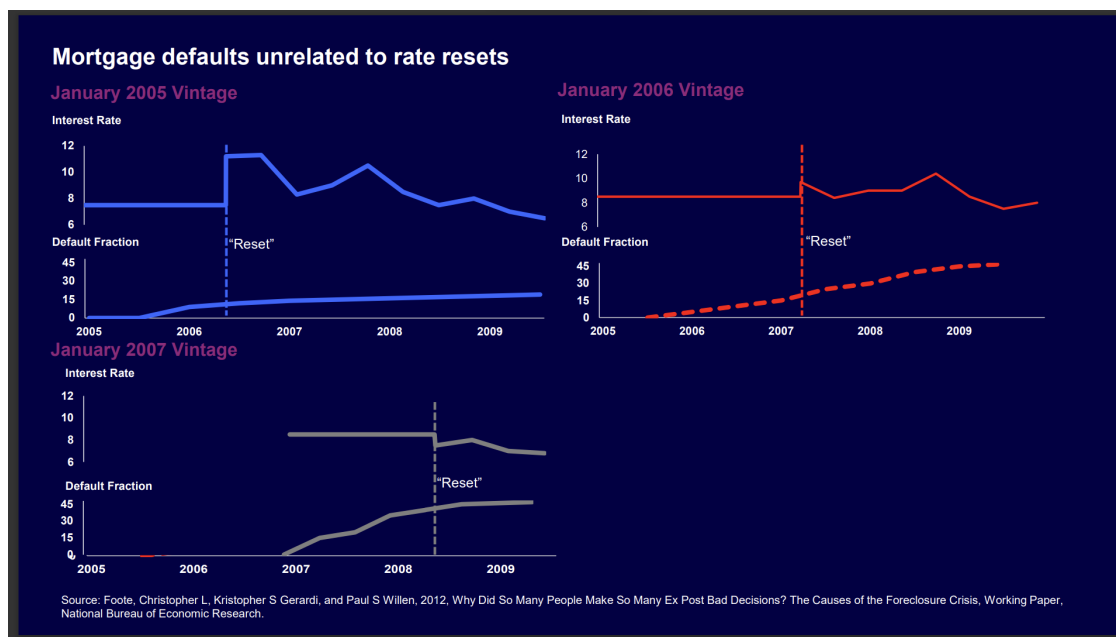


Figure 31. Lecture: Hypothesis 1: Bad Behavior, Part II 1:59.

5.4 Bad Behavior, Part III

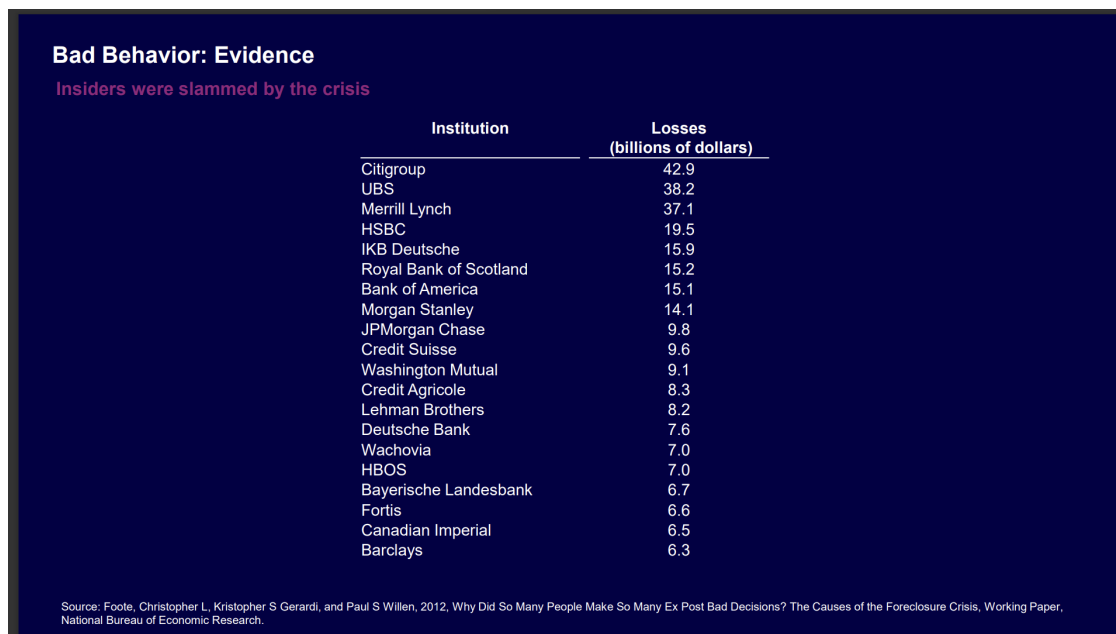


Figure 32. Lecture: Bad Behavior, Part III 1:20.

Bad Behavior: Evidence

Insiders had a lot of "skin in the game"

Entity	Loans	HELOC +2nds	Agency MBS	Non-Agency AAAs	CDOs (resi. subs)	Residential Subs	Total Exposure
<i>Billion dollars</i>							
US Banks/Thriffs	2,020	869	852	383	90	0	4,214
GSEs/FHLB	444	0	741	308	0	0	1,493
Broker/Dealers	0	0	49	100	130	24	303
REITs	0	0	82	10	0	0	92
Hedge Funds	0	0	50	51	0	24	125
Money Managers	0	0	494	225	0	24	743
Insurance Cos	0	0	856	125	65	24	1,070
Overseas	0	0	689	413	45	24	1,171
Financial Guarantors	0	62	0	0	100	0	162
Others	461	185	550	21	45	0	1,262
Total	2,925	1,116	4,362	1,636	476	121	10,680

Source: Foote, Christopher L., Kristopher S Gerardi, and Paul S Willen, 2012, Why Did So Many People Make So Many Ex Post Bad Decisions? The Causes of the Foreclosure Crisis, Working Paper, National Bureau of Economic Research.

Figure 33. Lecture: Bad Behavior, Part III 3:00.

5.5 Government Failure, Part I

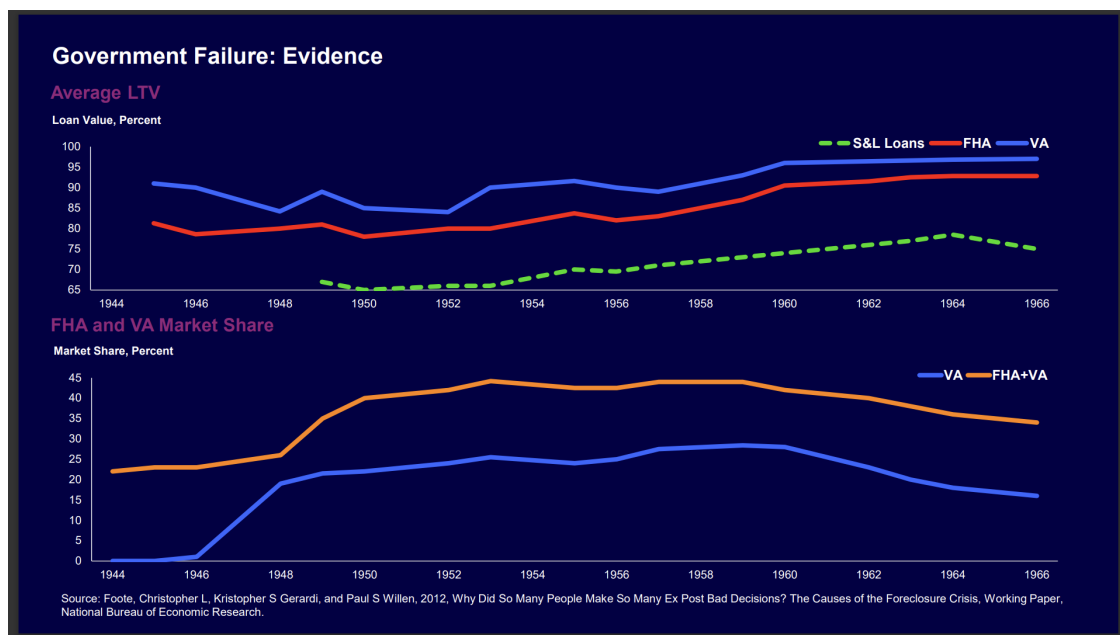


Figure 34. Lecture: Government Failure, Part I 2:22.

5.6 Hypothesis 2: Government Failure, Part II

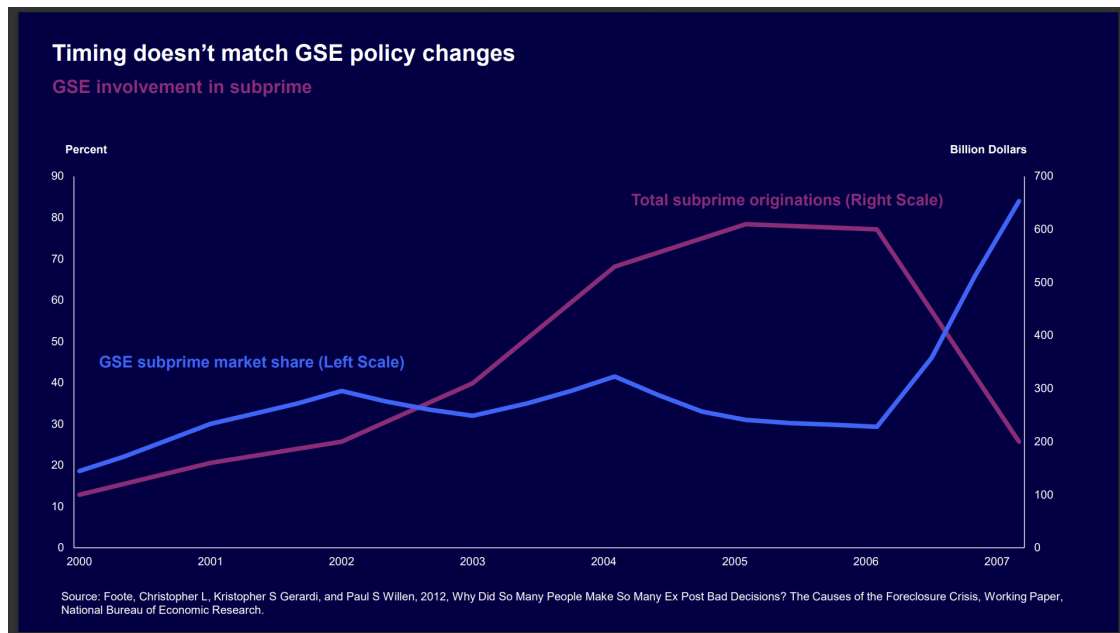


Figure 35. Lecture: Hypothesis 2: Government Failure, Part II 3:33.

5.7 Hypothesis 3: Bubble Thinking

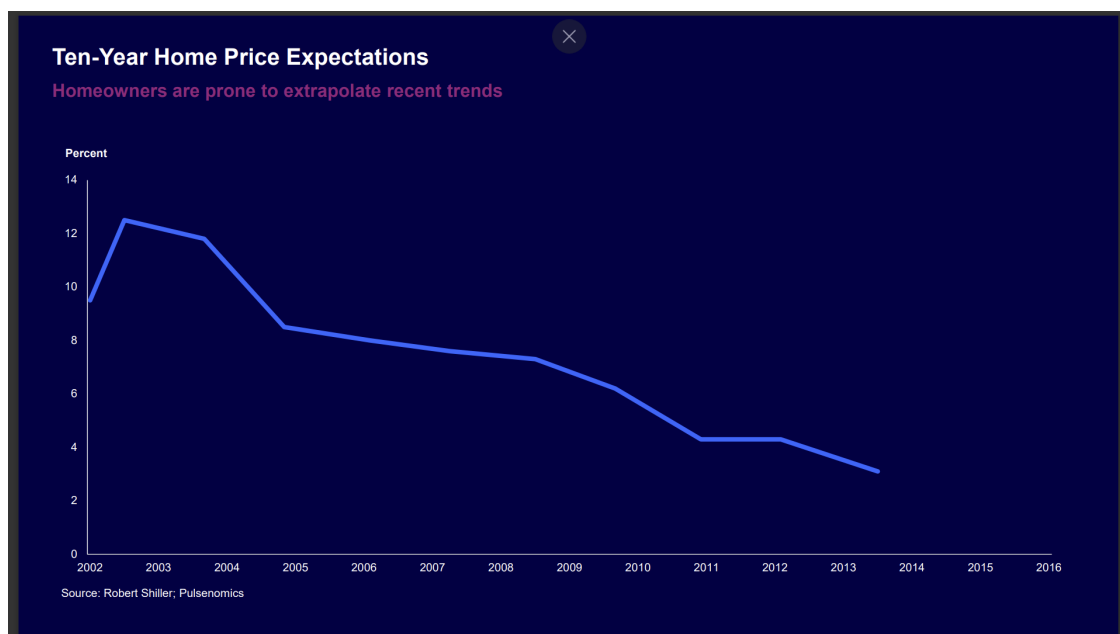


Figure 36. Lecture: Hypothesis 3: Bubble Thinking 2:00.

6 Anxiety, Part I

6.1 Some Notable Events in 2007

6.2 ABX

DEFINITION: ABX-HE, or "asset-backed index - home equity", is an index of credit default swaps (CDS) written on subprime mortgage securitizations. The price of the ABX index is a measure of the perceived value of subprime securities with various ratings. The return (or spread) on the ABX is a risk premium for subprime.

6.3 ABX Data

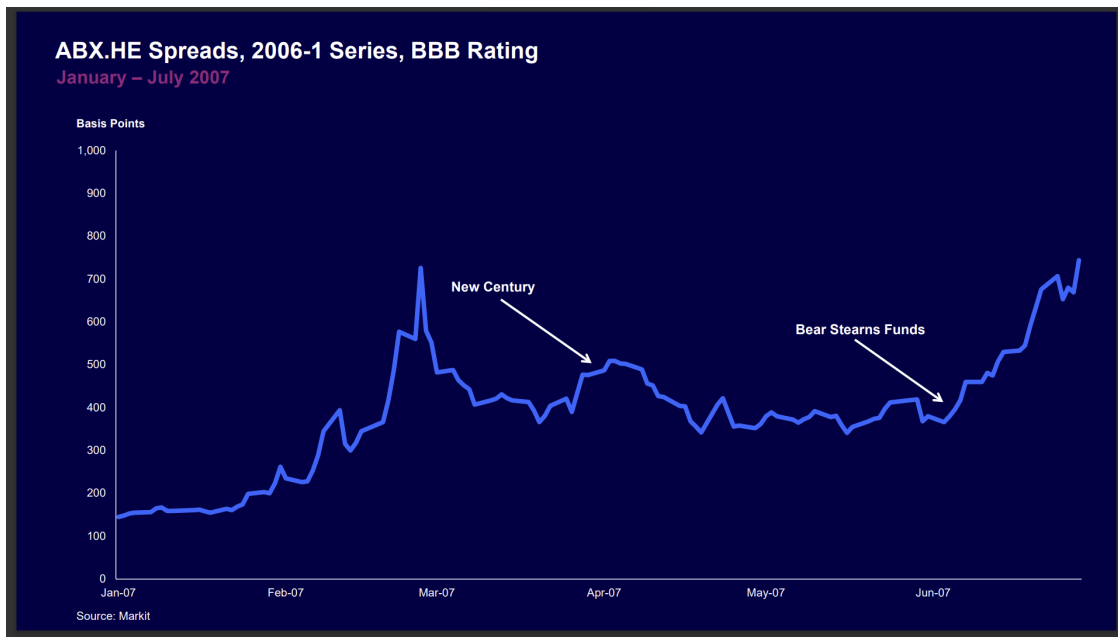


Figure 37. Lecture: ABX Data 1:00.

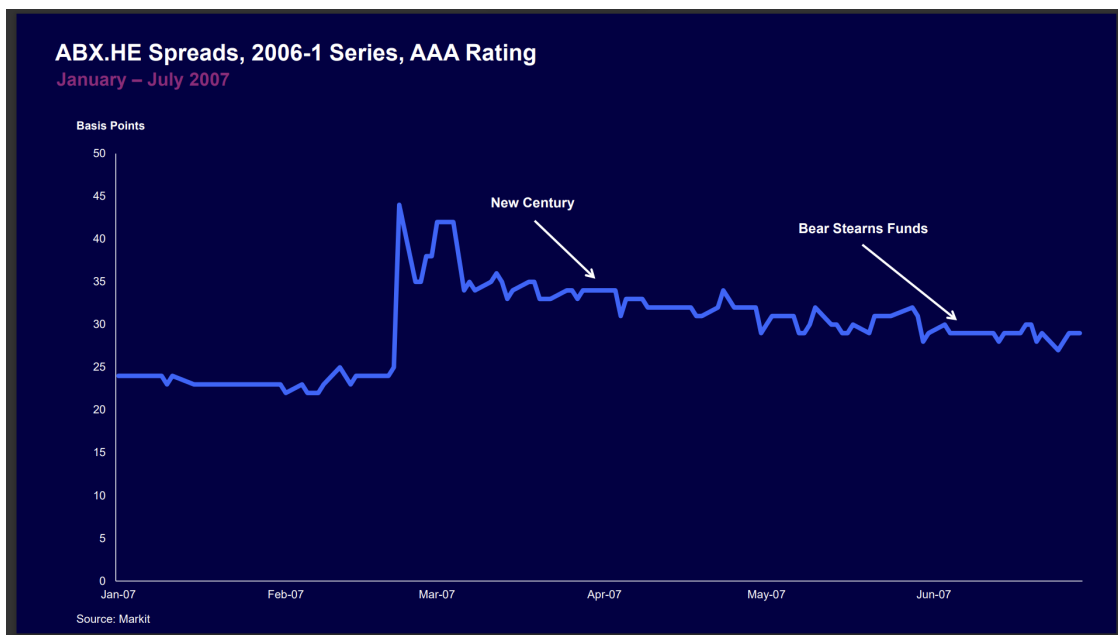


Figure 38. Lecture: ABX Data 4:00.

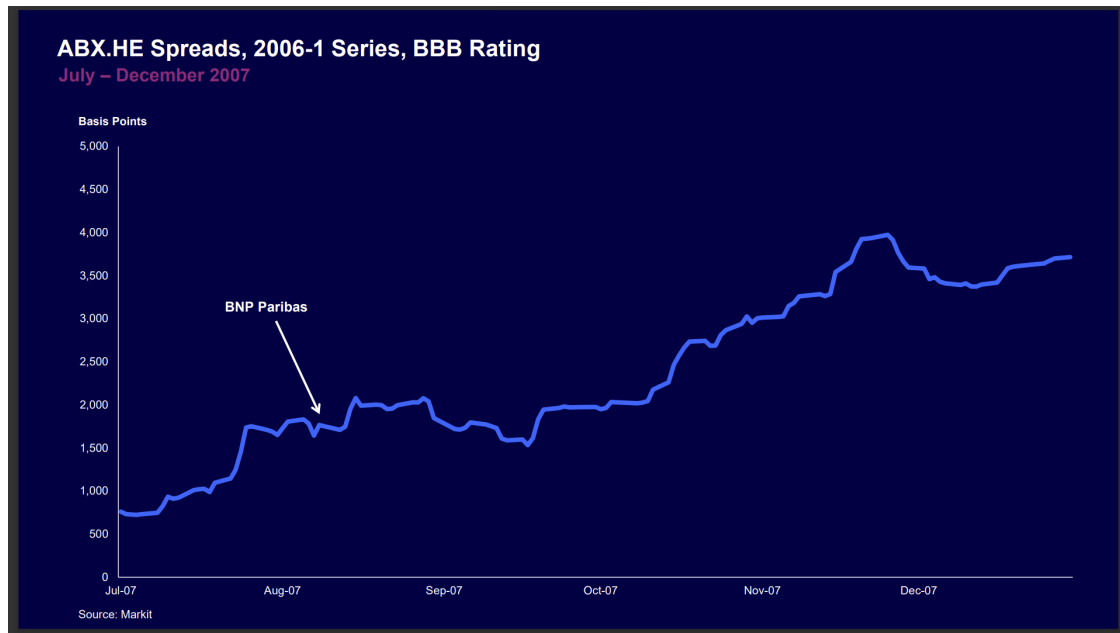


Figure 39. Lecture: ABX Data 5:00.

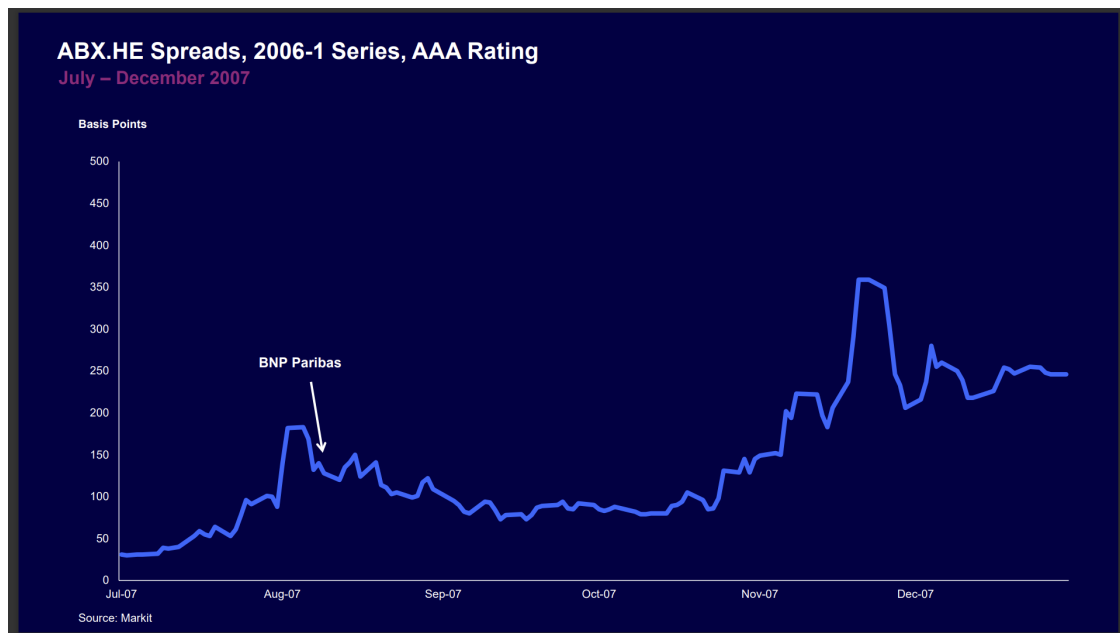


Figure 40. Lecture: ABX Data 6:00.

6.4 Anxiety Spreads to Interbank Markets

6.5 LIBOR-OIS

DEFINITION: The London Interbank Offered Rate (LIBOR) is a measure of the interest rates that banks charge each other for unsecured dollar funding. The Overnight Index Swap (OIS) is a fixed-floating interest-rate swap.

FACTS: Because the amounts owed daily are small and counterparties must continuously post collateral for expected payments, the fixed leg of this swap is considered to be a good proxy for risk-free interest rates. The LIBOR-OIS spread is thus a good measure for the riskiness of banks' unsecured borrowing. Historically, this spread was very small (around 10 basis points).

6.6 LIBOR-OIS Data

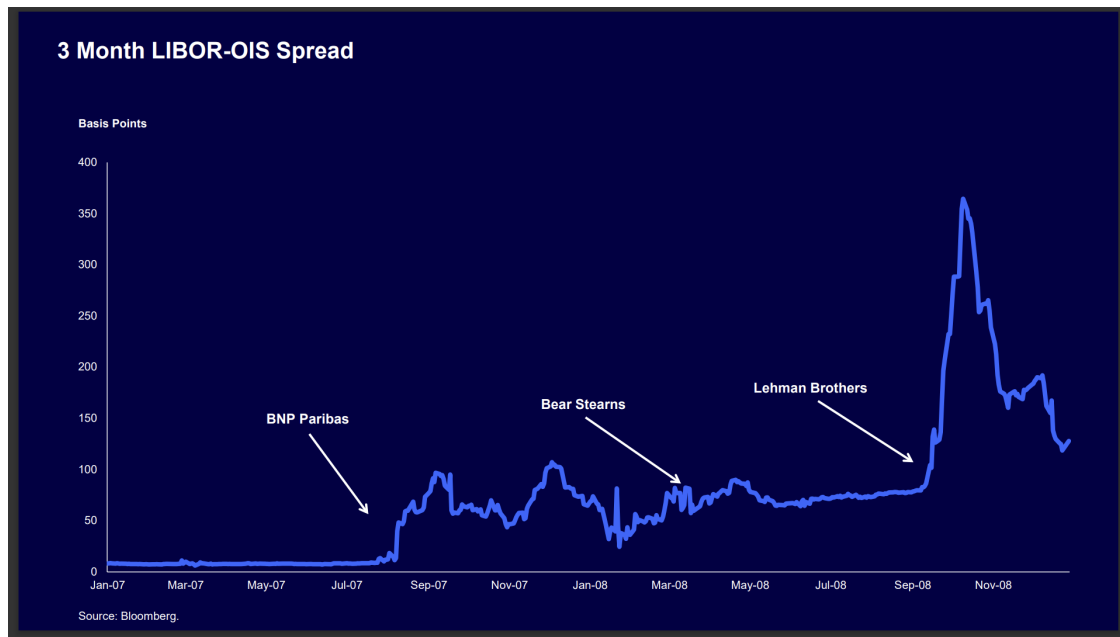


Figure 41. Lecture: LIBOR-OIS Data 2:00.

6.7 Lecture: Asset-Backed Commercial Paper

FACTS: BNP Paribas's announcement created enormous amount of anxiety. That anxiety led to pressure. On the inter-bank markets that lasted for the 13 months, between BNP Pariba and the fall of Lehman. And then, after the bankruptcy of Lehman Brothers in September of 2008, there was an all out run on the banking system.

DEFINITION: Asset-backed commercial paper (ABCP) is primarily a method of maturity transformation – funding a pool of long-term assets with short-term liabilities.

IDEA: ABCP is designed to meet specific needs of investors (often money-market mutual funds), and includes credit enhancement and liquidity support to achieve this goal.

FACTS: ABCP may appear similar to securitization, but there are many differences:

- Investments can be revolving and fluctuate in size
- Conduits may invest in various asset types
- Typically engage in maturity transformation, with backup liquidity support
- No scheduled amortization of assets and liabilities

6.8 ABCP Data

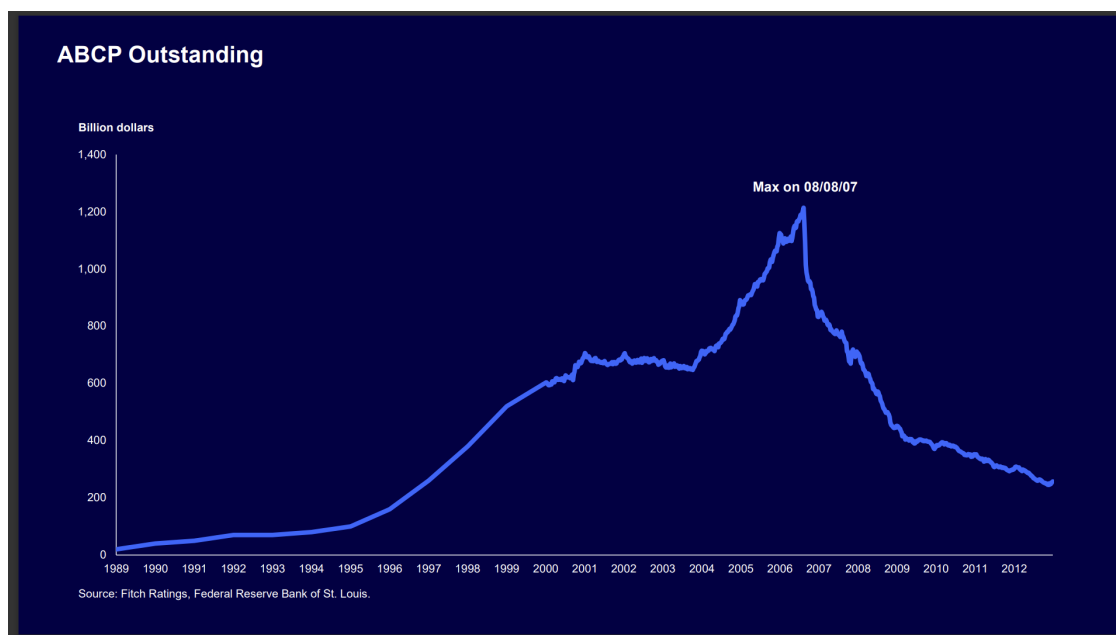


Figure 42. Lecture: ABCP Data 1:00.

FACTS: This was a business that many types of institutions were engaged in. And they were engaged in with a variety of different techniques

ABCP Program Types

ABCP programs by program type, as of July 2007

Program Type	Assets	Number of Programs	Market Share by Outstanding (%)	Programs with Extendible Paper (%)	Programs with Credit Support (%)	Programs with Large US Bank Sponsors (%)	Programs with Small US Bank Sponsors (%)	Programs with Foreign Bank Sponsors (%)	Programs with Nonbank Sponsors (%)
Multi-seller	Diversified receivables and loans	98	45	19	30	19	4	57	20
Nonmortgage single seller	Credit-car receivables and auto loans	40	11	62	28	8	10	8	74
Mortgage single seller	Mortgages and mortgage-backed securities (MBS)	11	2	67	0	9	9	0	82
Securities arbitrage	Highly rated long-term securities, including MBS	35	13	9	17	14	11	66	9
Structured investment vehicle	Highly rated long-term securities, including MBS	35	7	0	0	11	0	23	66
CDO	Highly rated long-term securities, including MBS	36	4	25	0	0	3	3	94
Hybrid and other N.A.		84	18	20	10	5	2	12	81
Total		339	100	24	16	10	5	30	55

Figure 43. Lecture: ABCP Data 3:00.

6.9 ABCP Runs, Part I

DEFINITION: ABCP “run” is a week when the program does not issue new CP despite having at least 10% of outstanding CP mature.

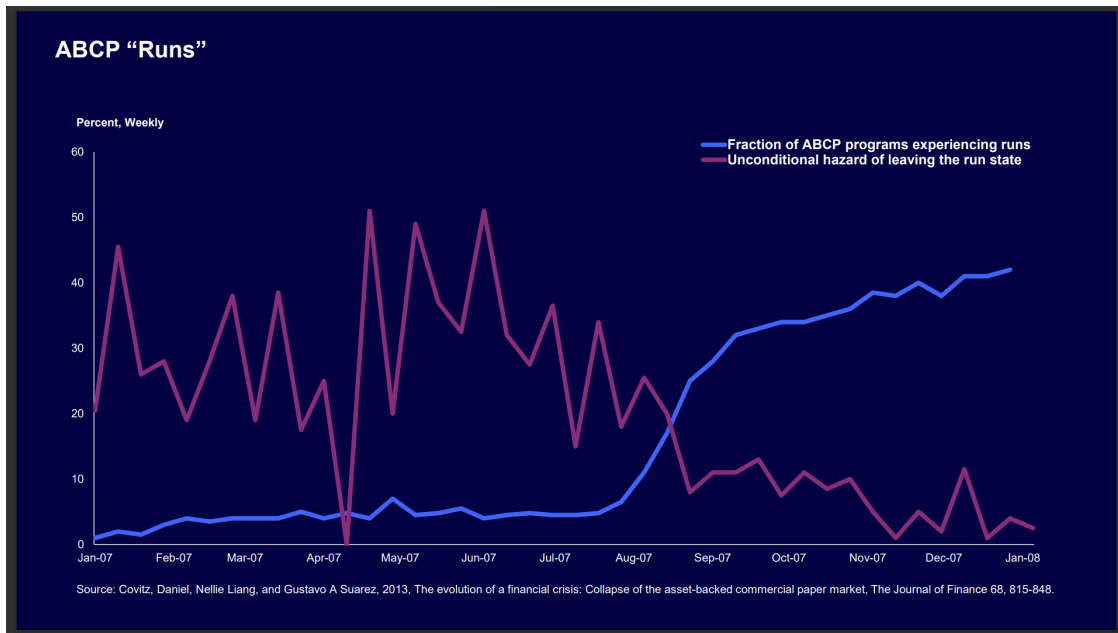


Figure 44. Lecture: ABCP Runs, Part I 3:00.

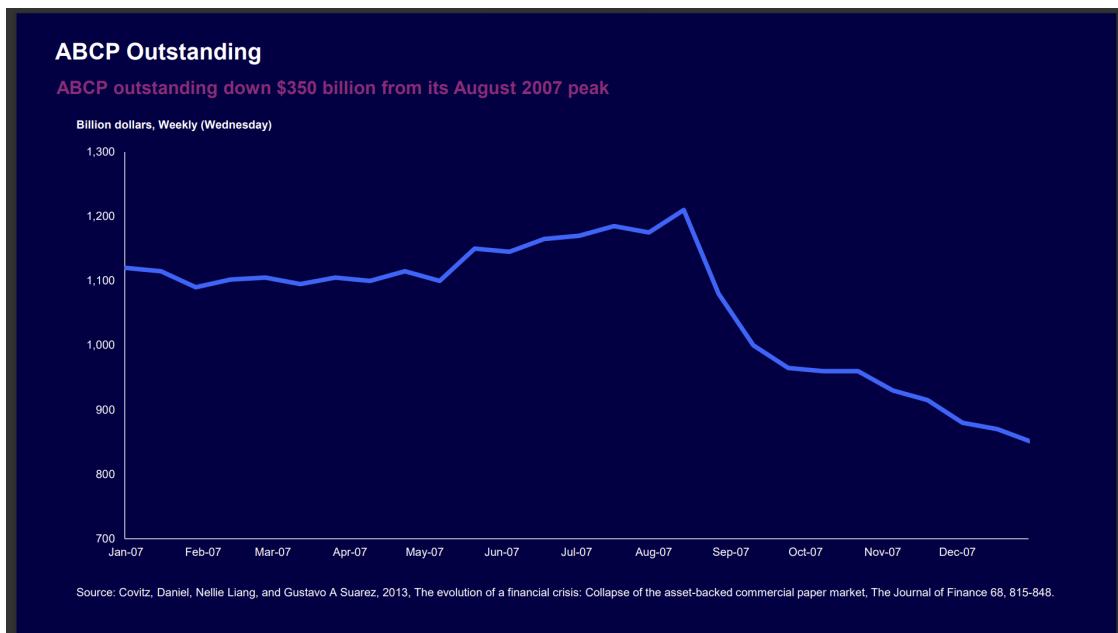


Figure 45. Lecture: ABCP Runs, Part I 5:00.

6.10 ABCP Runs, Part II

DEFINITION: Pressure on interbank markets: financial institutions used to be receiving funding largely from money market mutual funds and other lenders that are restricted to only purchasing short-term debt, and now they need to run into the markets to find a source of financing, either secured or unsecured, that is different.

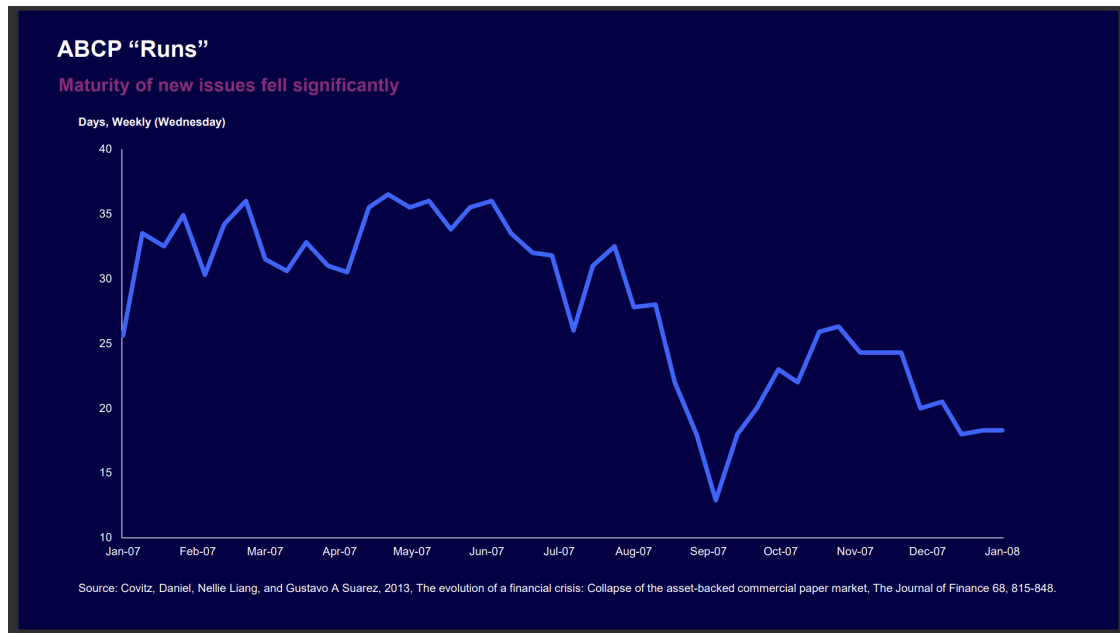


Figure 46. Lecture: ABCP Runs, Part II 2:00.

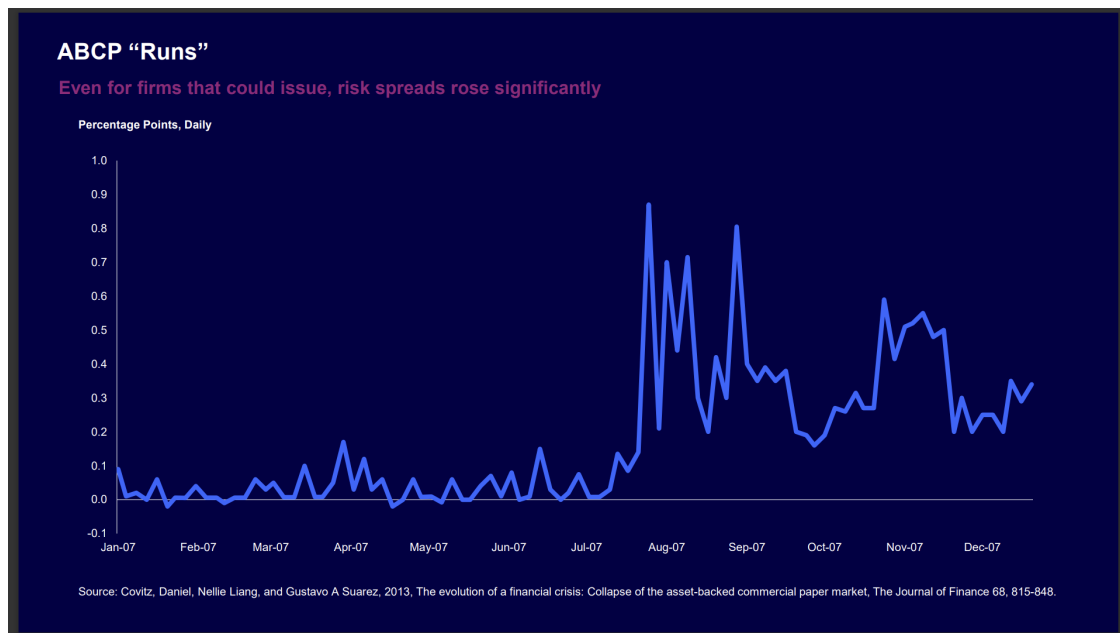


Figure 47. Lecture: ABCP Runs, Part II 3:00.

7 Anxiety, Part II

7.1 Northern Rock: Narrative

7.2 Northern Rock and the Global Financial Crisis

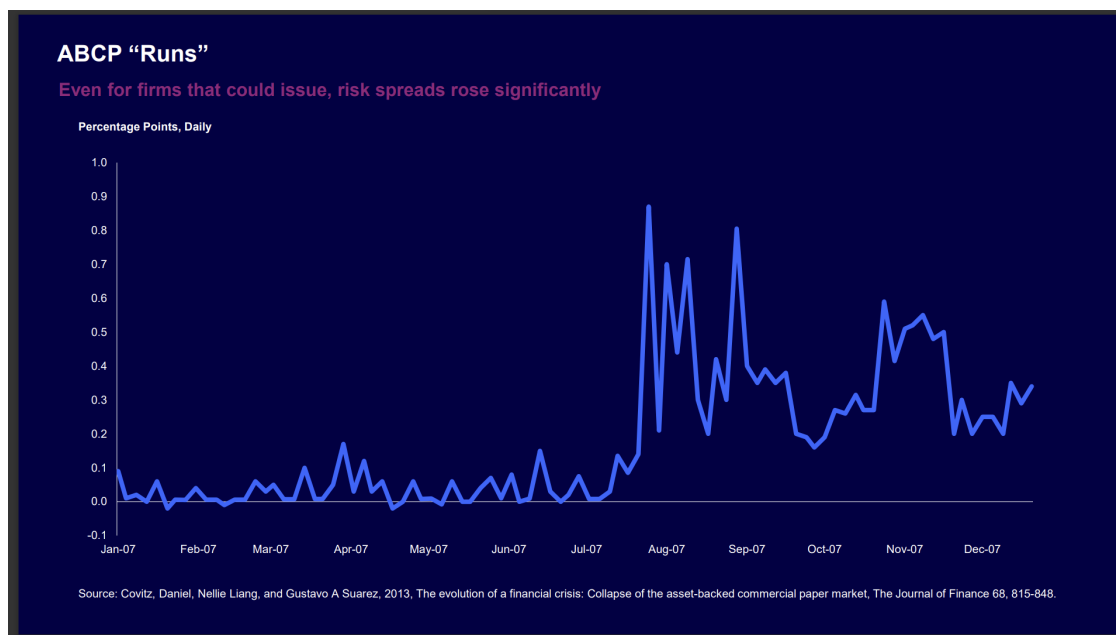


Figure 48. Lecture: ABCP Runs, Part II 3:00.

7.3 The "Monolines"

DEFINITION: Since they began in the 1970s, bond insurers' main business was to provide credit enhancement to municipal bond offerings. This insurance was provided by specialized companies that do not sell other types of insurance products – hence the name "monolines". In the roaring 2000s, they expanded into insurance provision for structured products (securitizations and CDOs).

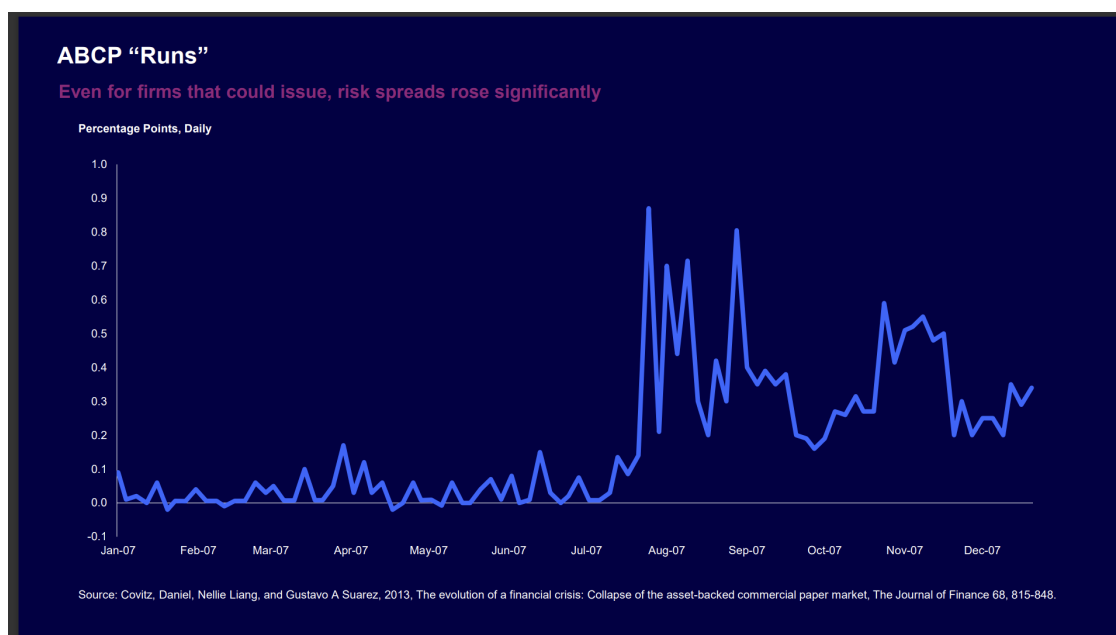


Figure 49.

7.4 Auction Rate Securities

IDEA: The traditional way to sell securities is with a primary offering, followed by the management of a secondary market. For some types of securities, the secondary markets are so illiquid that investors are scared away. Auction rate securities (ARS) are a solution to illiquidity, with a broker-dealer holding periodic auctions of long-term bonds to reset the interest rates based on demand. Historically, broker-dealers' reputational concerns meant that they provided a backstop to the market, holding paper to manage short-term liquidity disruptions. ARS are used for a very quiet type of securities: mostly student-loan pools and municipals. The municipal securities were given credit enhancement by the monolines.

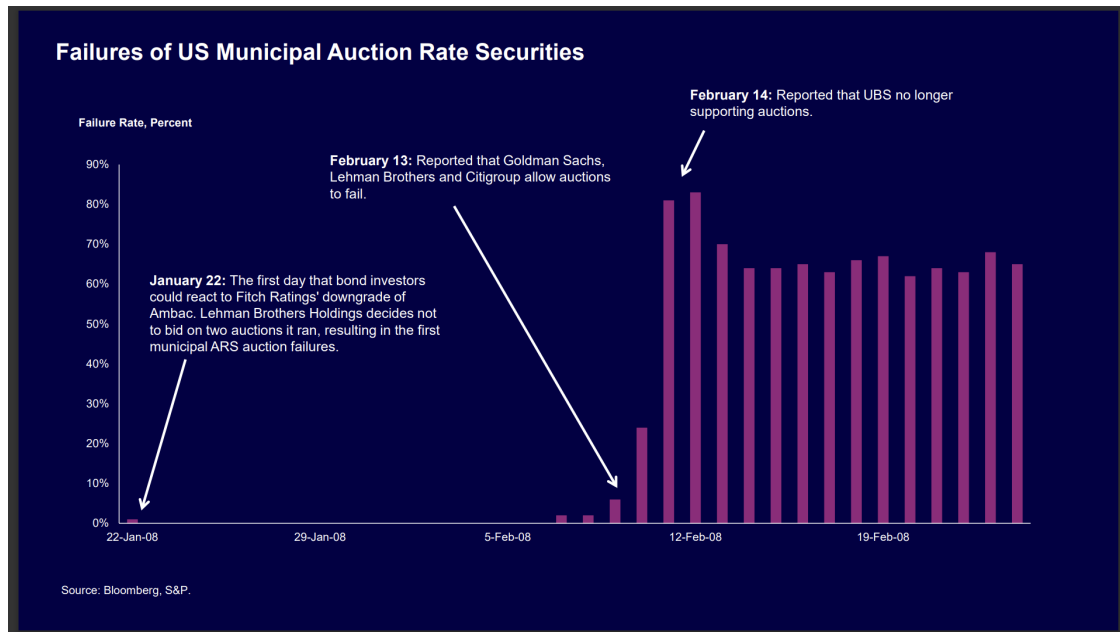


Figure 50.

7.5 Bear Stearns, Part I

7.6 Bear Stearns, Part II

7.7 Bear Stearns, Part III

8 Panic

8.1 Shadow Banking

8.2 Repo

DEFINITION: A sale and repurchase agreement ("repo") is a short-term deposit of cash receives interest, and is backed by collateral. Depositor takes legal ownership of the collateral. Carved out of Bankruptcy Code; unilateral termination by non-defaulting party. Two types of repo: bilateral and tri-party. Both types caused trouble in the crisis. Collateral may be "rehypothecated". Collateral value typically exceeds the amount of cash deposited, this is called a haircut.



Figure 51.

Financing of Broker-Dealer Financial Instruments

A New Type of Banking

Million Dollars	May 31, 2008	May 31, 2008	May 31, 2008	June 27, 2008	Feb. 29, 2008	2 nd Quarter Total
	Morgan Stanley	Goldman Sachs	Lehman	Merrill Lynch	Bear Stearns	Total
Financial Instruments Owned	390,393	411,194	269,409	288,925	141,104	1,501,025
of which pledged	194,492	158,363	123,031	80,537	76,903	633,326
% owned financial instruments pledged	50%	39%	46%	28%	55%	42%

Figure 52.

8.3 The GSEs

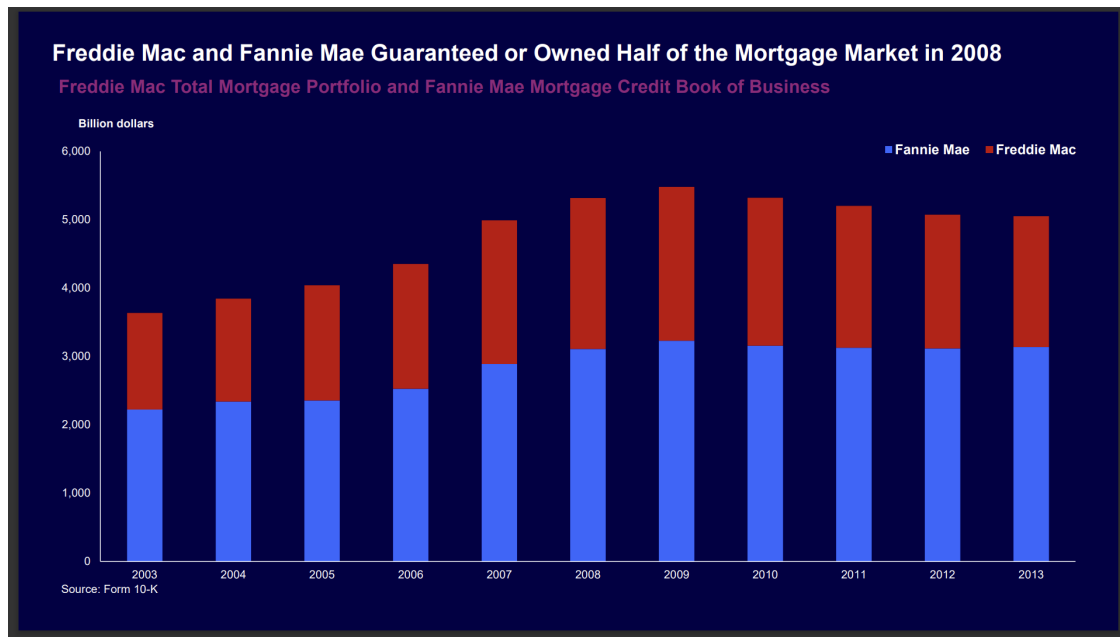


Figure 53.

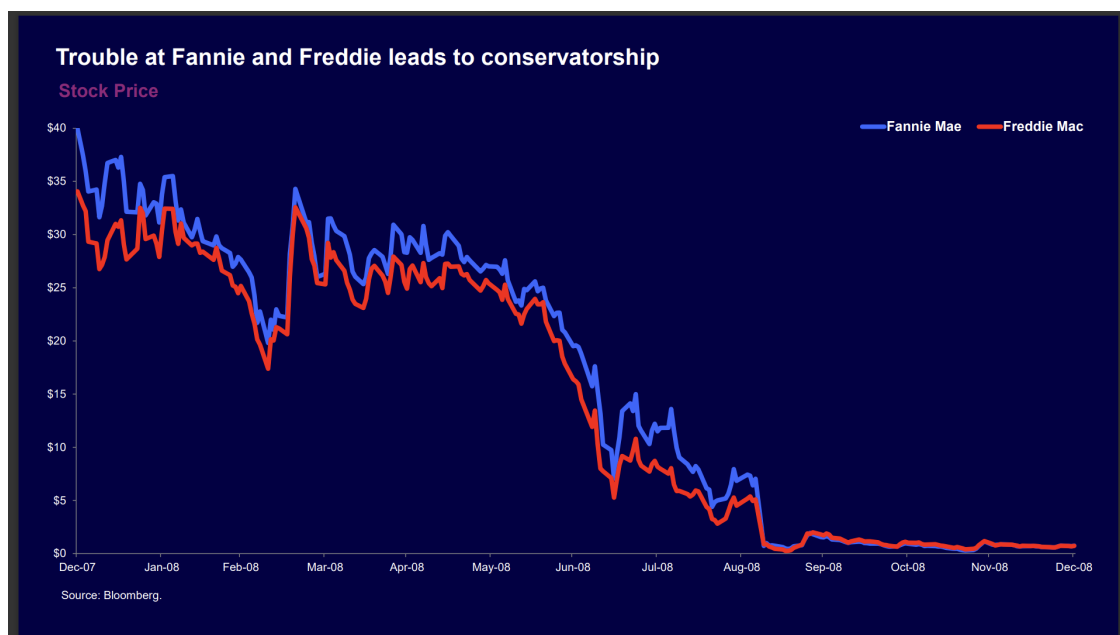


Figure 54.

8.4 Lehman Brothers

FACT: On September 10, 2008, Lehman reported \$28 billion in shareholder equity, \$4 billion higher than a year earlier. But it was simply impossible to know if this equity cushion was accurate.

FACT: Lehman reported \$54 billion in real estate assets. Some market participants thought the true value was closer to half of that, which would effectively wipe out Lehman's equity. At the same time, Lehman's counterparties in derivatives, commercial paper, and repo were pulling back, shortening terms, and demanding more collateral.

FACT: Over the weekend of 9/12 – 9/14, the U.S. government tried unsuccessfully to arrange a private rescue for Lehman. The government insisted there would be no public money spent on the rescue. Bank of America chose to buy

Merrill Lynch instead of Lehman. On Saturday, Barclays agreed to buy Lehman, but by Sunday the deal was effectively blocked by UK regulators. Without sufficient liquidity to operate the next day, and otherwise out of options, Lehman filed for bankruptcy early in the morning on September 15.

8.5 MMMFs

DEFINITION: Money-Market Mutual Funds (MMMFs) are a specific type of investment company that is only permitted to own a narrow range of securities. In return for accepting this narrow investment range, they had the right (at this time) to report “stable values” for their share prices. On September 16, 2008, Reserve Primary Fund “broke the buck” due to exposure to Lehman Brothers commercial paper. This led to a run on many MMMFs – mostly by institutional investors – and then quickly to an explicit guarantee from the U.S. government. We really should have seen this coming – but we did not. Because MMMFs had significant problems in August 2007 as a result of the Asset-Backed Commercial Paper (ABCP) runs.

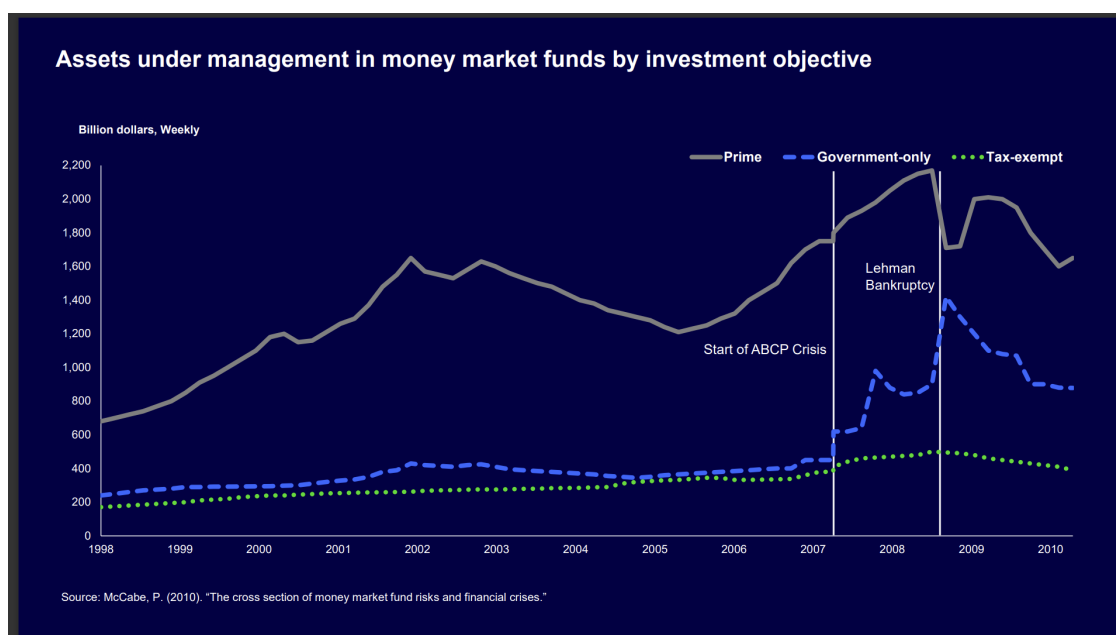


Figure 55.

8.6 AIG

DEFINITION: Main weaknesses:

- Credit-default-swap (CDS) mark-to-market losses and collateral calls.
- cash collateral investment losses in securities lending business.
- Funding pressure in CP and repo.
- Ratings downgrade triggers additional collateral calls
- Liquidity puts on CDOs.

After Lehman, markets are in turmoil and no private rescue is possible. Fed led rescue of \$85 billion, later supplemented by more from Fed and TARP

8.7 The Run on Repo

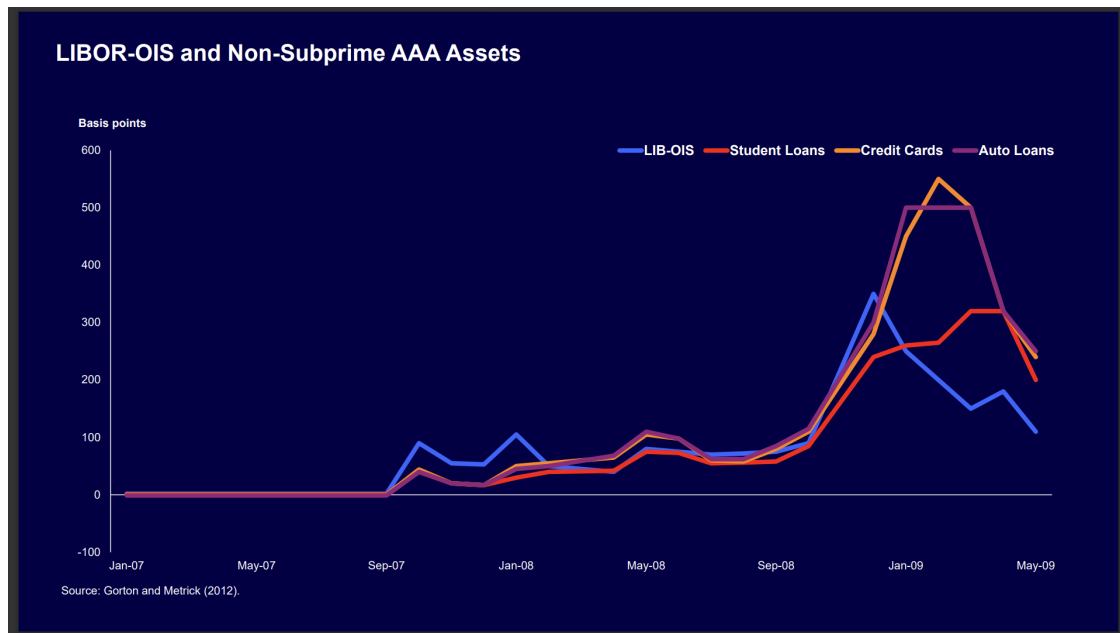


Figure 56.

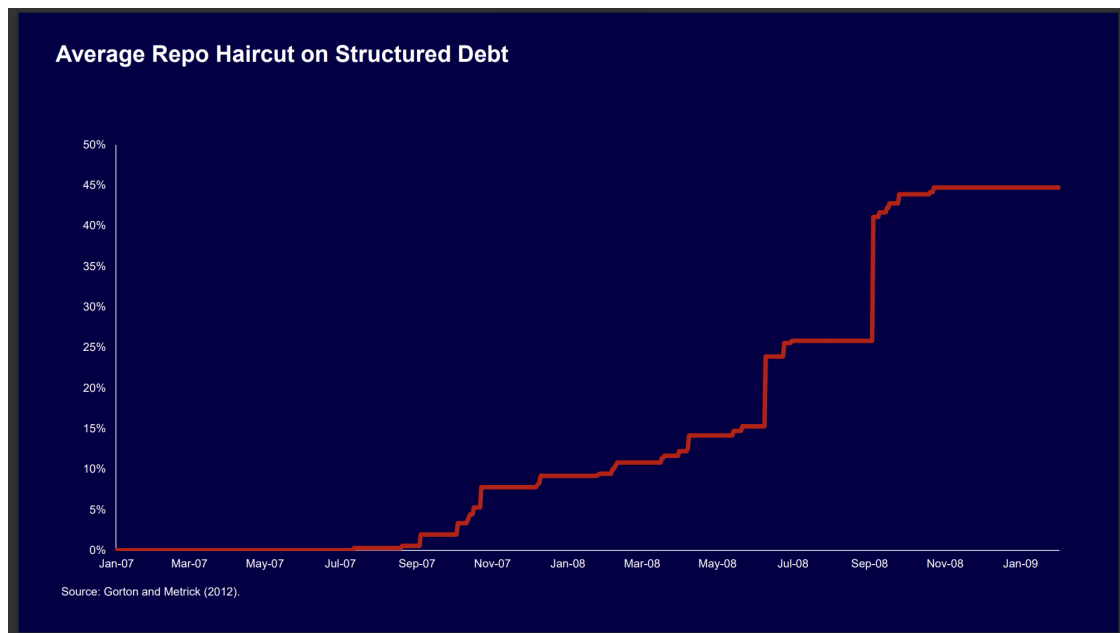


Figure 57.

9 Our Response to the Crisis

9.1 United States Financial Capacity

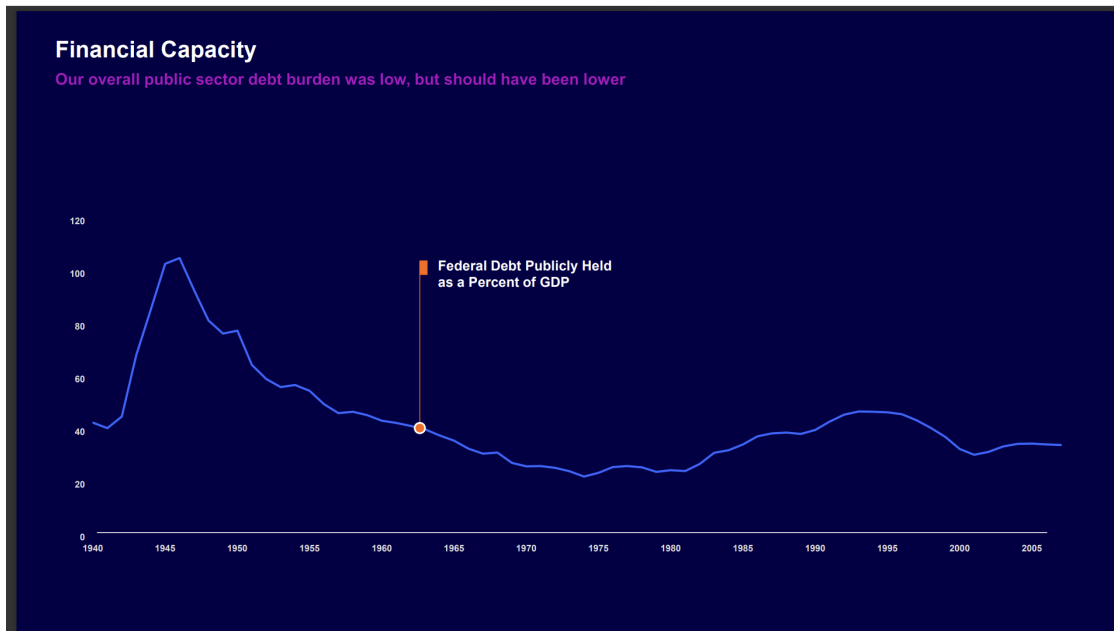


Figure 58.

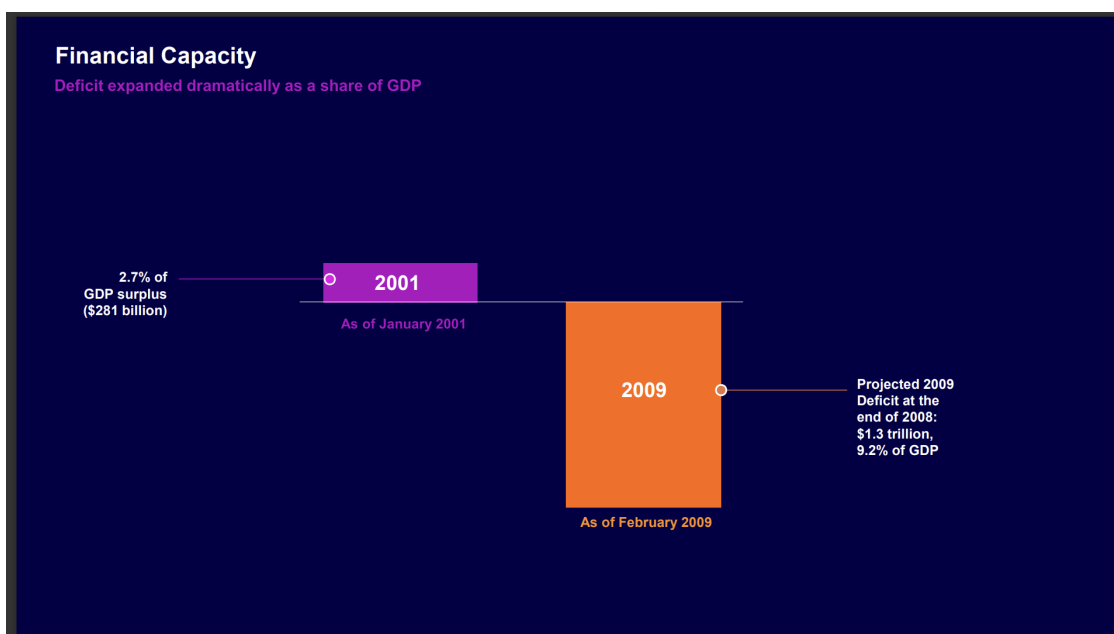


Figure 59.

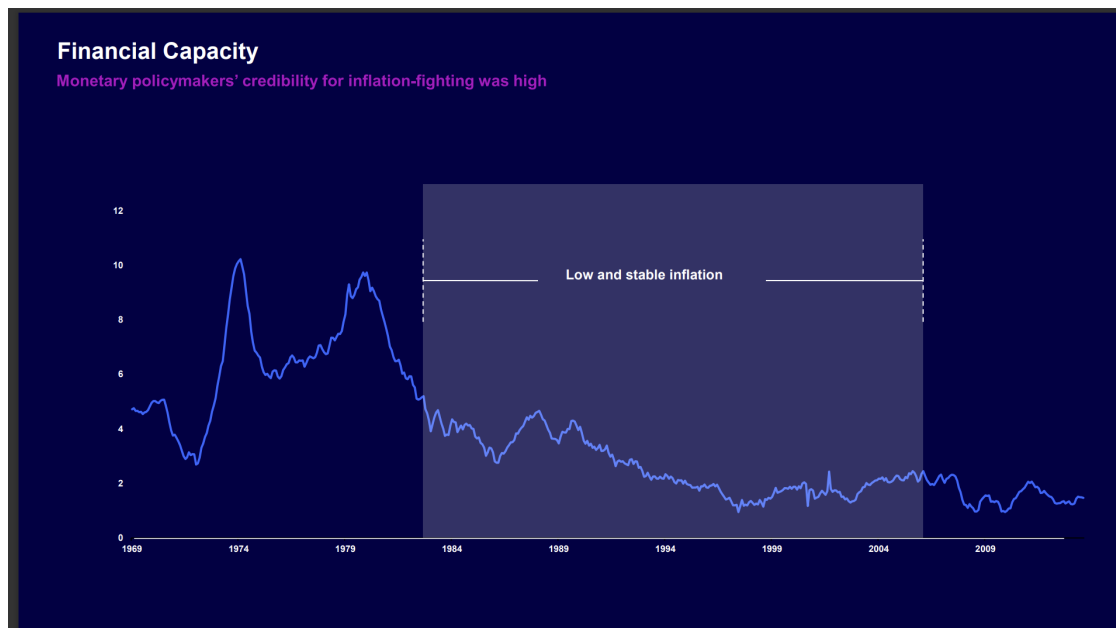


Figure 60.

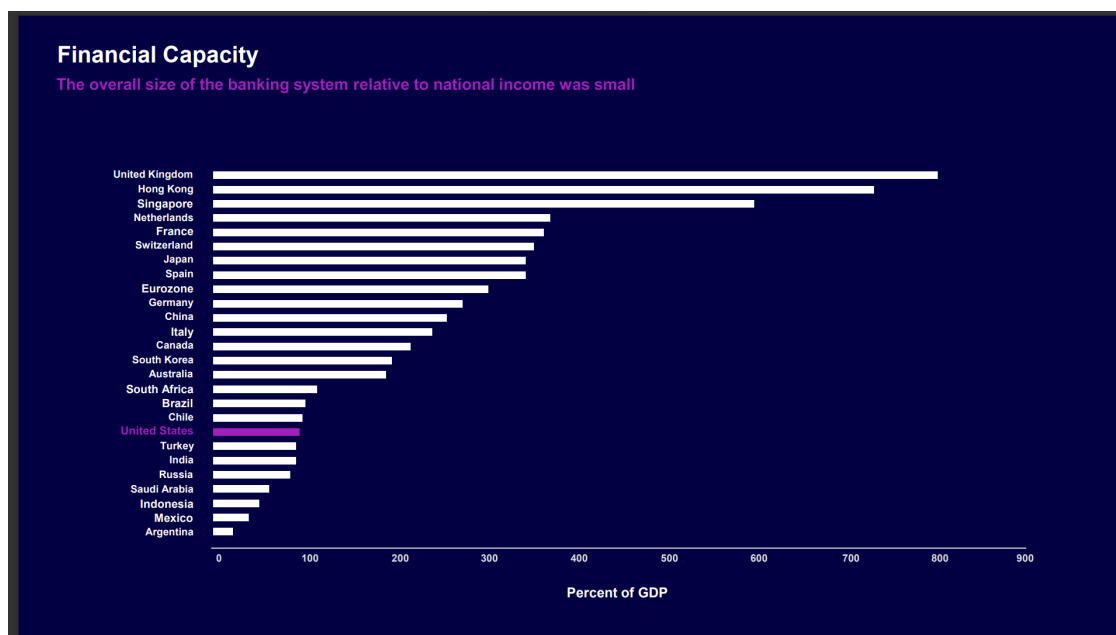


Figure 61.

Weak Fire Fighting Tools: Capital and Resolution

Capital: No Treasury, FDIC, or Fed authority to inject capital

Resolution Authority: FDIC could safely wind down failing banks but could not wind down complex banks or shadow banks

Weak Fire Fighting Tools: Liquidity

Existing: Discount window lending to commercial banks

Emergency Authorities: 13(3) authority to lend to non banks in "unusual and exigent circumstances" No authority to purchase financial assets other than Treasuries and Agencies

Weak Fire Fighting Tools: Guarantees

Existing: FDIC-insured deposits up to \$100k Not Included: Deposits above \$100k, business transaction accounts, foreign deposits, repo, commercial paper for banks Any liabilities of shadow banks Emergency Authorities: FDIC's systemic risk

exception. Treasury's Exchange Stabilization Fund

9.2 Phase 1: Letting it Burn - Framework for Crisis Resolution



Figure 62.

IDEA: Main policy choices:

- How far to let the financial fires burn?
- How fast to escalate in deploying protection?
- to break the panic once it starts?
- How to conduct the triage, where to draw the periphery of support?
- How to get the financial system working again?
- How to balance the moral hazard risk in intervention with the economic costs of financial collapse?

IDEA: Policy Considerations:

- The reality of operating in extreme uncertainty
- Hard to know what works; a lot of it is about confidence
- We were living with the constant and acute fear that the crisis would get away from us
- Our strategy was ultimately to decide to do too much, rather than risk doing too little

IDEA: The Framework for Crisis Resolution:

- Fiscal:
 - Stimulus large relative to fall in private demand
 - Quick and sustained
- Monetary:
 - Negative real rates
 - Sustained to help facilitate deleveraging
- Financial:
 - Classic lender of last resort provision of liquidity
 - Guarantees
 - Support for funding markets
 - Capital
 - Resolution and restructuring



Figure 63.

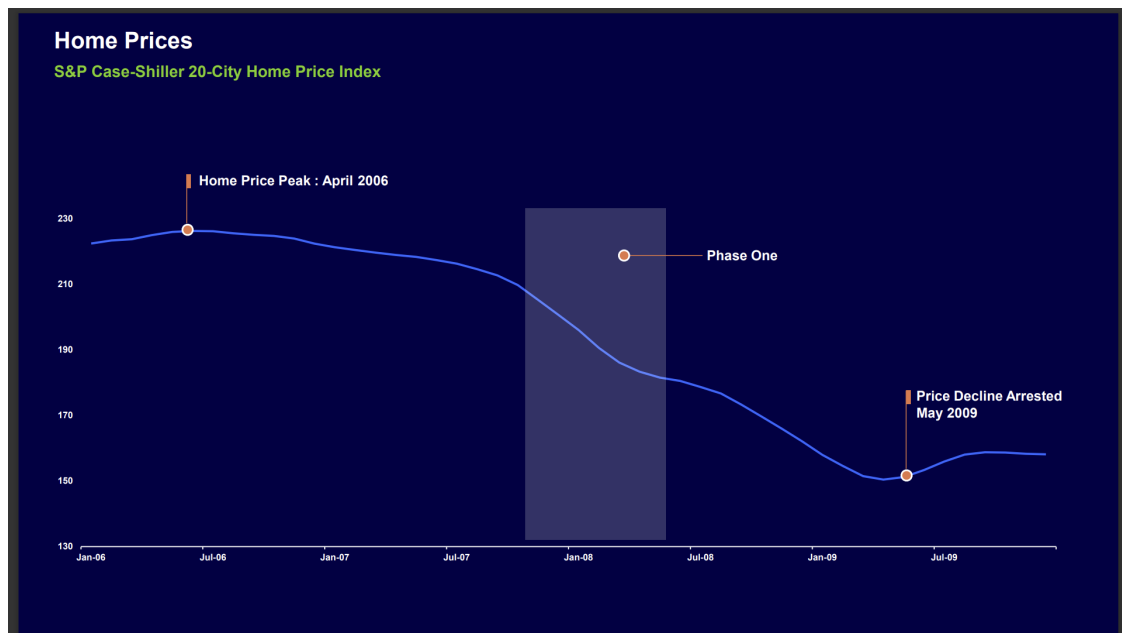


Figure 64.



Figure 65.

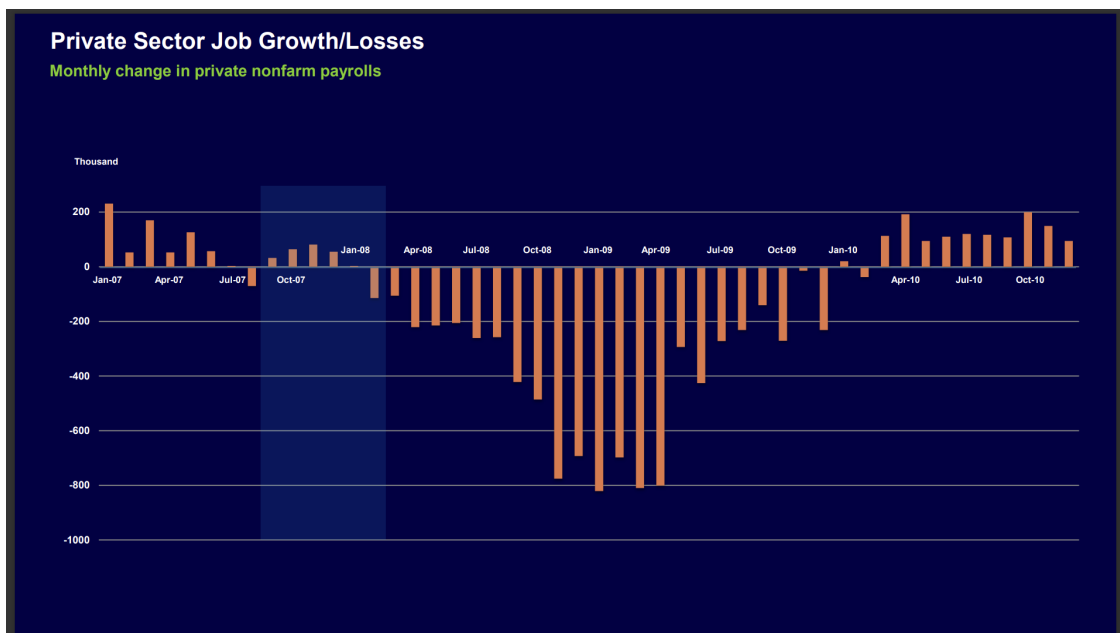


Figure 66.

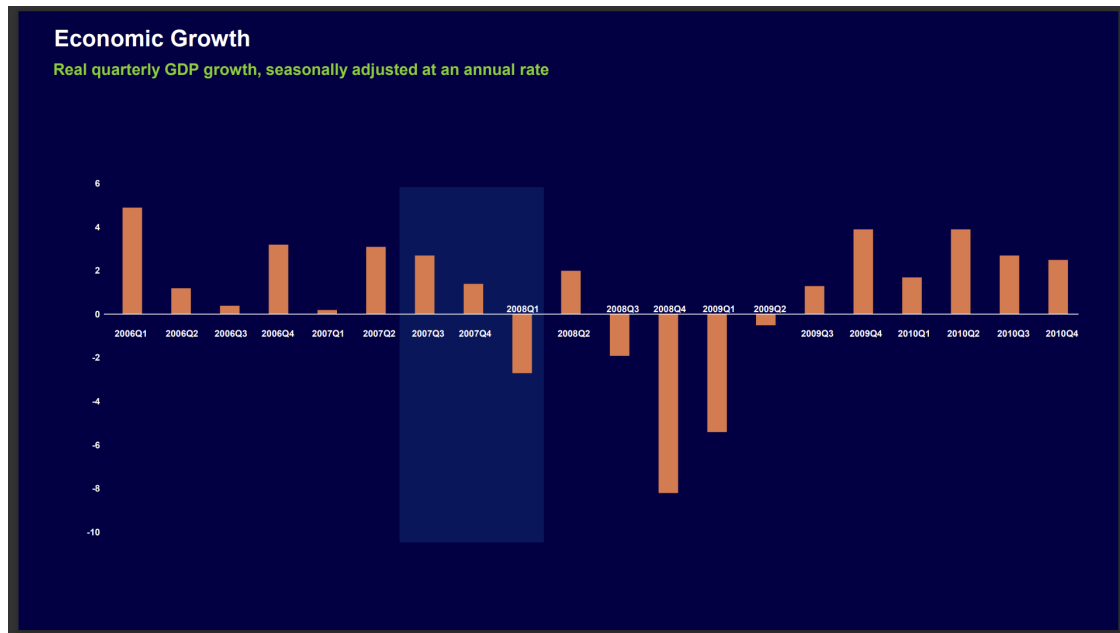


Figure 67.

IDEA:Actions:

- Standard monetary policy: lowered interest rates aggressively
- Get liquidity into the banking system by reducing the stigma of the discount window
- Liquidity swaps for foreign central banks
- Fiscal stimulus through modest tax rebates

IDEA:But failed in:

- didn't lend beyond banks
- didn't finance the securities the market didn't want to finance
- allowed the weaker non-banks to fail

More than two dozen mortgage lenders failed, including New Century in April 2007

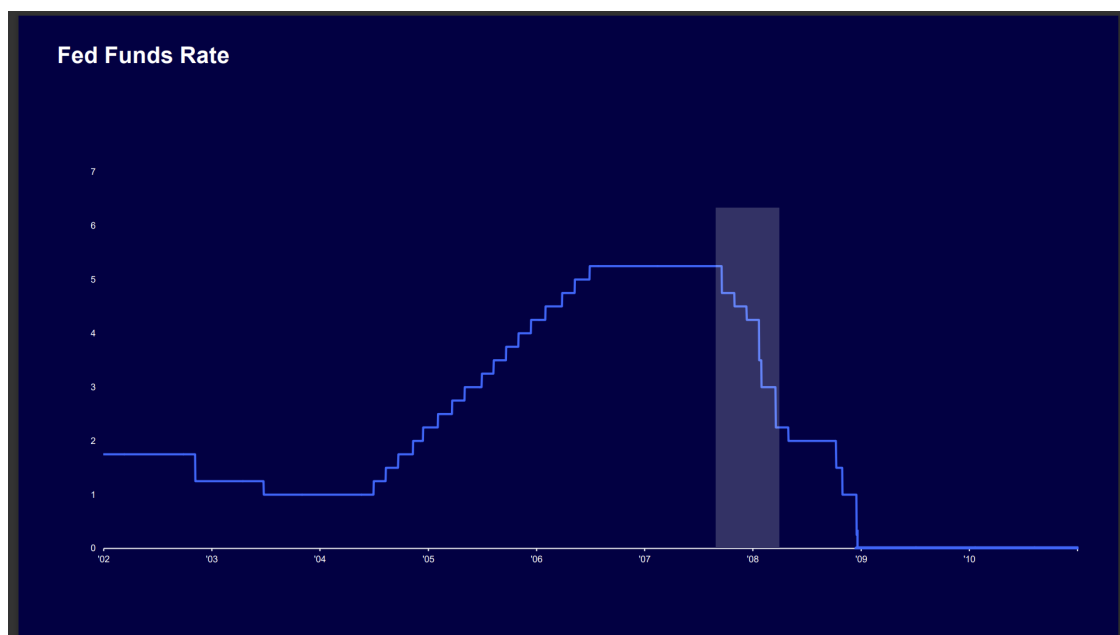


Figure 68.

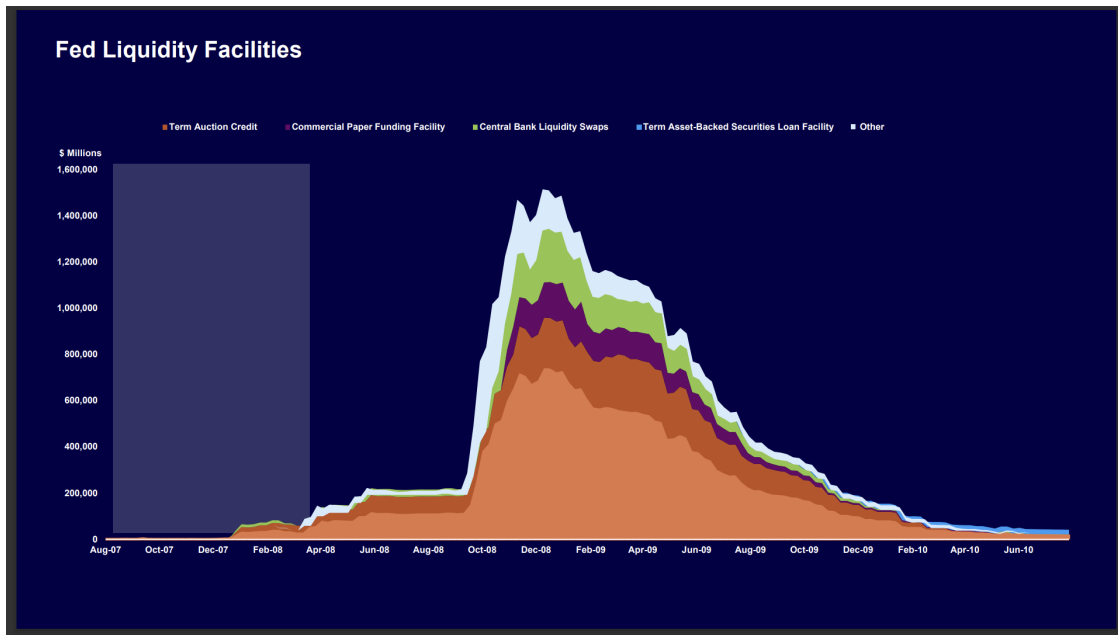


Figure 69.

9.3 Phase 2: Early Escalation - The Policy Response



Figure 70.

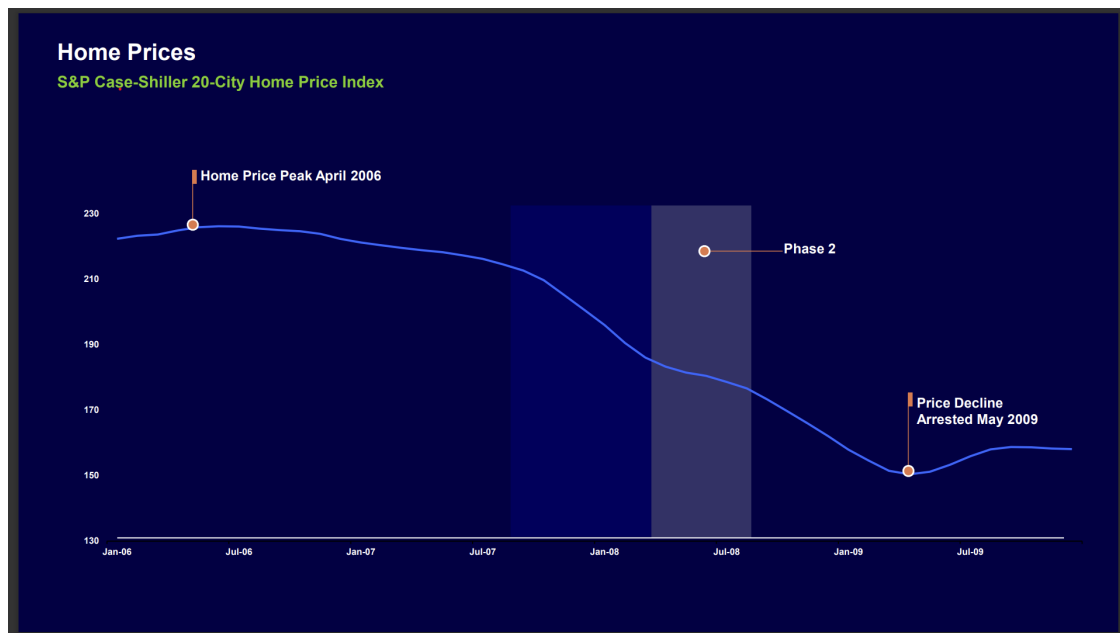


Figure 71.



Figure 72.



Figure 73.

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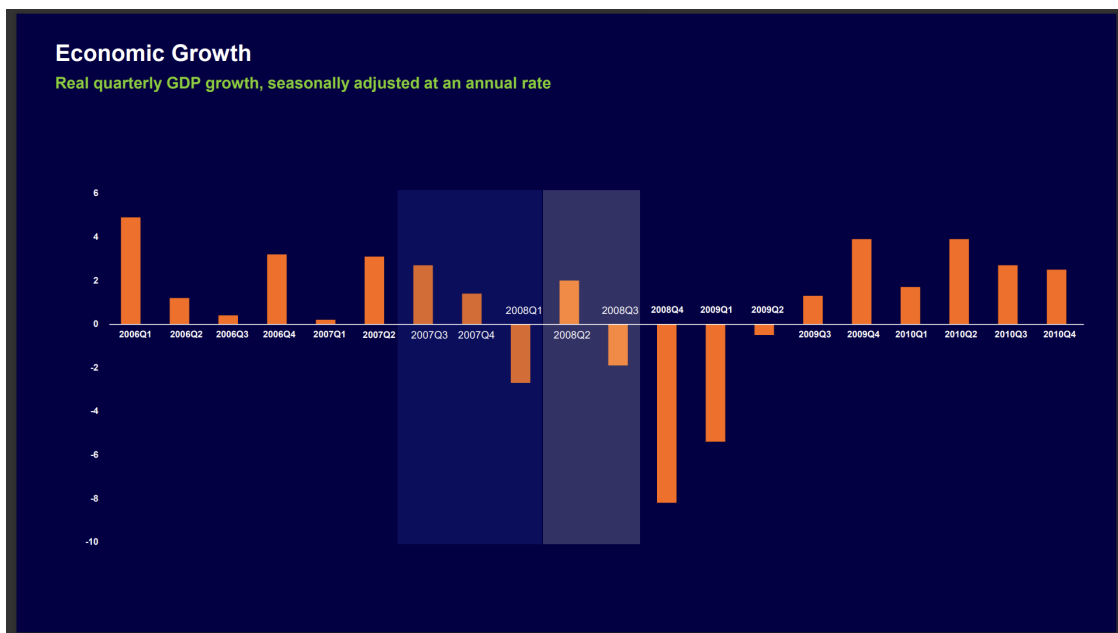


Figure 74.

IDEA: What We Did: Early Escalation

- Broke the line past banks—lent directly to nonbanks PDCF, TSLF, Bear Stearns
- Went to Congress for Fannie/Freddie bazooka, prepared for war

All this helped slow the burn

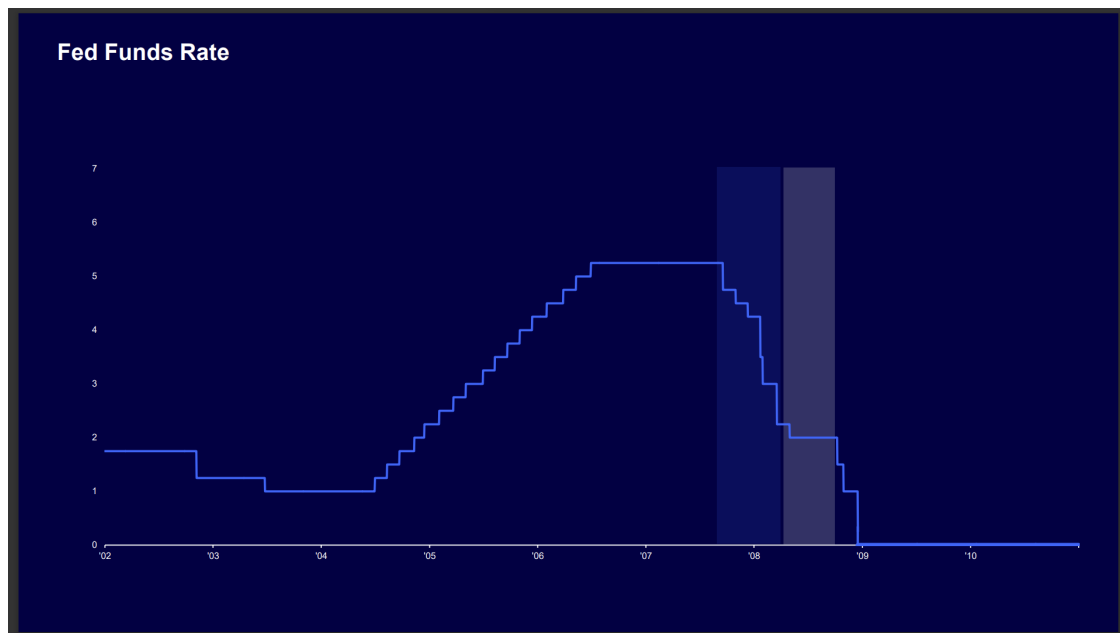


Figure 75.

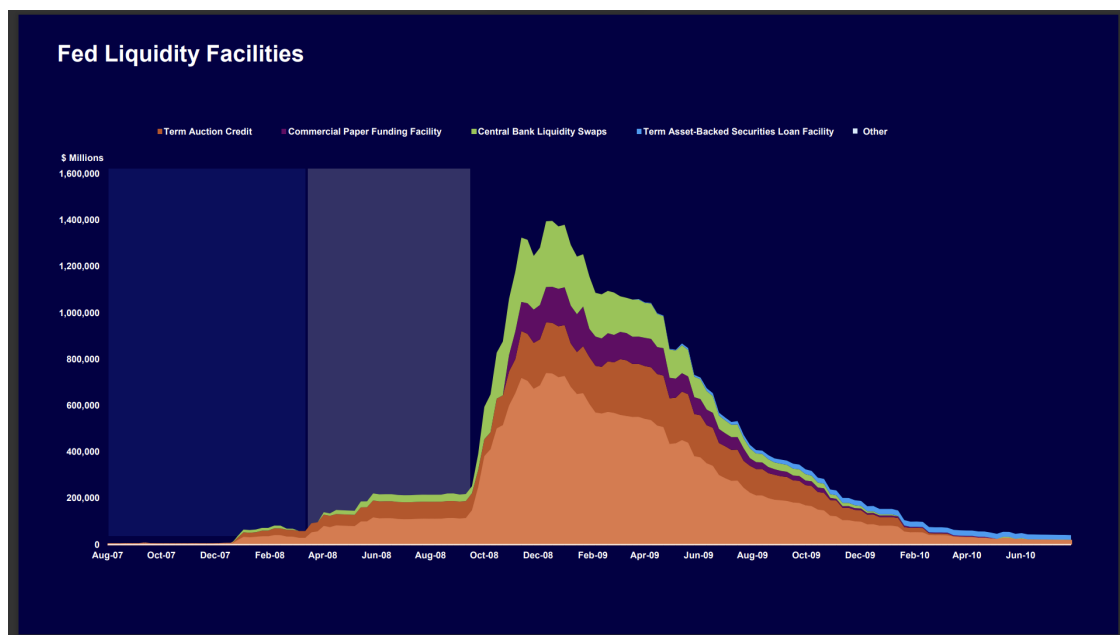


Figure 76.

9.4 Phase 3: Breaking the Panic - Market Developments

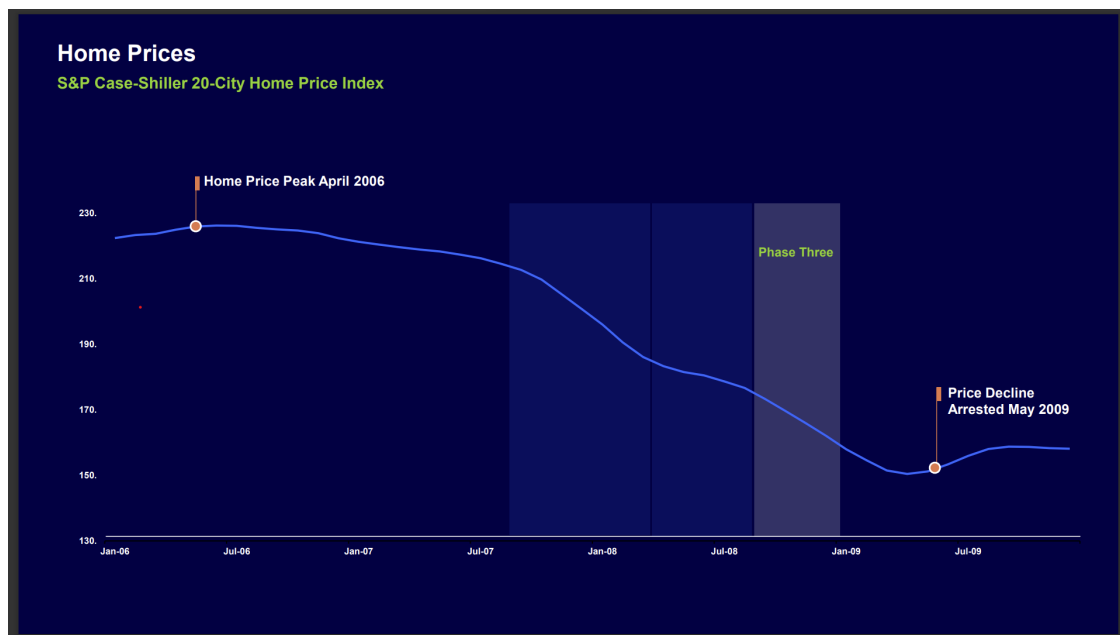


Figure 77.



Figure 78.

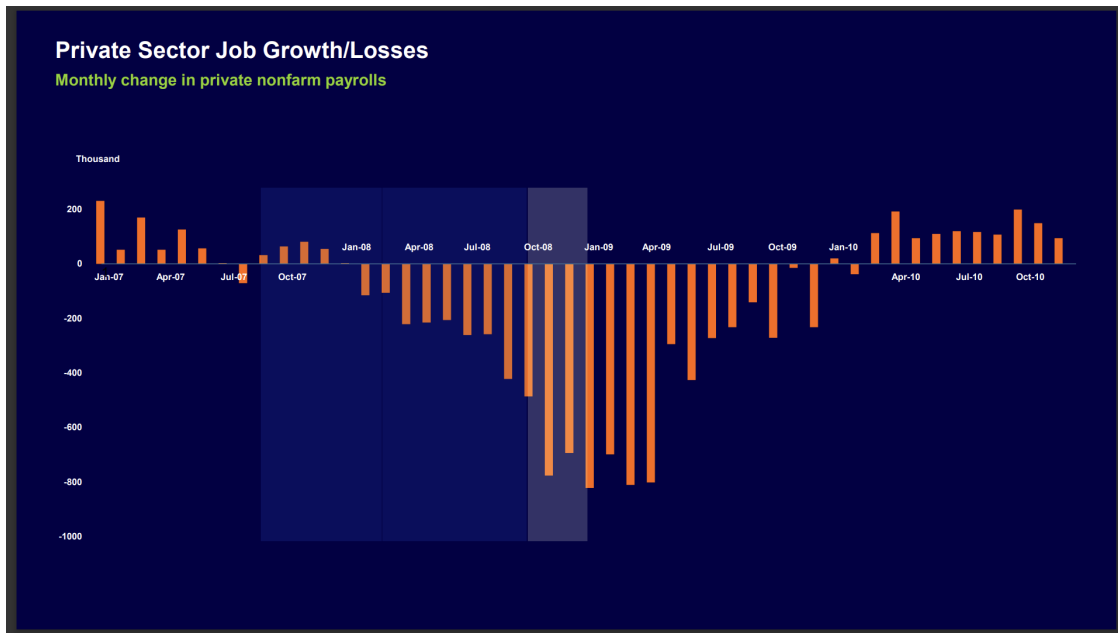


Figure 79.

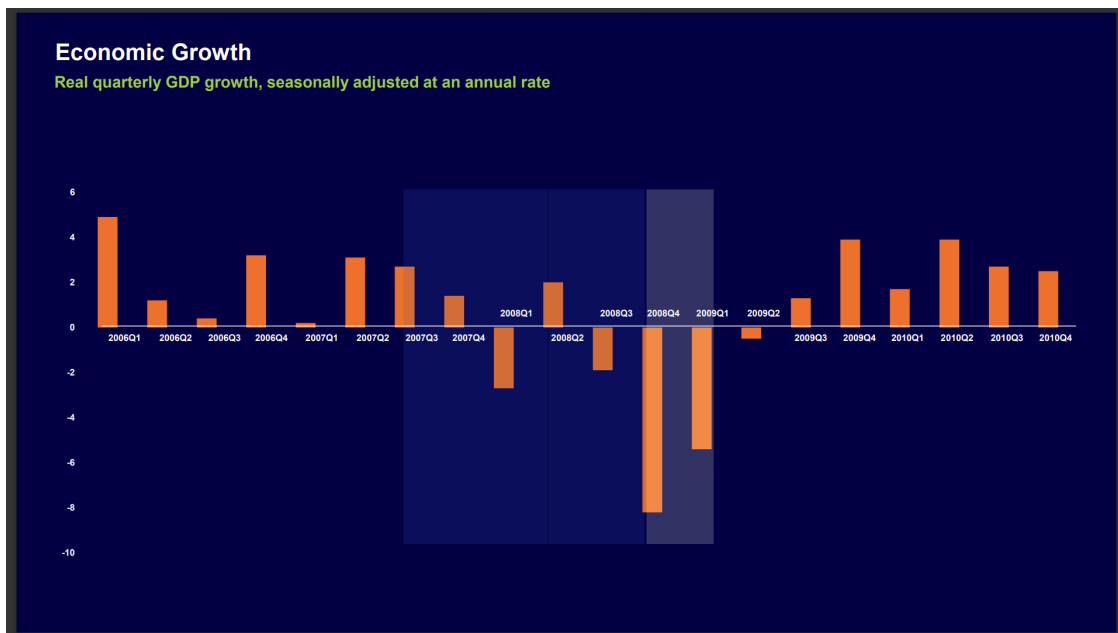


Figure 80.

IDEA:What We Did: Powell Doctrine to Break the Panic

- Dramatically expanded liquidity into critical markets (CP and ABCP) and shadow institutions
- Provided additional support to systemic institutions: AIG, Citigroup, and Bank of America
- Provided guarantees to \$3.4 trillion money fund industry and the entire banking system
- Injected capital into more than half of the banking system by assets
- Deployed unlimited central bank swap lines to support the global system
- Provided bridge financing to major auto companies

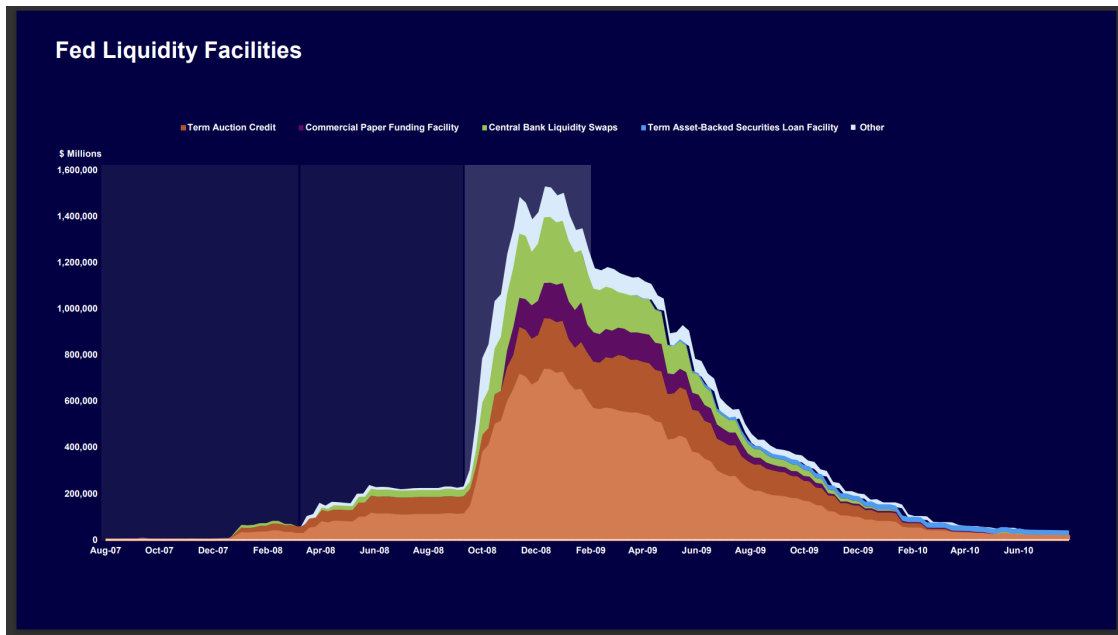


Figure 81.

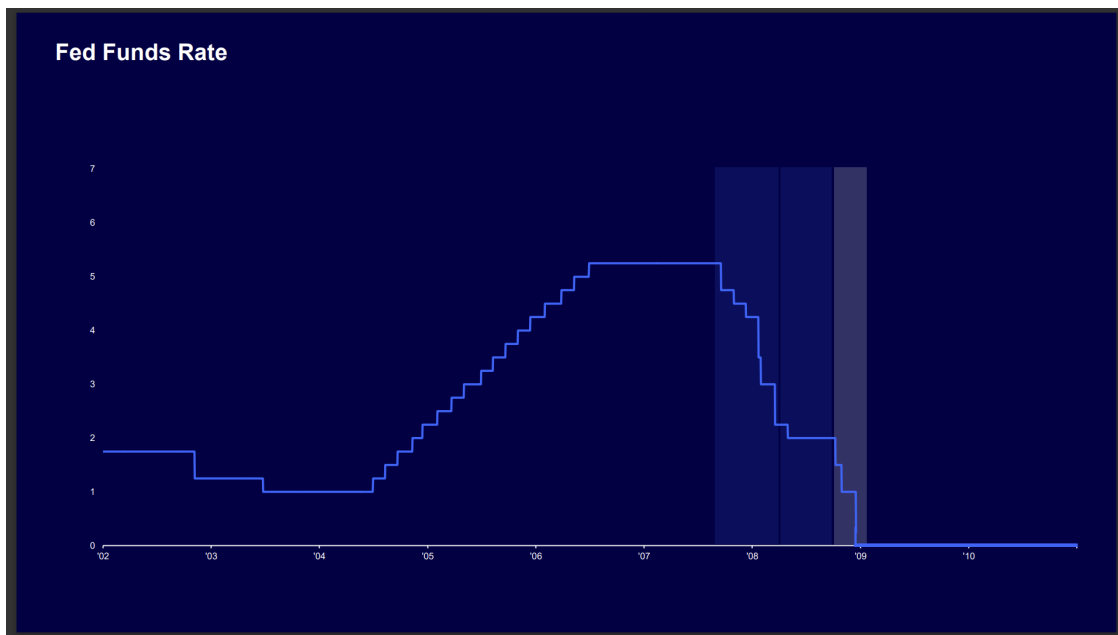


Figure 82.

9.5 Triage in a Panic

In a period of four weeks in September 2008, the United States:

FACT:

- Put Fannie and Freddie into conservatorship, guaranteeing their creditors but forcing their equity holders to absorb losses.
- Helped encourage the acquisition of a failing investment bank (Merrill Lynch) by Bank of America.
- Failed to find a buyer for Lehman Brothers, another failing investment bank.
- Acted to prevent the failure of a large global insurance company, AIG.
- Imposed losses on the creditors of a large failing bank (WaMu) in the context of facilitating its merger with JP Morgan.

9.6 Phase 4: Resolution - Market Developments



Figure 83.

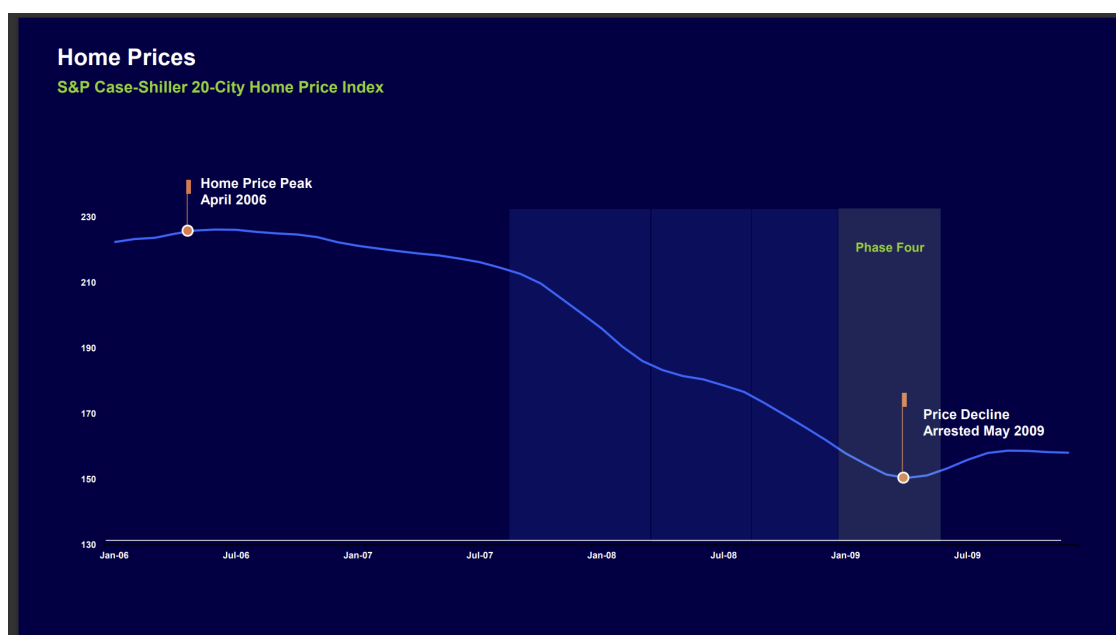


Figure 84.



Figure 85.

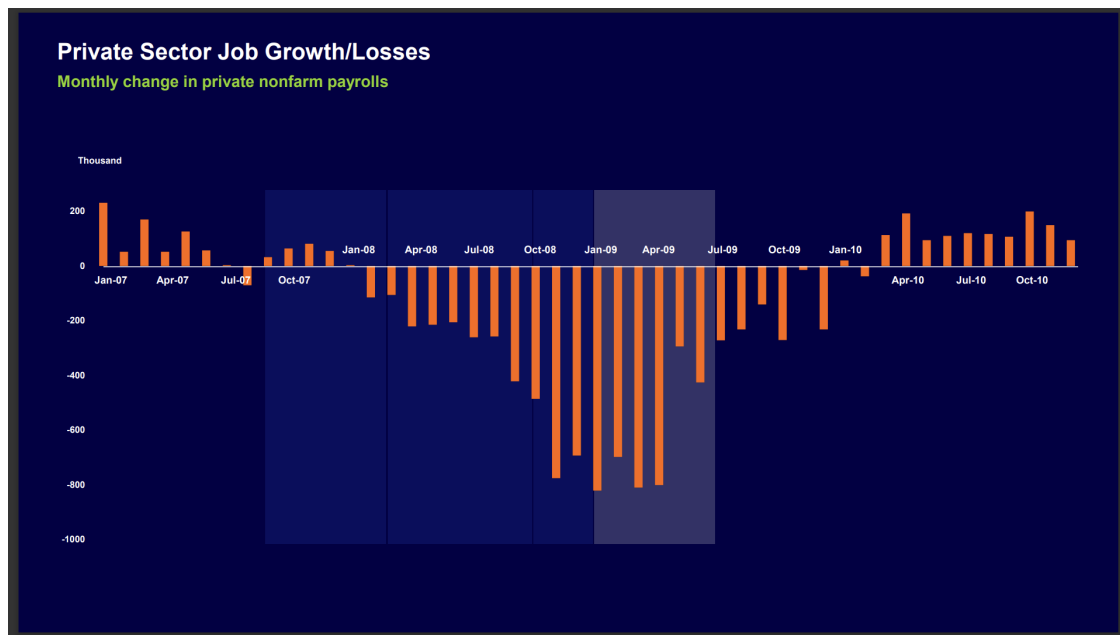


Figure 86.

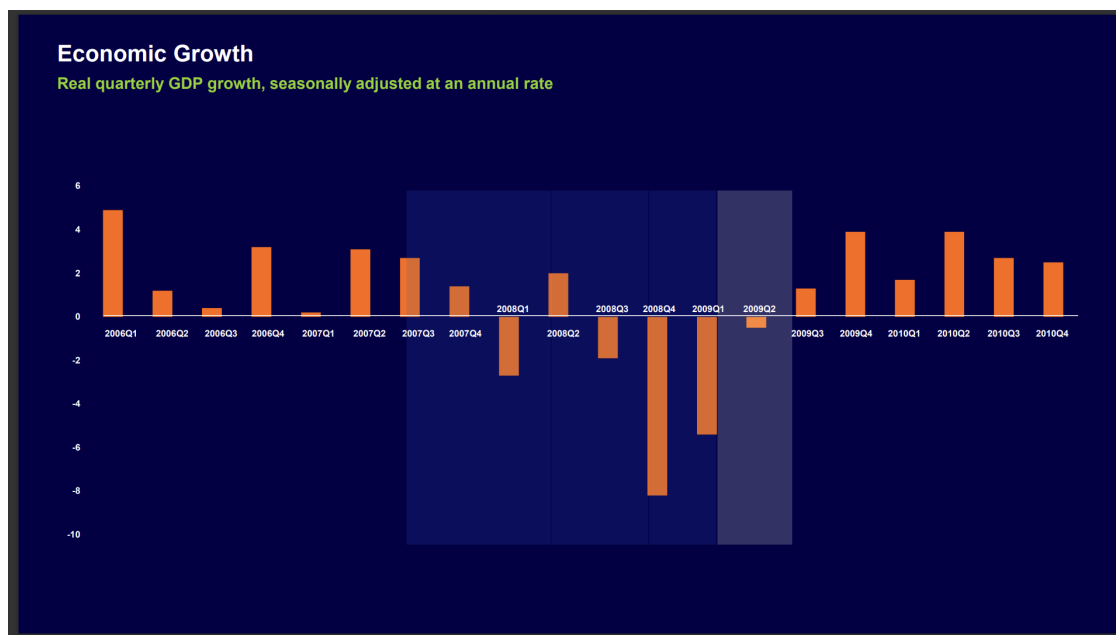


Figure 87.

9.7 Phase 4: Resolution: What We Did

FACT:

- Powell doctrine to revive growth—fiscal, monetary, financial
- \$800 billion fiscal stimulus
- Continued monetary stimulus: zero bound and quantitative easing
- Coordinated global Keynesian response
- Stress test to restore confidence and recapitalize the system
- Expanded scope of backstop to credit markets
- Hardened the guarantee of the banking system (BHCs and GSEs too)

9.8 The Stress Test

IDEA: The Stress Test: How and Why Did It Work?

- Transparency, firm by firm, about losses in the extreme event.
- Device for triage, for determining nationalization.
- Tool for recapitalizing the financial system.
- Helped maximize the chance that capital would come from the private sector.

Why did it work?

- Loss estimates were credible.
- We hardened the guarantees on liabilities.
- Fiscal and monetary policy escalation got traction.
- Confidence improved by global cooperation and massive mobilization of dollar-based financial support for EM.

We put enough money in the window.

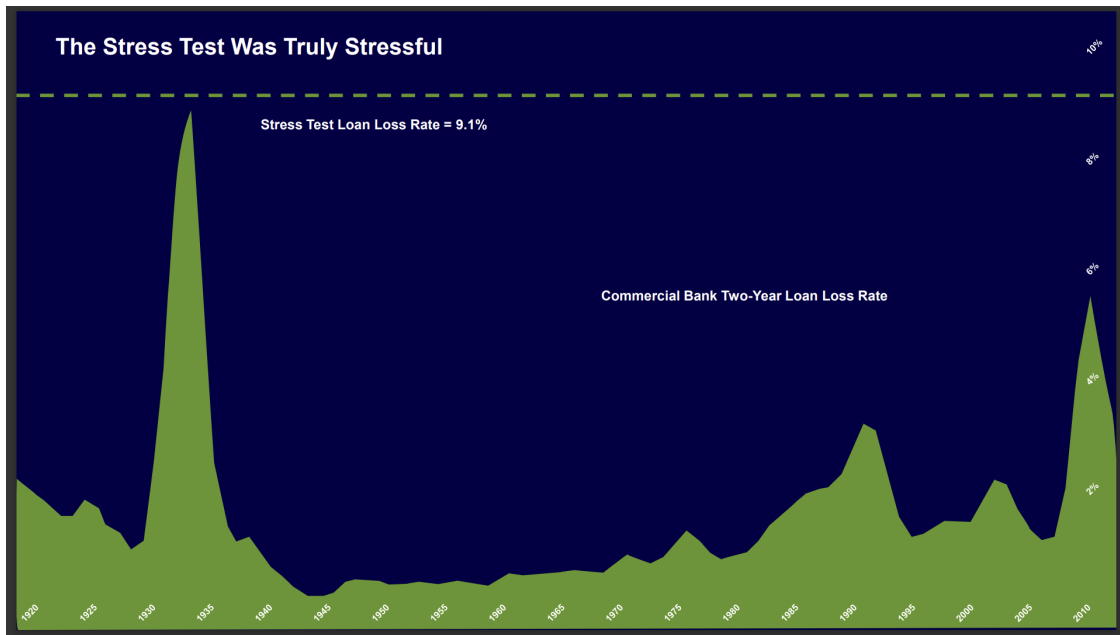


Figure 88.



Figure 89.

9.9 Housing Response

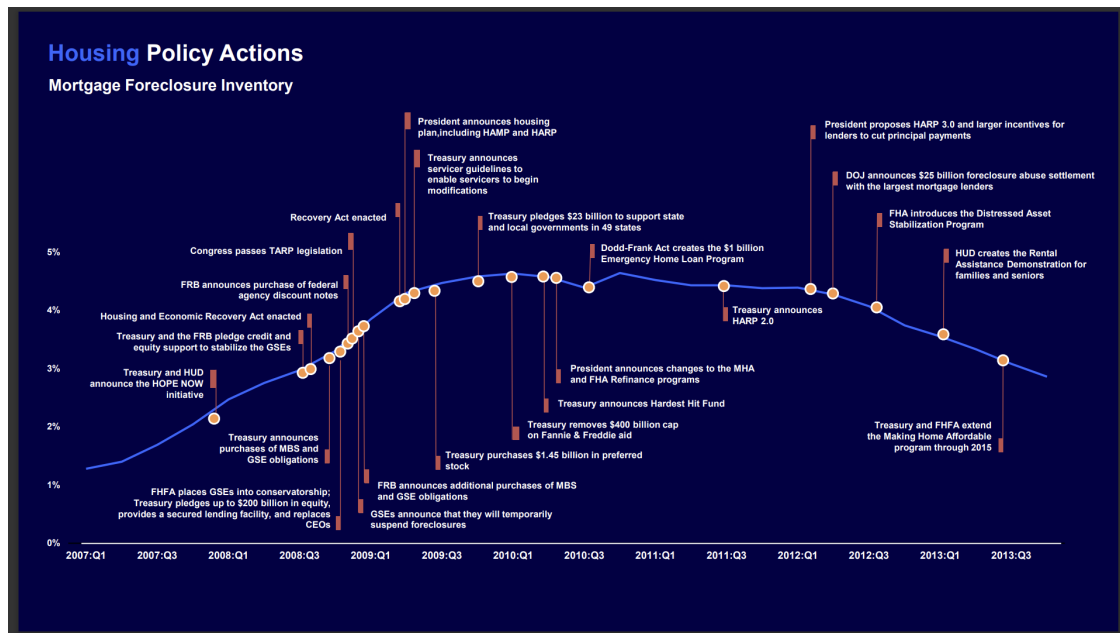


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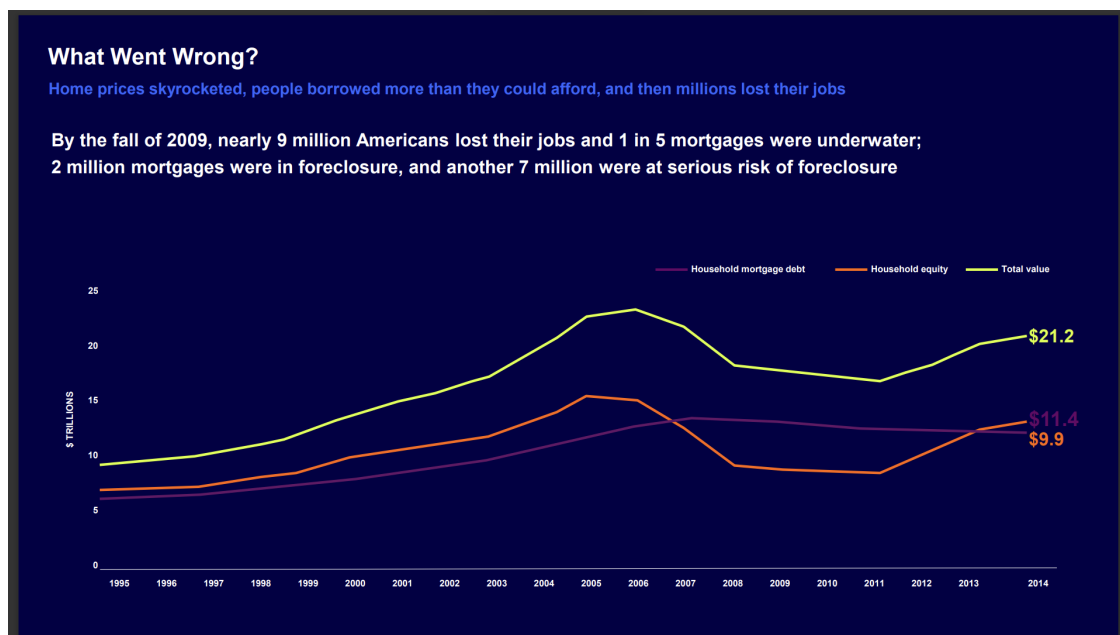


Figure 91.

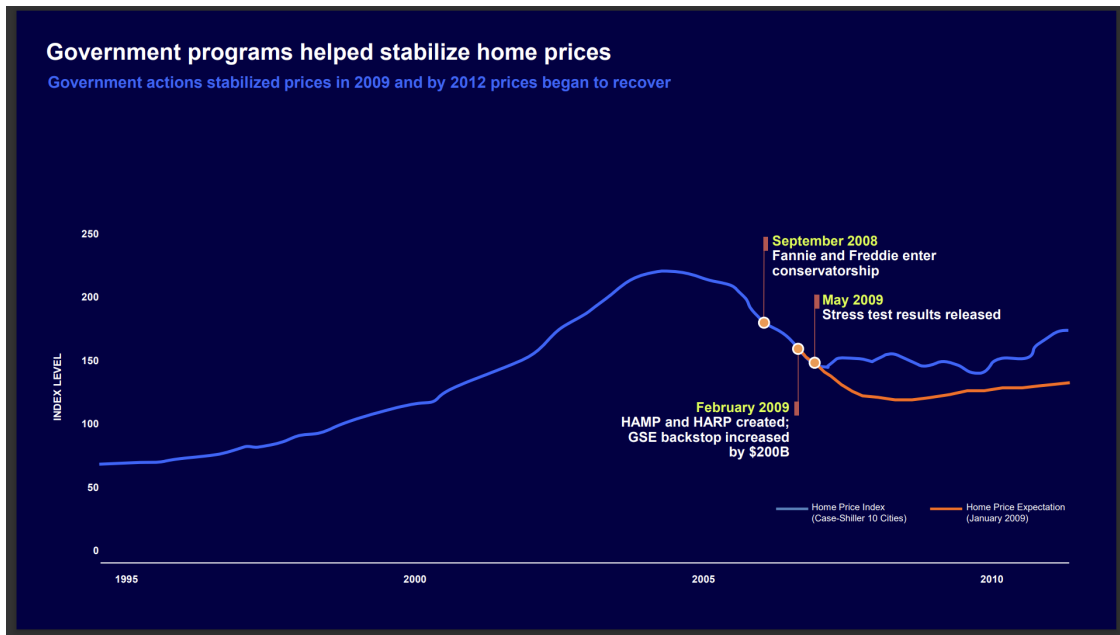


Figure 92.

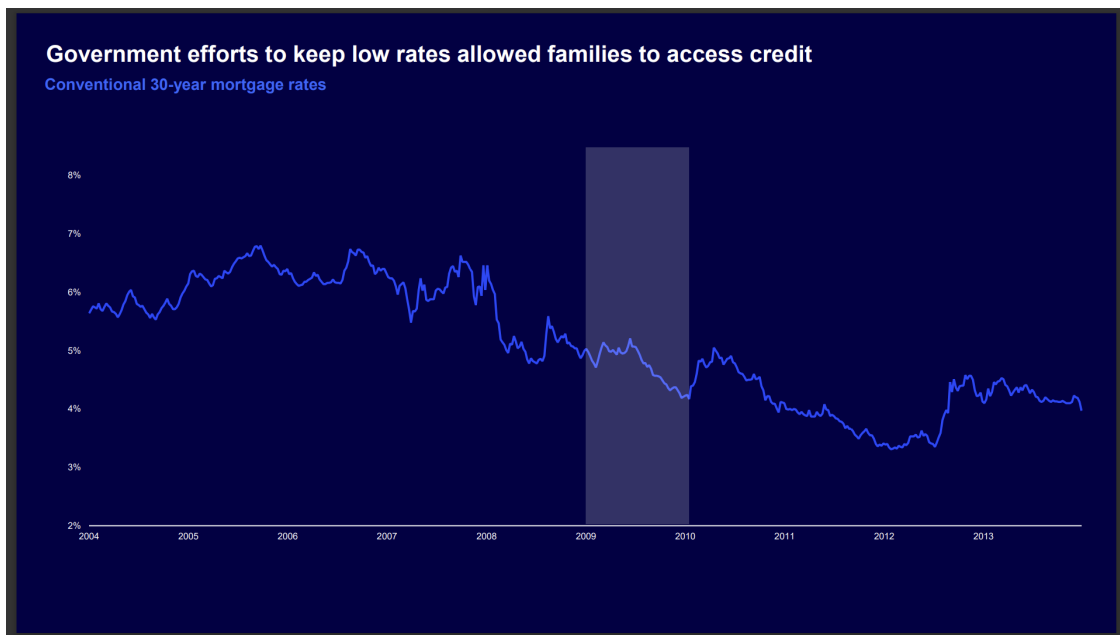


Figure 93.

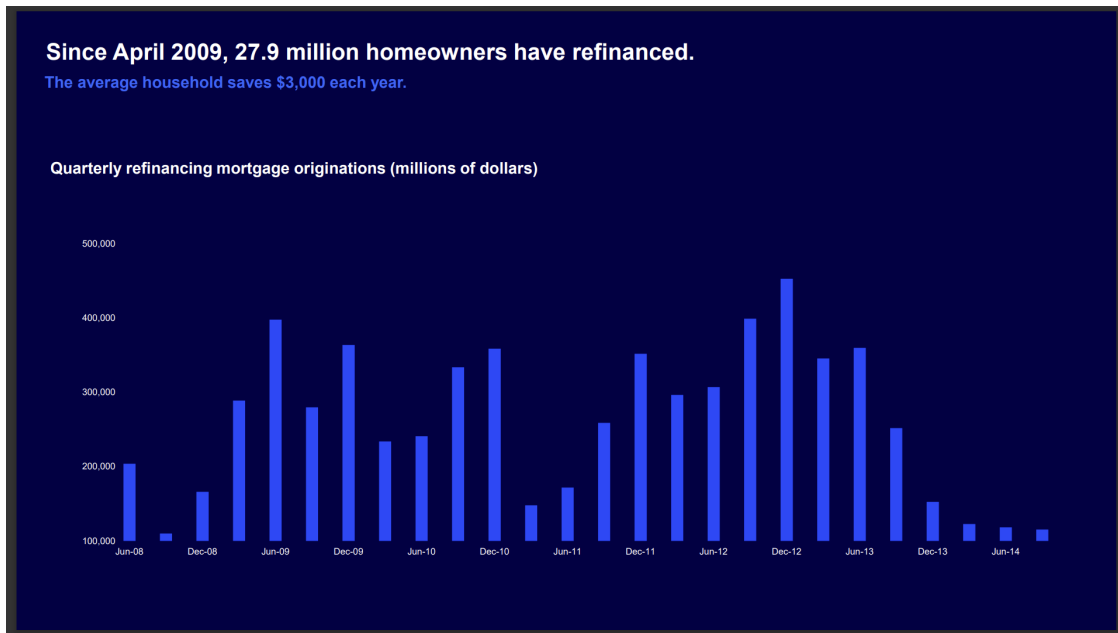


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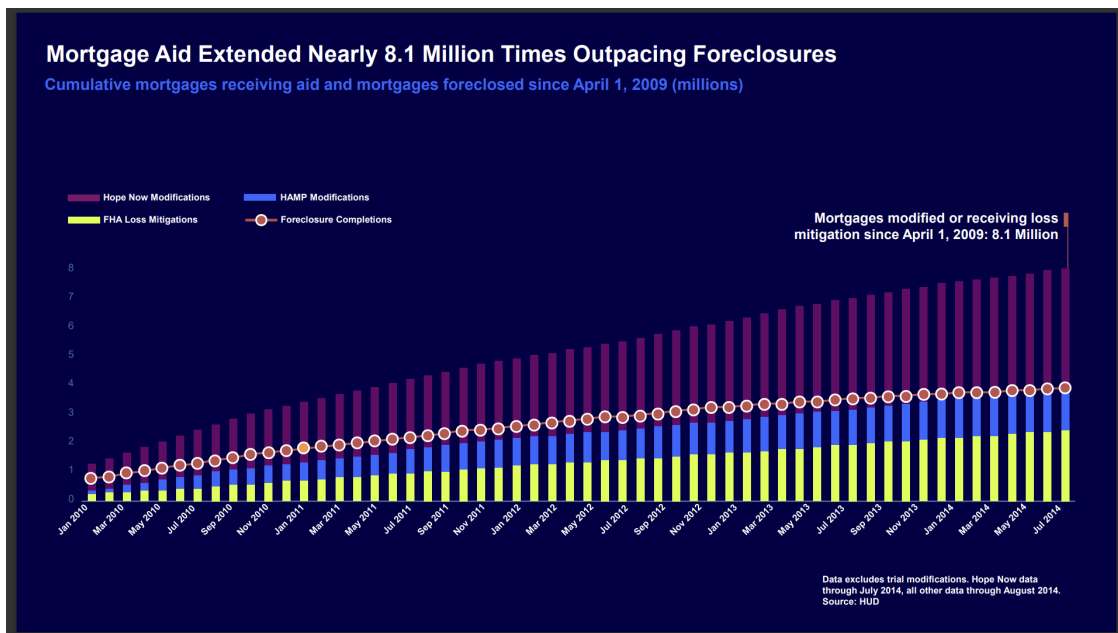


Figure 95.

IDEA:We Could Not:

- Discharge homeowner debt in bankruptcy—"cramdown" legislation failed twice
- Create a "Home Owners' Loan Corporation"—exceptionally complicated and Congressional action was needed
- Start another mortgage refinancing program—again, required legislation
- Undertake widespread mortgage debt forgiveness—Administration pursued targeted "principal reduction" but we couldn't force FHFA to use its authority

We needed greater resources and authority to alleviate the pain of homeowners

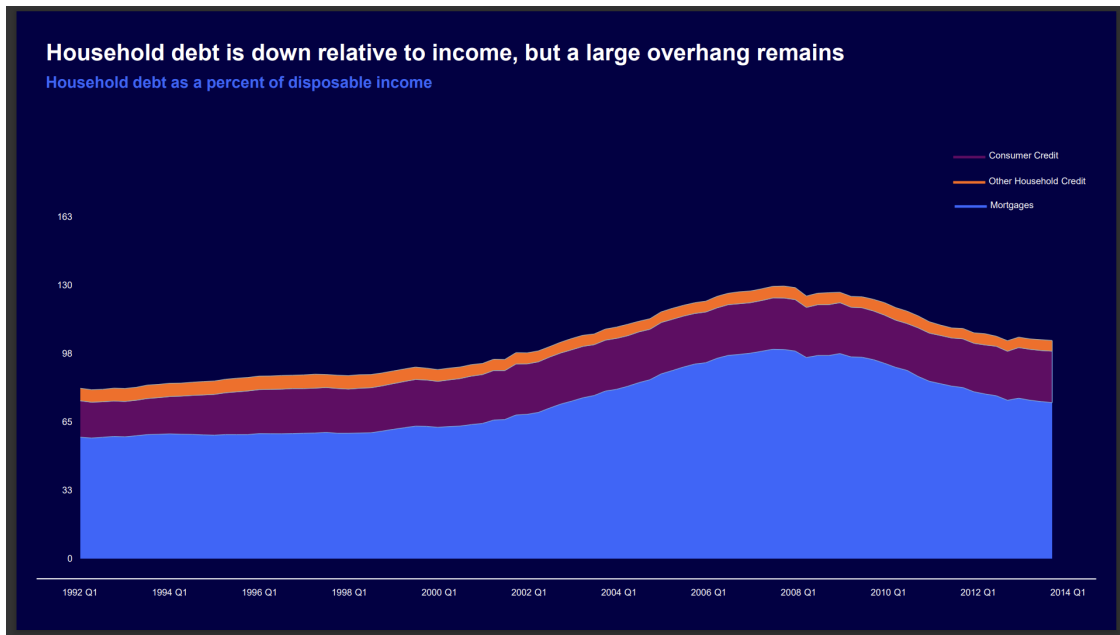


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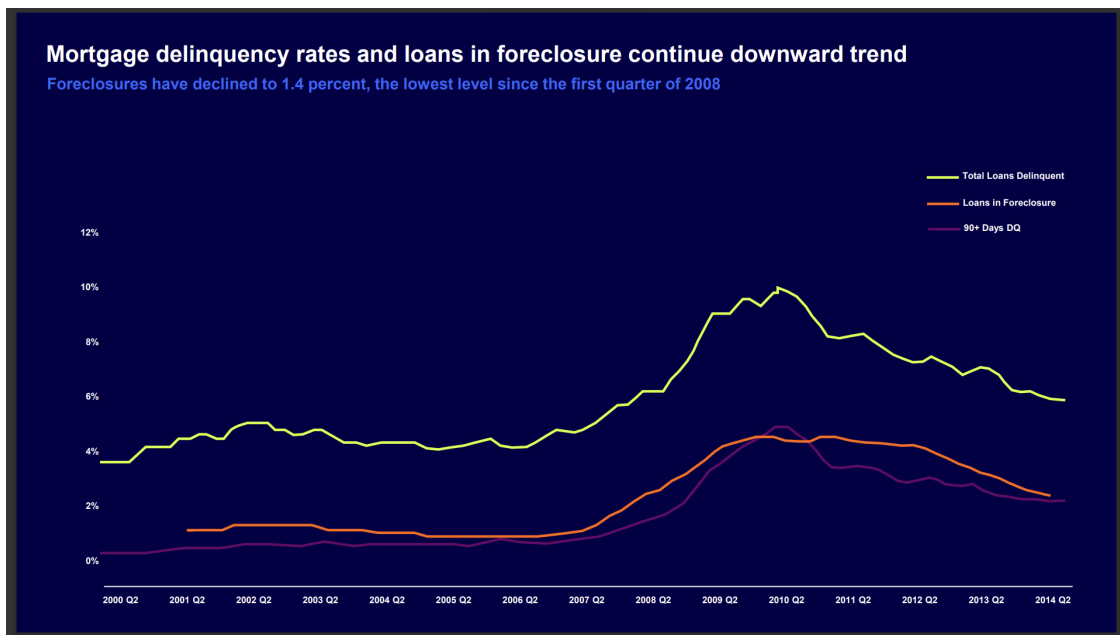


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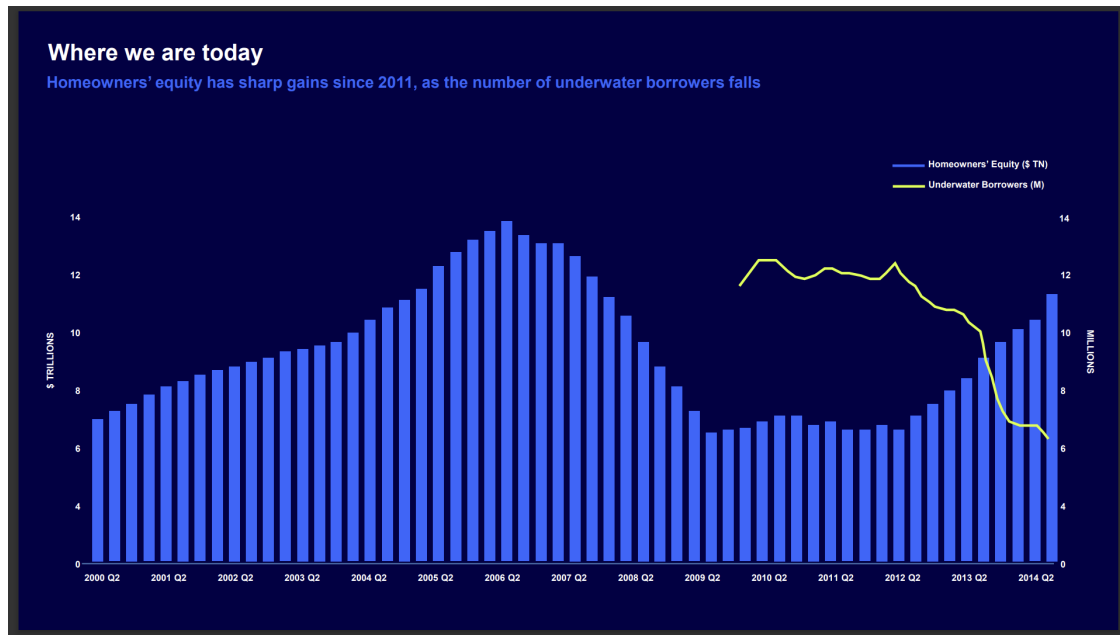


Figure 98.

10 Responding to Future Crises

10.1 What Have We Learned? Outcomes: Prisms 1-5

FACT: Economic cost of the crisis:

- Real GDP fell more than 4% and took more than 3 years to return to the pre-crisis peak
- Public debt has increased by more than 30% of GDP, mostly due to cyclical forces
- At market bottom, \$15 trillion in household wealth had disappeared
- But these costs would have been much greater in the absence of an effective financial rescue

IDEA: Policy failures:

Failures Ex Ante:

- Home equity was too thin
- Capital cushions were too thin—and, more importantly, too narrow in scope
- Too much issuance of short-term, deposit-like liabilities, without regulatory constraints on leverage and without access to the safety net

Failures In the Crisis

- We escalated too late, mostly because of limitations of authority
- Fiscal policy turned too tight too soon
- Mortgage restructuring authority, resources, and incentives were too weak

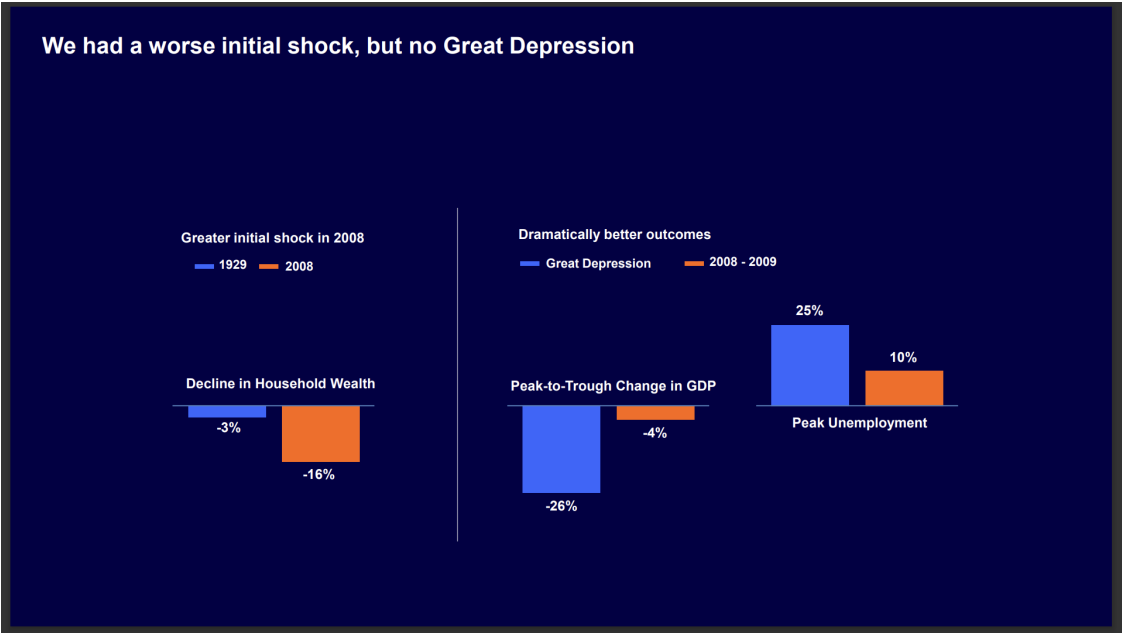


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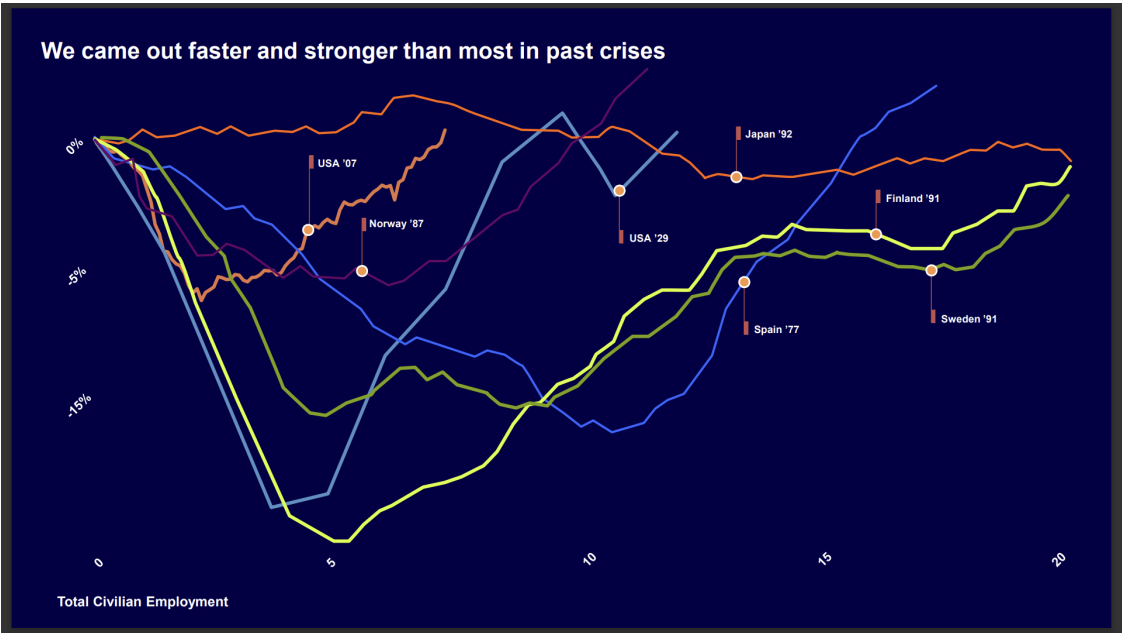


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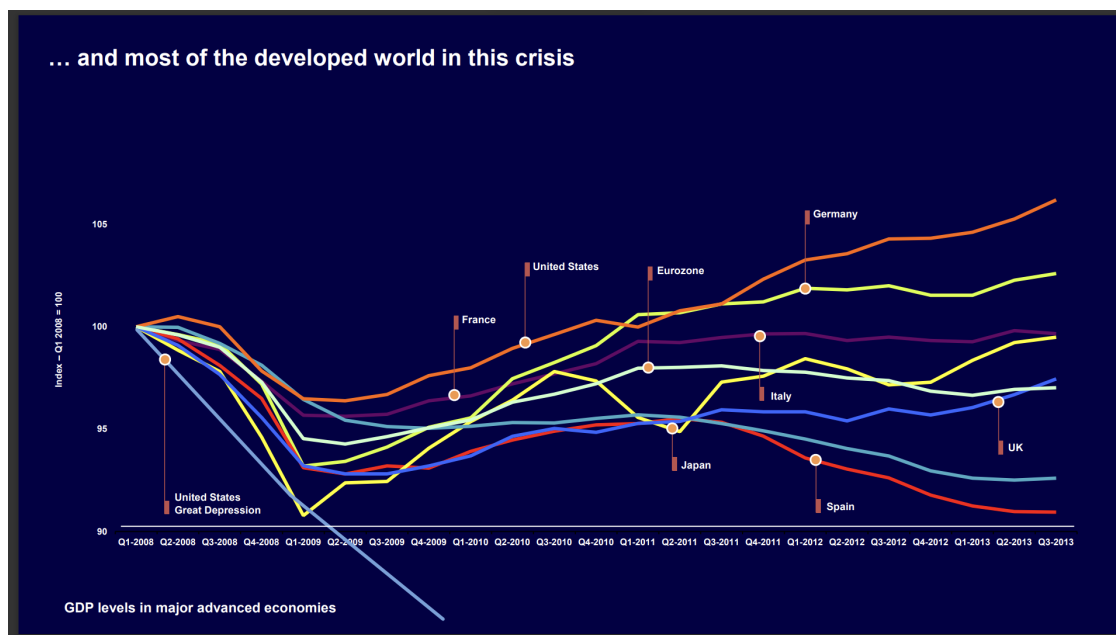


Figure 101.



Figure 102.



Figure 103.

10.2 What Have We Learned? Outcomes: Prisms 6-10

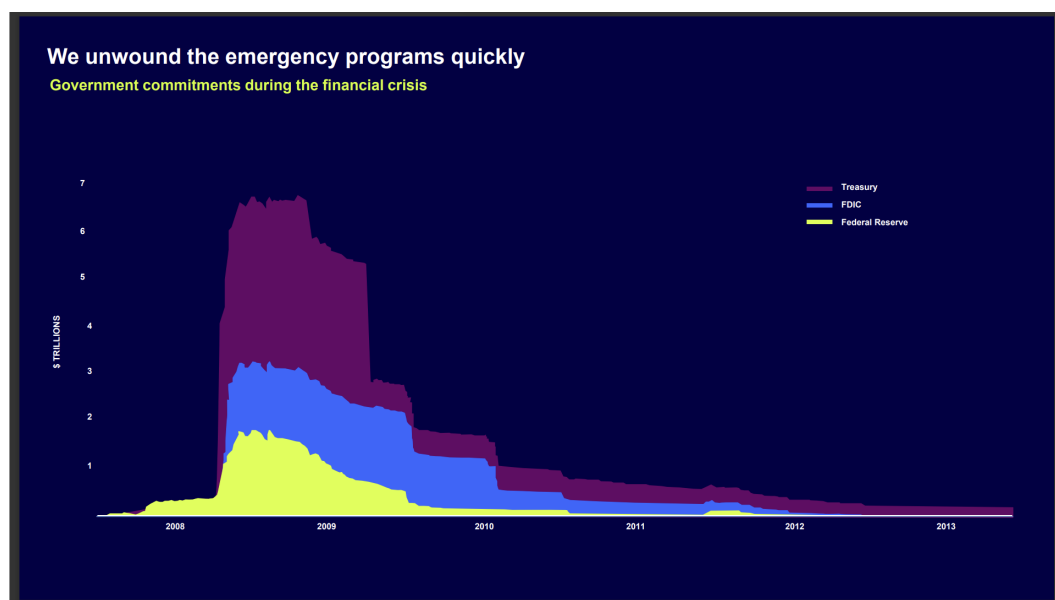


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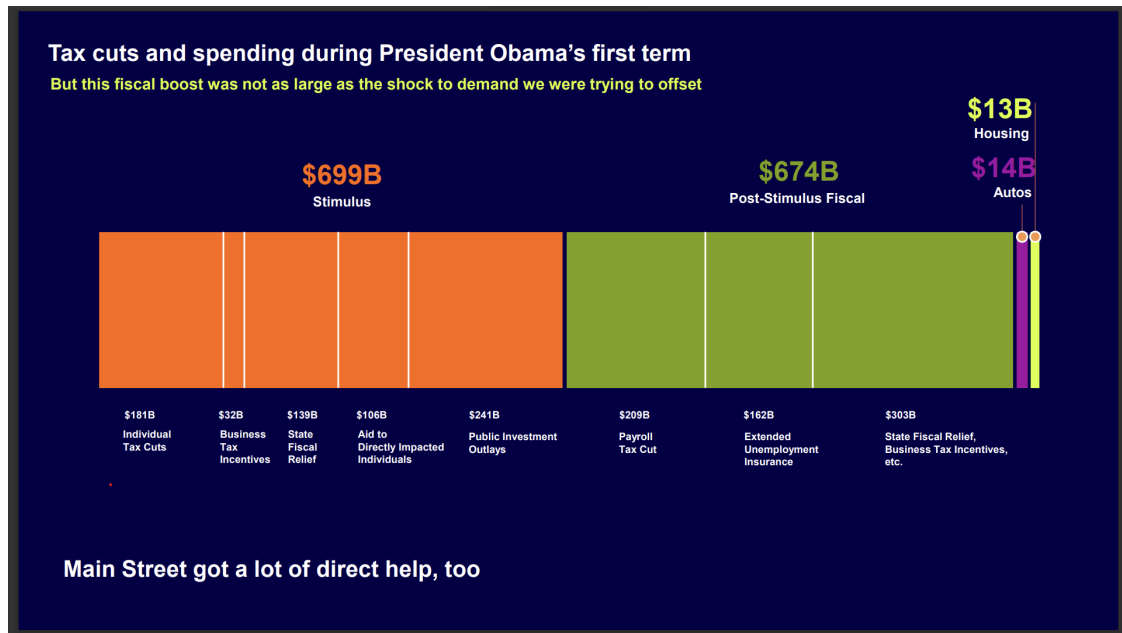


Figure 105.



Figure 106.



Figure 107.

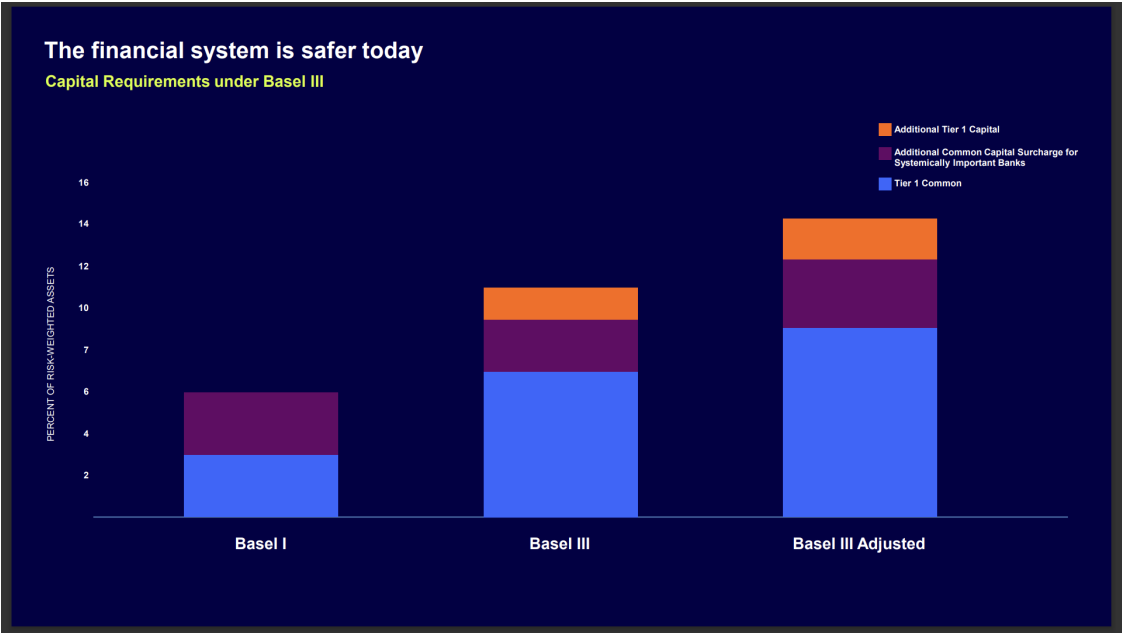


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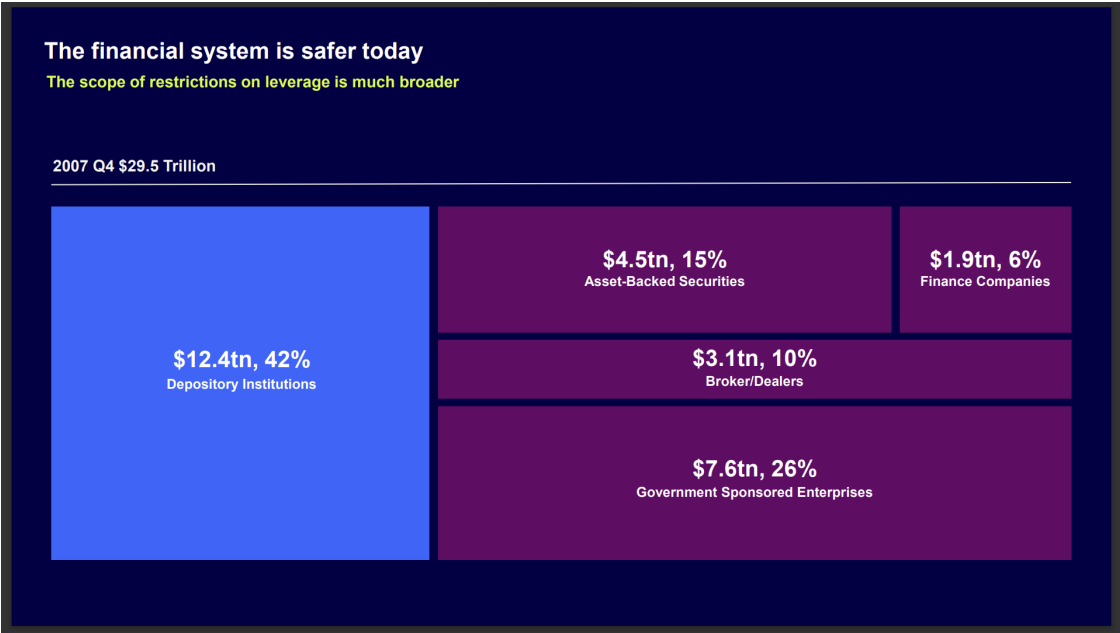


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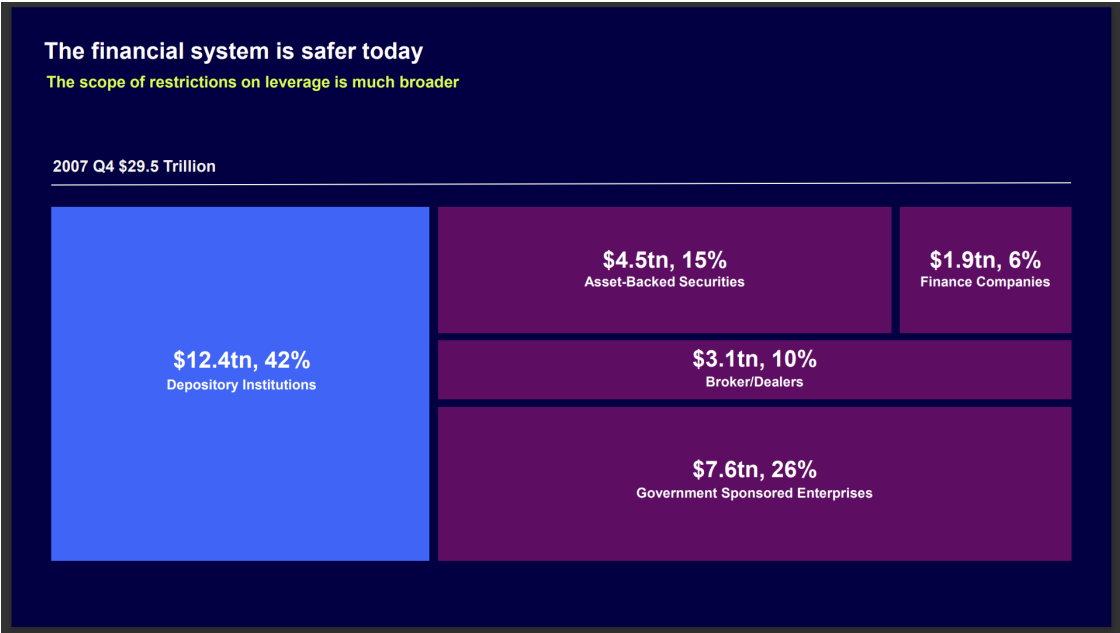


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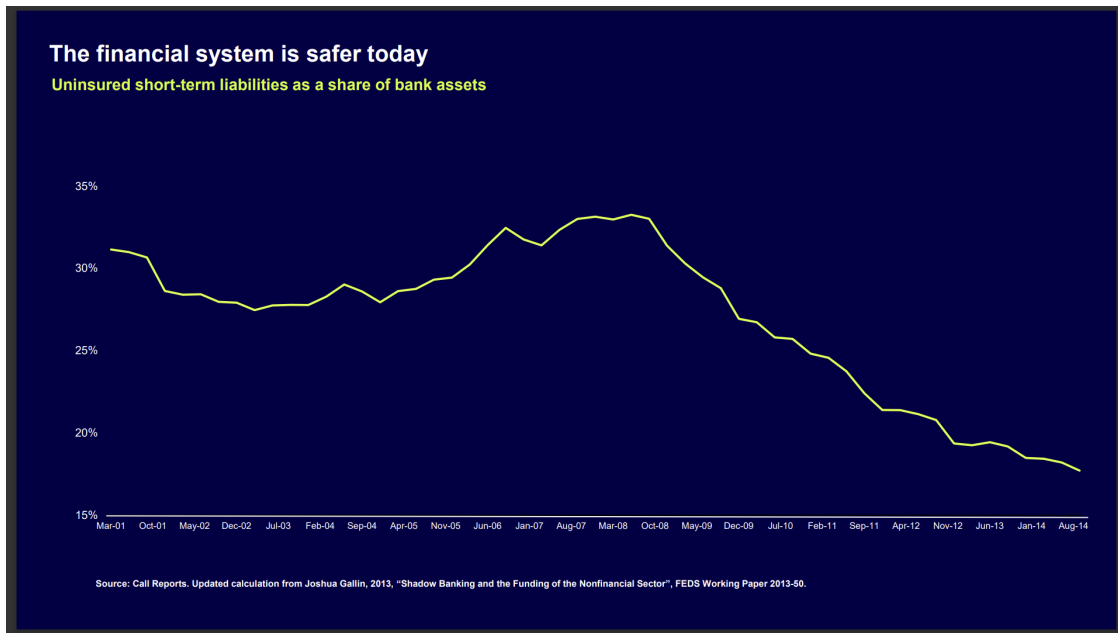


Figure 111.

IDEA: "Too big to fail" is less of a problem

- The largest institutions are subject to a systemic capital surcharge
- The FDIC has resolution authority to manage the failure of systemic institutions
- We imposed tougher concentration limits on the financial system
- The Fed's emergency lending authority has been curtailed, and the FDIC's emergency guarantee authority has been eliminated

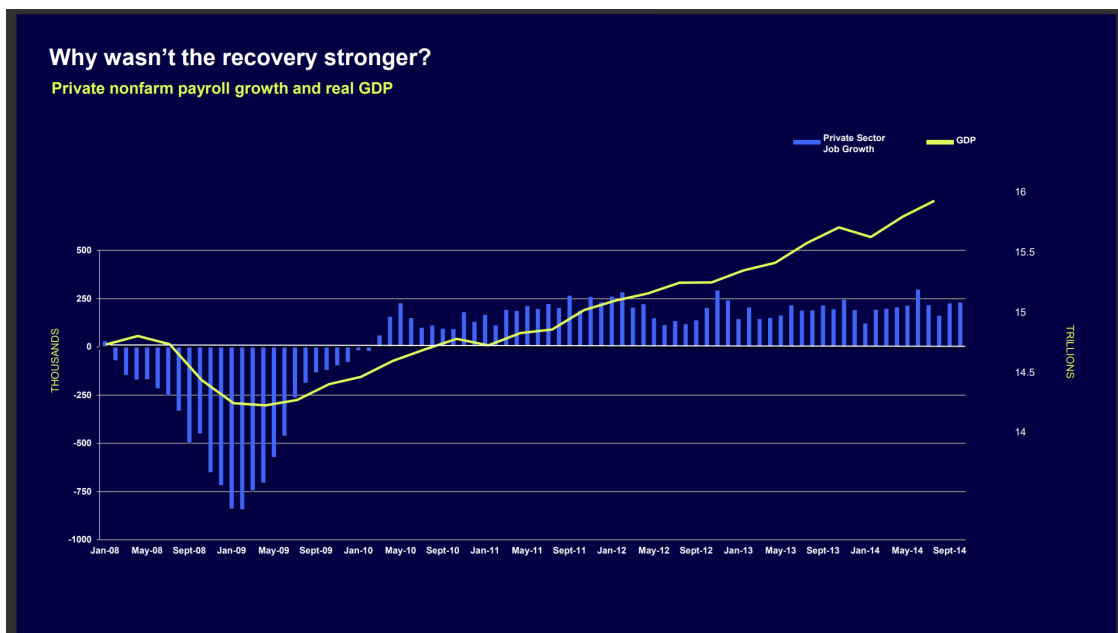


Figure 112.

IDEA: Why wasn't the recovery stronger?

- The natural effects of deleveraging after a crisis following a credit boom
- The self-inflicted wounds of premature fiscal austerity
- The external shocks from Europe, oil prices, and Japan

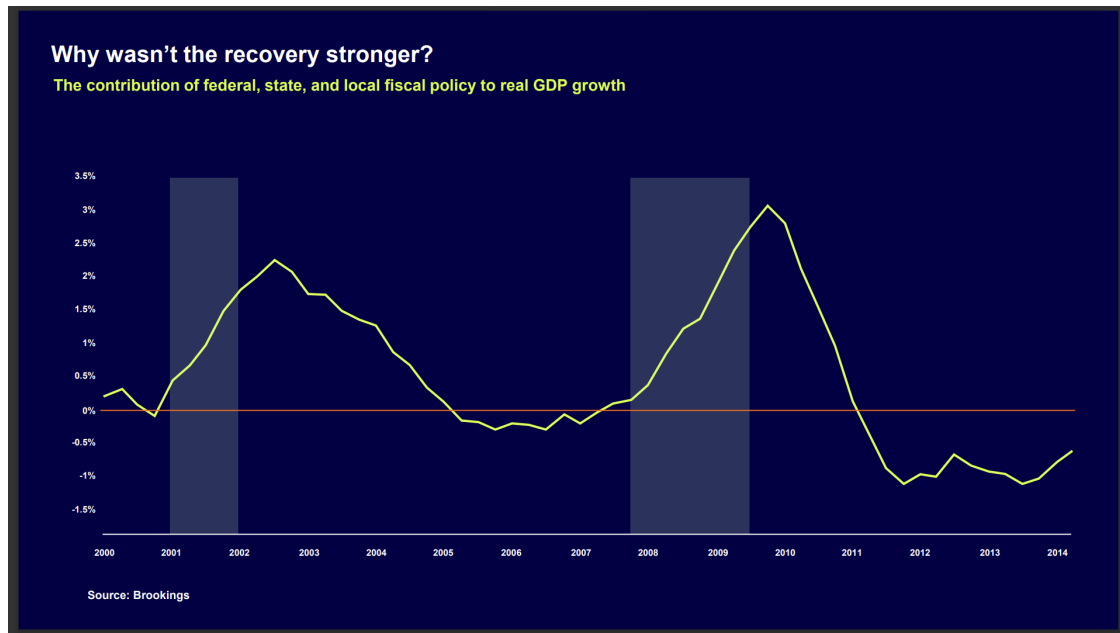


Figure 112.

IDEA:

- The economic costs of the crisis were brutal
- We had a worse initial shock, but no Great Depression
- We came out faster and stronger than most in past crises and most of the developed world in this crisis
- Credit stopped flowing, and we restarted it
- The taxpayer actually made a profit
- We unwound the emergency programs quickly
- Main Street got a lot of direct help, too
- We forced a lot of restructuring
- The financial system is safer today
- Deleveraging, premature austerity, and external shocks all hurt the recovery

10.3 Unfinished Business

IDEA:

- Lots of economic damage remains from the crisis
- The housing finance system is still broken
- Our firefighting tools are too weak
- The regulatory oversight structure is a mess

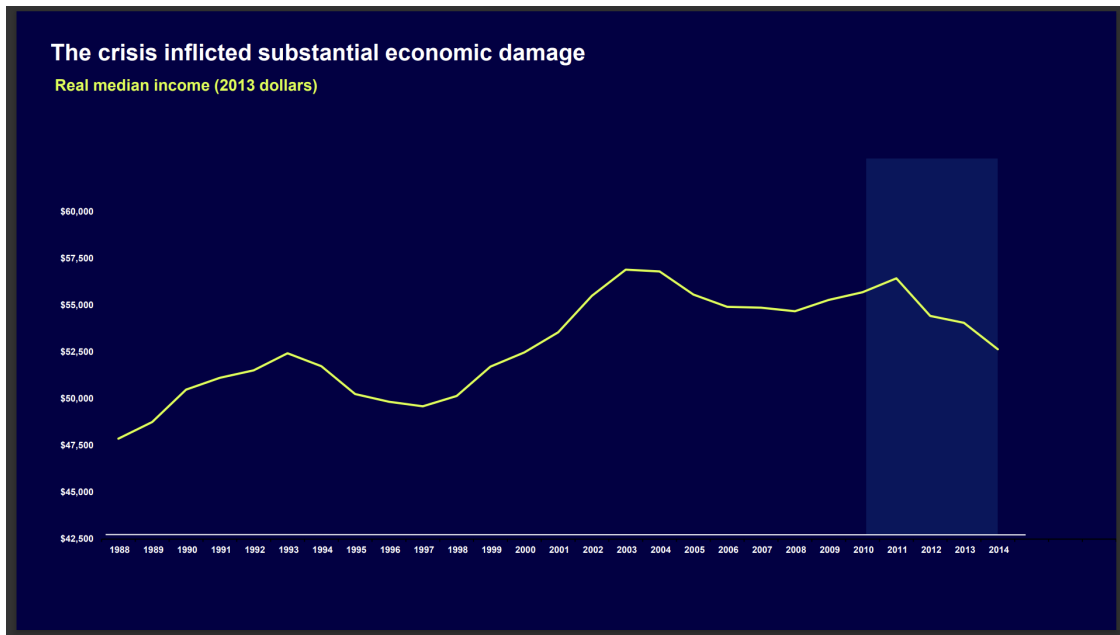


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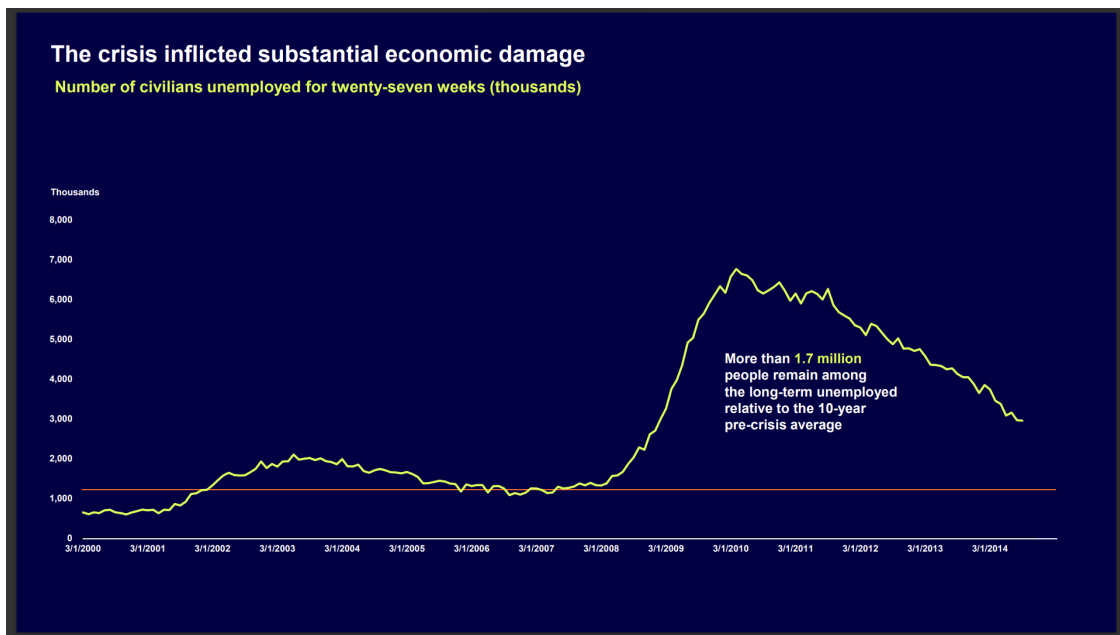


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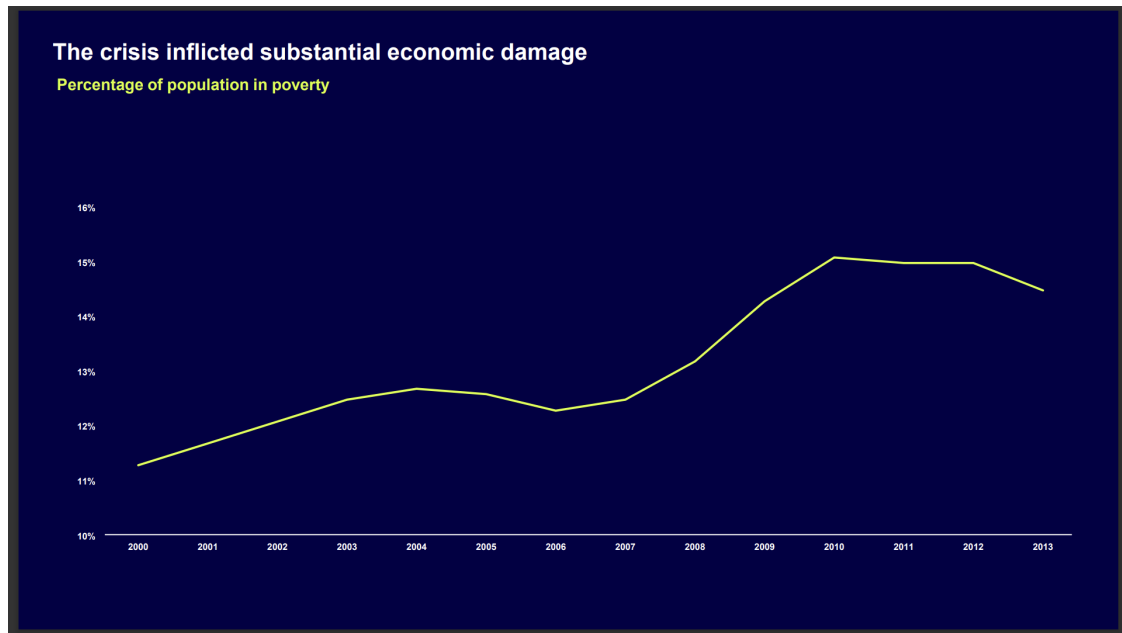


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Figure 116.

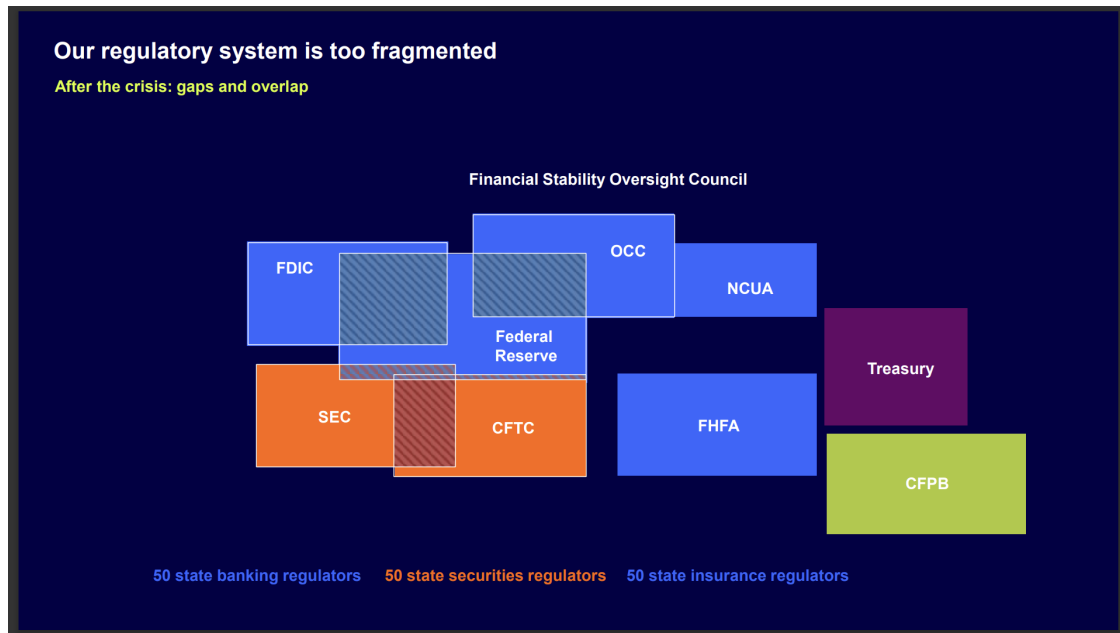


Figure 117.

IDEA: The costs of regulatory balkanization

- There's too much opportunity for regulatory arbitrage
- Regulators and regulations are too slow to adapt to changing markets and institutions
- The rules are too complex
- There's little accountability
- There's too much regulatory capture

IDEA: Principles to govern a streamlined system

- Functional specialization
- More budgetary independence from Congress
- Stronger enforcement
- Better firefighting authority

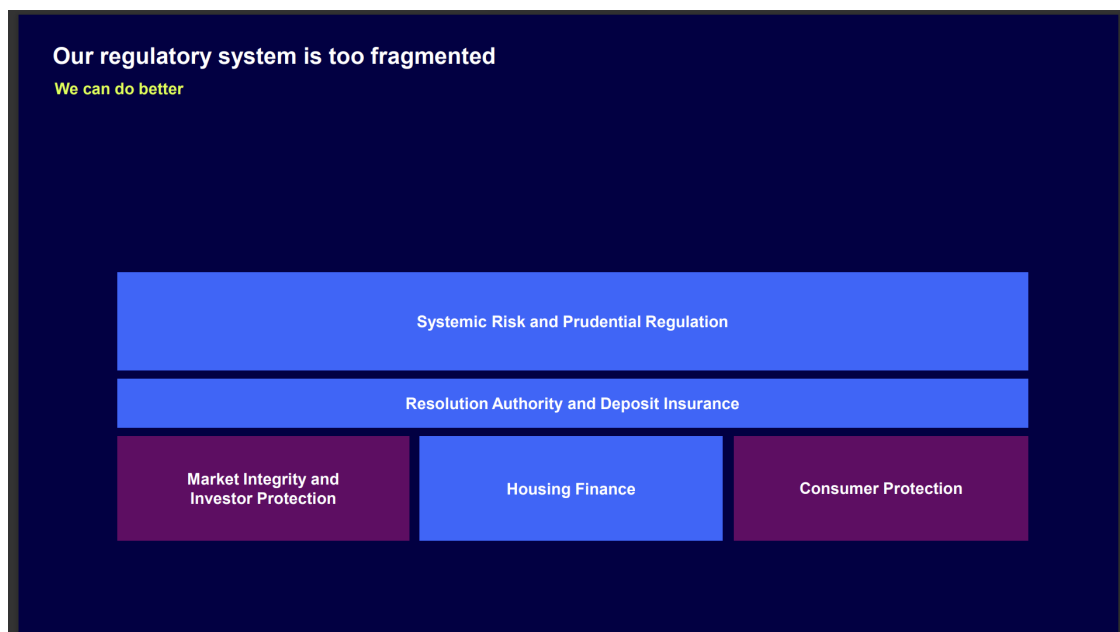


Figure 118.

10.4 Playbook: Rule 1-5

Rule 01:

If you want peace, prepare for war:

- The extreme crisis is unimaginable ex ante
- Manias are not preventable, and there's no over-the-horizon radar for preempting panics and crashes
- So design the system not to prevent failure, but to be safe for failure
- This means you need strong shock absorbers

Rule 02:

Design the shock absorbers for the extreme crisis and apply them broadly

- Impose capital, leverage, and funding requirements on bank-like institutions; systemic surcharge on the biggest
- Place margin requirements on repo and derivatives
- Design for the migration problem:
 - Use the regulated system to impose limits on the shadow system
 - Build in the ability to expand the perimeter of prudential regulation
- Require thick down payments for home mortgages
- Have clear accountability for the design and enforcement of these limits

Rule 03:

The supplemental arsenal of ex ante tools

- Don't let your banking system get too large relative to the size of your economy (size and concentration limits)
- Don't let your "banks" borrow too much in foreign currency
- Transparency
- Deterrence and enforcement

Rule 04 Maintain strong financial capacity and a powerful firefighting arsenal

- Shock absorbers will not save you from the risk of the extreme crisis, so you need firefighting tools and the capacity to use them
- The key sources of financial strength are:
 - Relatively low public debt levels
 - A credible central bank, with a strong record on inflation
- Your firefighting arsenal must include:
 - Liquidity for vulnerable parts of the system
 - Guarantee authority for banks and bank-like institutions
 - Capital and resolution authority

Rule 05 When the fire starts, let it burn—but build a firebreak around the core

- At the beginning of a crisis, there is a fog of diagnosis—idiosyncratic or systemic?
- To get it right, let the fire burn—and let the system adjust
- Escalate slowly at the beginning, but accelerate quickly when necessary
- And set the perimeter of protection so that you capture the solvent and the systemic, but not the insolvent and the peripheral

10.5 Playbook: Rule 6-10

Rule 06:

When it's time to escalate, apply the Powell Doctrine

- Use overwhelming force—fiscal, monetary, and financial
 - Fiscal: as much as you can get
 - Monetary: expansive throughout crisis and deleveraging
 - Financial: liquidity, guarantees, capital, resolution
- None of these are powerful enough to work in isolation
- Together they must be powerful enough to remove risk of depression

Rule 07:

Plan for the long war

- Crises that follow booms entail long periods of deleveraging
- That makes growth weak and fragile
- Need sustained period of monetary policy accommodation to keep real rates negative
- And sustained fiscal support
- Avoid applying the brakes too early

Rule 08:

Impose conditions on the rescue

- Bagehot: Lend freely at a penalty rate
- Ideally, limit the scope of the rescue to the core of the financial system
- Price it to be valuable in the panic, but expensive when the panic breaks
- No naked, unconditional guarantees
- The guarantees should cover catastrophic risk, but not all risk
- Put the non-viable firms into resolution
- Accelerate restructuring and recapitalization of the core of the system to fuel the recovery
- Do the hard stuff early and quickly

Rule 09:

Distinguish the systemic from the idiosyncratic

- Faced with a systemic panic, do the opposite of what makes sense in a typical crisis
- Rather than imposing haircuts, you have to guarantee liabilities
 - Rather than worrying about the potential costs of each specific intervention, you must try to limit the risk of broader failures and the greater costs they will entail
- You have to be prepared to take more potential risk if you want to limit actual risk to the taxpayer
- If you misdiagnose the systemic crisis as a normal crisis, then you will end up having to nationalize everything and socialize more risk

Rule 10:

Keep some perspective on moral hazard

- You can't design a financial system or an effective financial rescue that eliminates moral hazard risk—you can wound moral hazard but you can't kill it
- You can't break a panic without creating moral hazard
- In a systemic crisis, the failure of small firms can jeopardize the stability of the system
- Without a firewall of guarantees around the core of the system, you won't be able to get private capital to come in, and you won't be able to allow failure and facilitate restructuring
- Preserving ambiguity around when you escalate and how broad to extend liquidity support buys you some market discipline
- Systemic capital surcharge and limits on size and/or consolidation are necessary

11 Europe in the Global Financial Crisis

11.1 The Euro: Background

FACT:

- Evolved from previous pan-European institutions, with its formal name change and major steps towards union from the Maastricht Treaty in 1993.
- Monetary union commenced in 1999 with the euro as an accounting unit, and was completed with the introduction of physical euros in 2002.
- This single currency is managed by the European Central Bank
- As of January 1, 2015, there were 28 members of the EU, of which 19 use the euro ("eurozone").
- There are several other countries that are part of the "single market" of the European economic area (EEA), or that use the euro as their currency without being part of the EU.

Eurozone: January 1, 2015

Welcome Lithuania!

Sveiki atvyke!



Figure 119.

11.2 European Growth and Unemployment

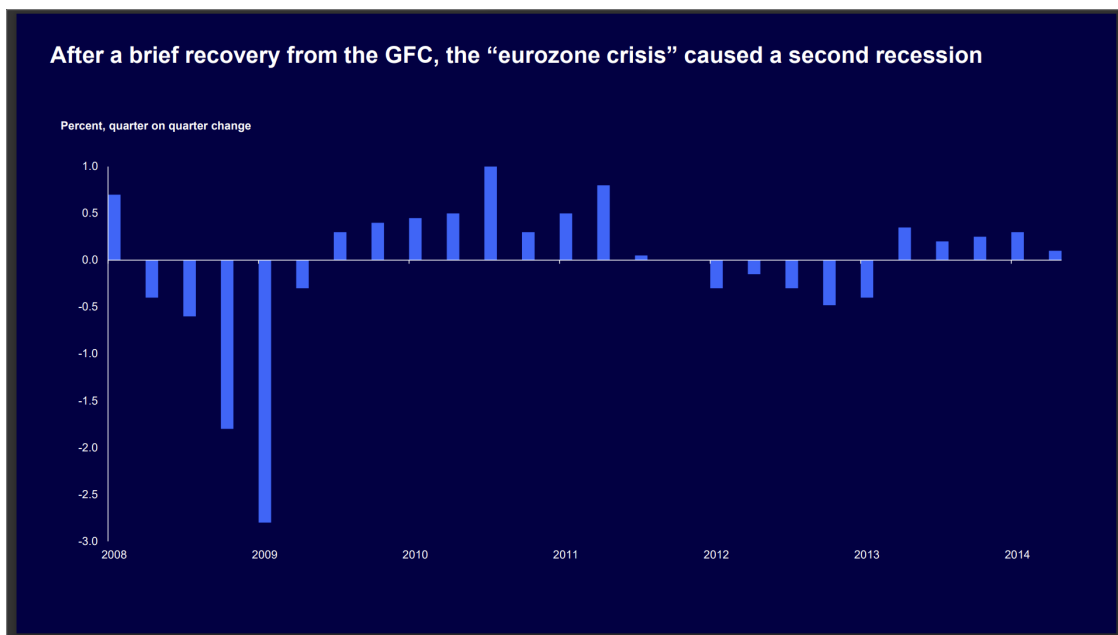


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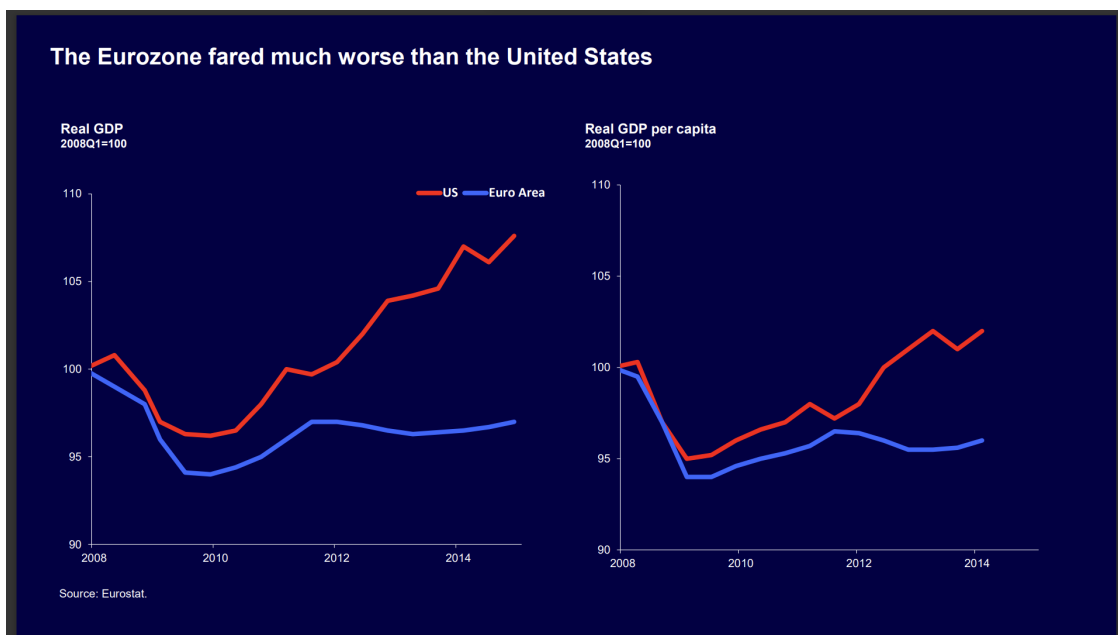


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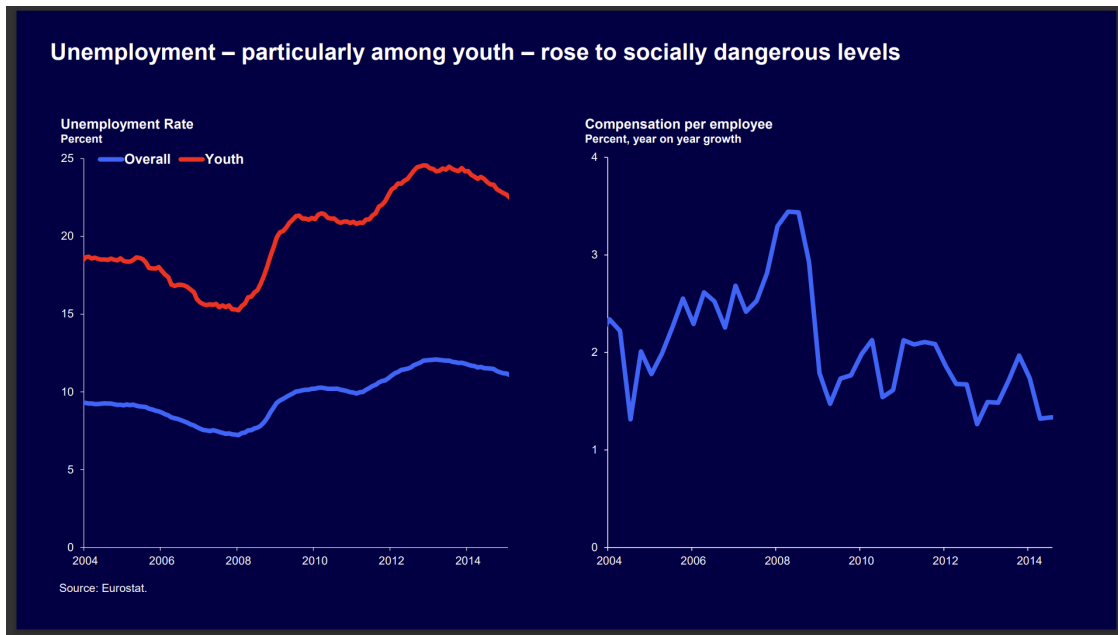


Figure 122.

11.3 Three Connected Crises

11.4 Timeline

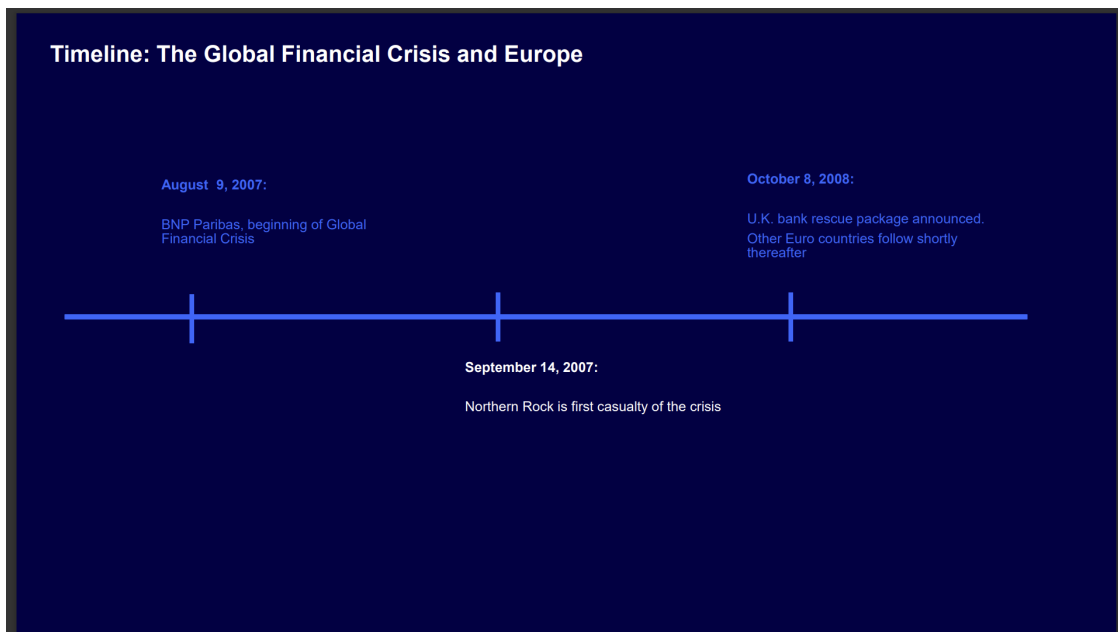


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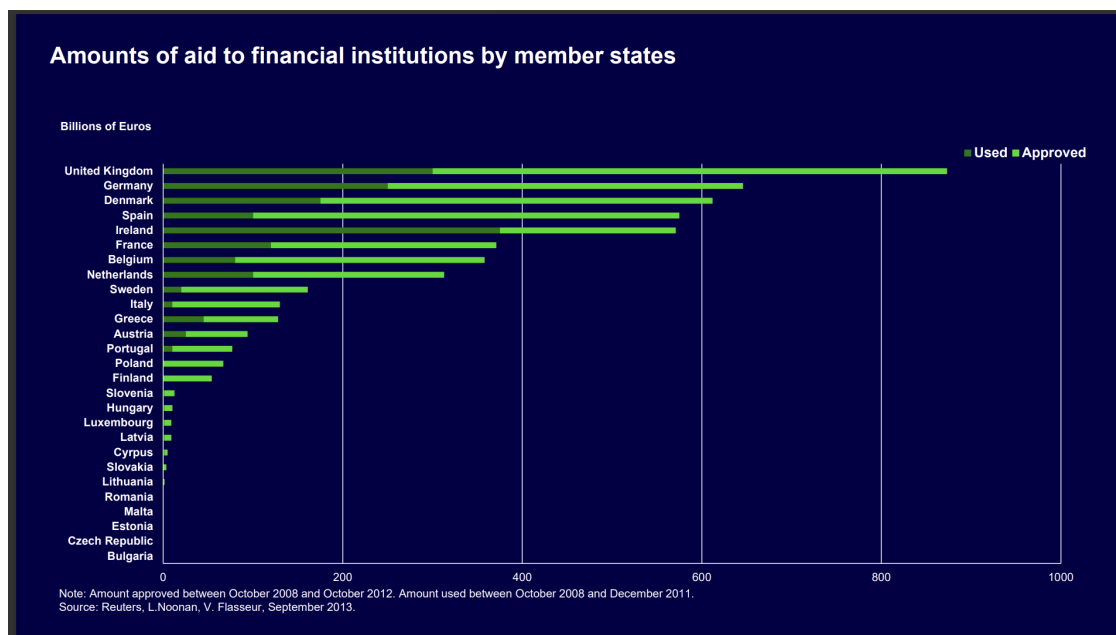


Figure 124.

11.5 Ireland



Figure 125.

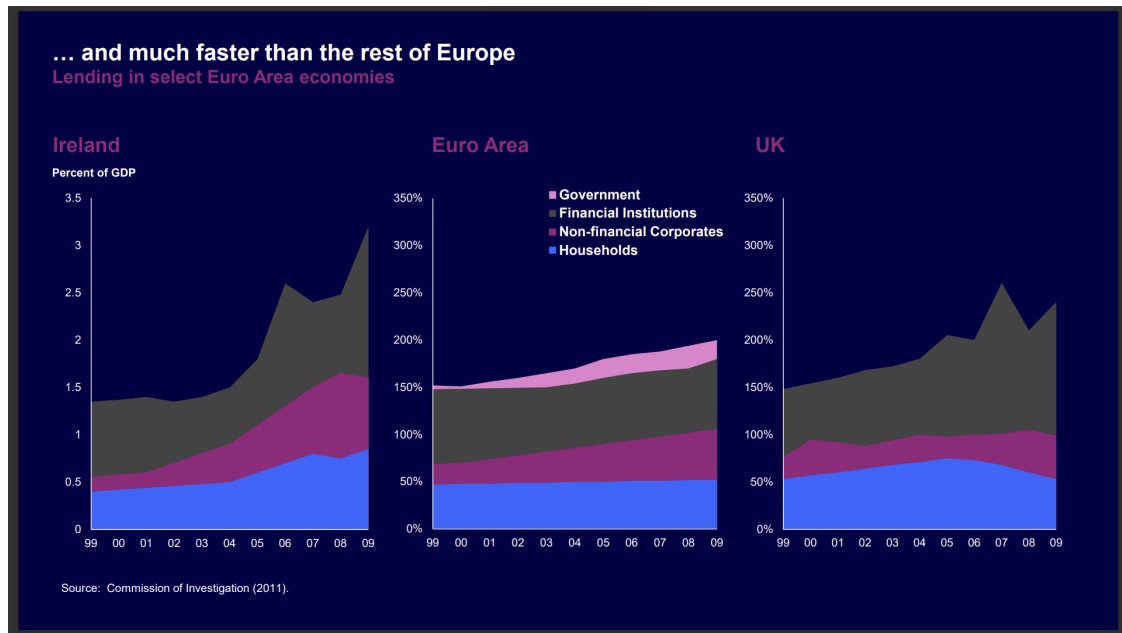


Figure 126.

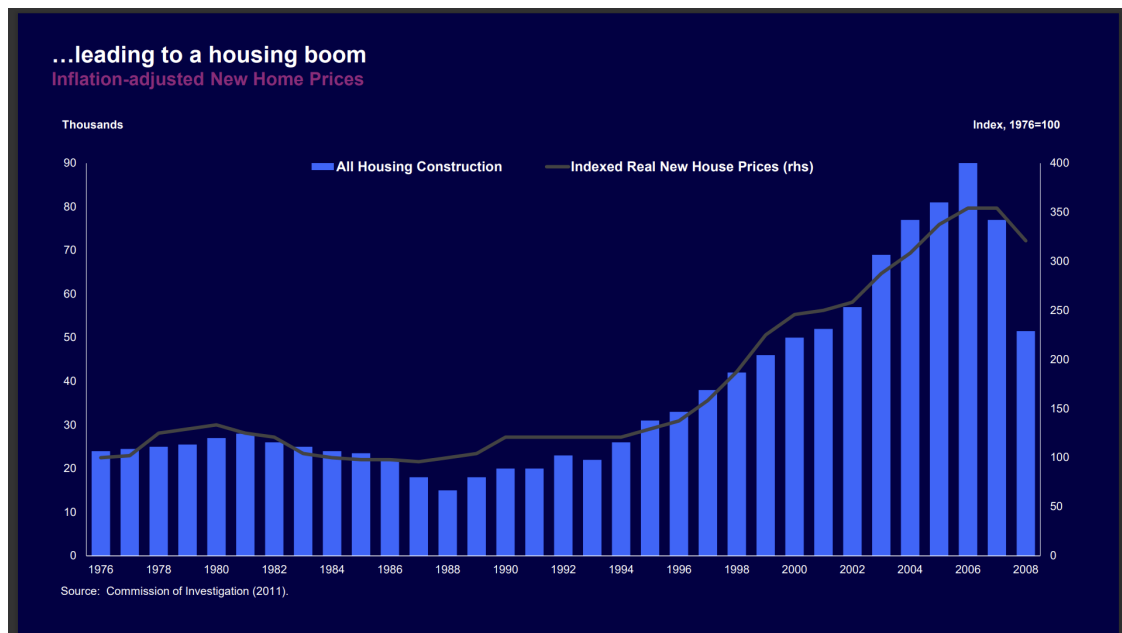


Figure 127.

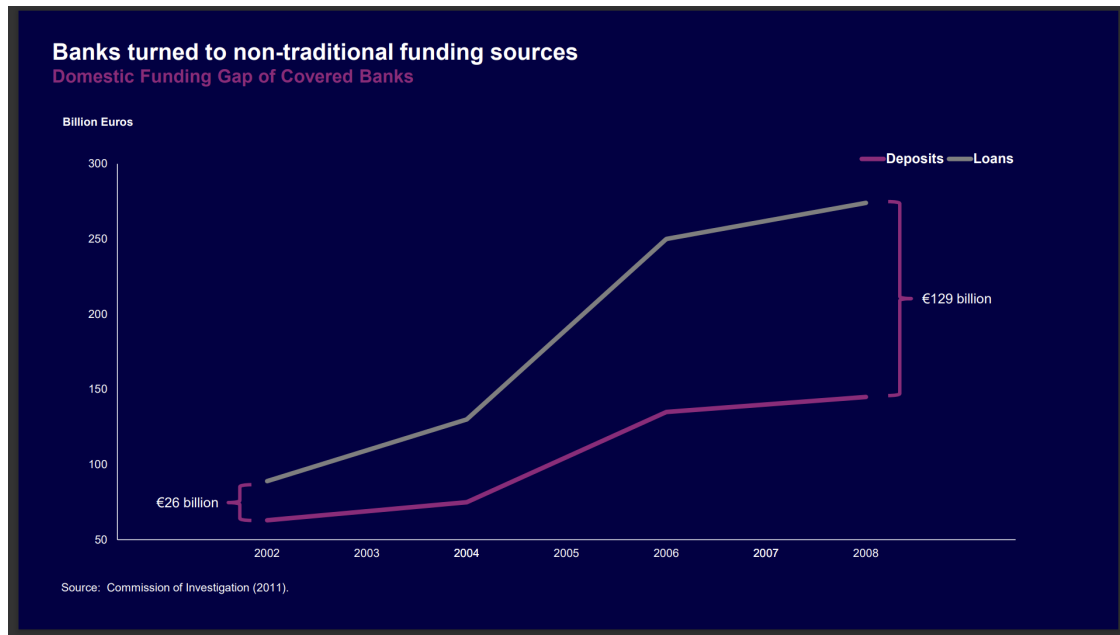


Figure 128.

IDEA:

- Shortly after the Lehman failure, Irish banks were on the precipice, unable to fund the external gap built up during the boom.
- In September 2008, the Irish government made the most powerful response of any nation during the GFC: guaranteeing all liabilities for seven “covered banks” – a total of €375 billion, or about two times Irish GDP.
- It was clear to markets that €375 billion was just too big of a burden for the Irish government.
- Ultimately, Ireland required an international bailout in November 2010.

11.6 Iceland

FACT: Tiny country of about 320,000 people, less than 10% the population of Connecticut Dominated by three banks: Glitnir, Landsbanki, and Kaupthing collectively, “GKL”)

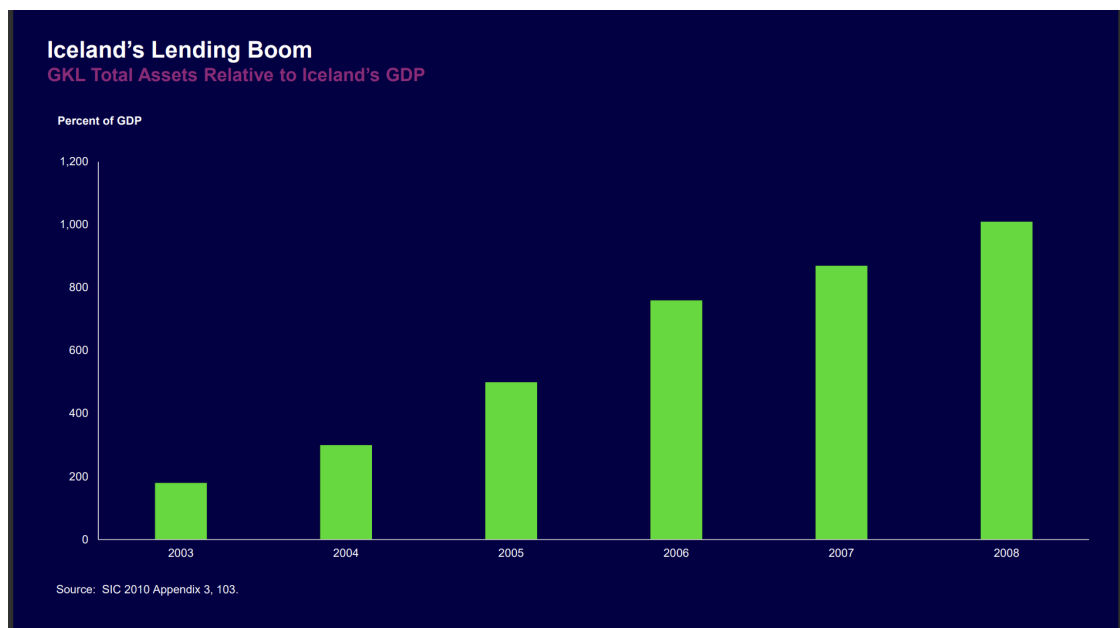


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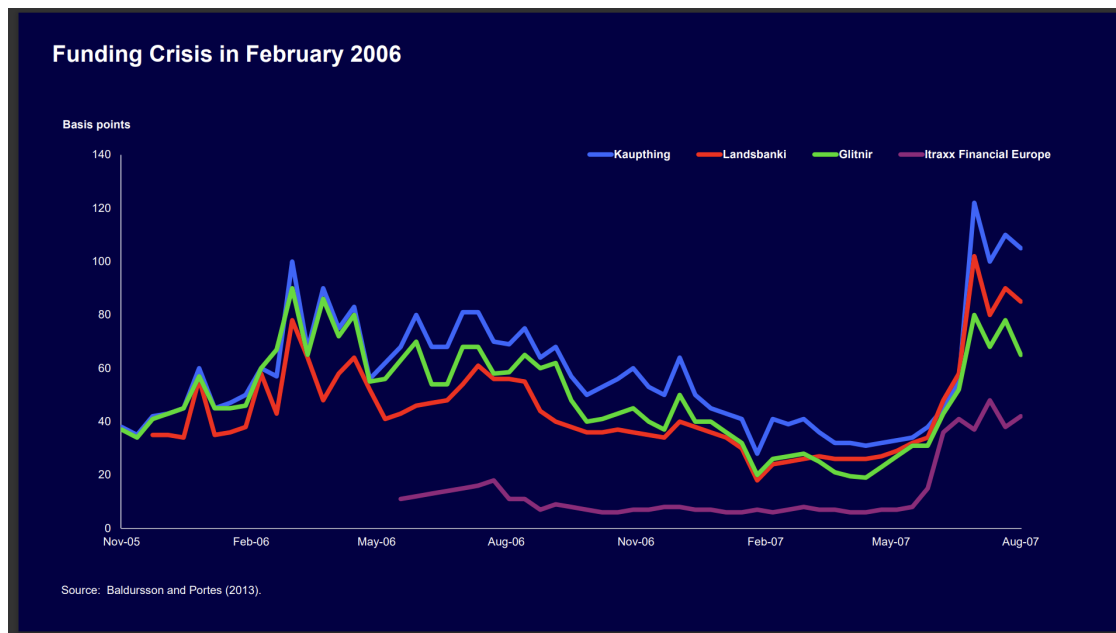


Figure 130.

FACT:

- After the Northern Rock run of September 2007, there was increased pressure on GKL, and the online accounts began a slow run (jog?).
- After Lehman, the jog turned to a sprint, and GKL were unable to keep up their short-term funding.
- Iceland nationalized all three banks over the week of October 6.
- Iceland's deposit fund was insufficient to pay everyone back, and so they made the controversial decision to pay back only domestic customers. The UK and the Netherlands made their own depositors whole, but were not pleased.

12 The Eurozone Crisis

12.1 Eurozone Crisis: Overview

IDEA: Ireland and Spain Banking problems related to credit and housing booms

IDEA: Italy, Portugal and Greece Fiscal issues and sovereign debt

IDEA: Across the Eurozone Institutional and political aspects of the Union make it very difficult to respond to such a large economic shock: Slow price adjustment due to single currency Fiscal federalism too weak to support significant transfers

12.2 Eurozone Crisis: Timeline, Part I

IDEA:

- Weakness in periphery of the eurozone: Greece, Ireland, and Portugal.
- Crisis spreads to the larger countries of Italy and Spain.
- Anti-austerity political parties gain power and Greece repeatedly comes close "Grexit".

IDEA: Phase one:

- December 2009: Greece acknowledges a debt of €300 billion 113% of GDP: this far exceeds the eurozone limit of 60
- Early 2010: Greek debt and deficits continue to be revised Concerns about Greece and other heavily indebted

eurozone countries (notably Ireland and Portugal).

- The "Troika": These concerns are eventually validated in a series of IMF/ECB/EC bailouts:
 - Greece, €110 billion in May 2010
 - Ireland, € 85 billion in November 2010
 - Portugal, € 78 billion in May 2011

12.3 Eurozone Crisis: Timeline, Part II

IDEA:Phase two:

- July 2011:
 - Preliminary agreement of a second Greece bailout of €109 billion fails to stem concerns of contagion.
 - Spreads on Spanish and Italian sovereign debt increase
 - Italy and Spain respond with austerity policies.
- Early 2012: The financial situation is more stable, but a recession has taken hold Greece enters a "technical default" in March 2012, forcing haircuts on some bondholders.
- July 2012: ECB President Mario Draghi states that the central bank will do "whatever it takes to preserve the Euro".

12.4 Eurozone Crisis: Timeline, Part III

IDEA:Phase three: Inevitable political backlash to austerity-driven economic weakness. Anti-austerity and anti-Euro parties have gained power and pressed their governments to relax austerity conditions. This process is most advanced in Greece, where the Syriza party, led by Alex Tsipras, has teetered on the edge of leaving the eurozone.

12.5 The Economics of the Euro

IDEA: Why have the after-effects of the GFC been so much worse in the Eurozone than in the United States? The Eurozone is NOT an "optimal currency area"

IDEA: When a region has a single currency, it is not possible to adjust to regional shocks using monetary policy. In the absence of monetary policy, the adjustment process requires labor mobility, fiscal transfers, or nominal wage and price reductions (deflation). The United States can do the first two things. The eurozone has not been able to do them. That leaves an "internal price adjustment" – which is extremely difficult politically.

12.6 Ireland and Iceland Revisited, Part I

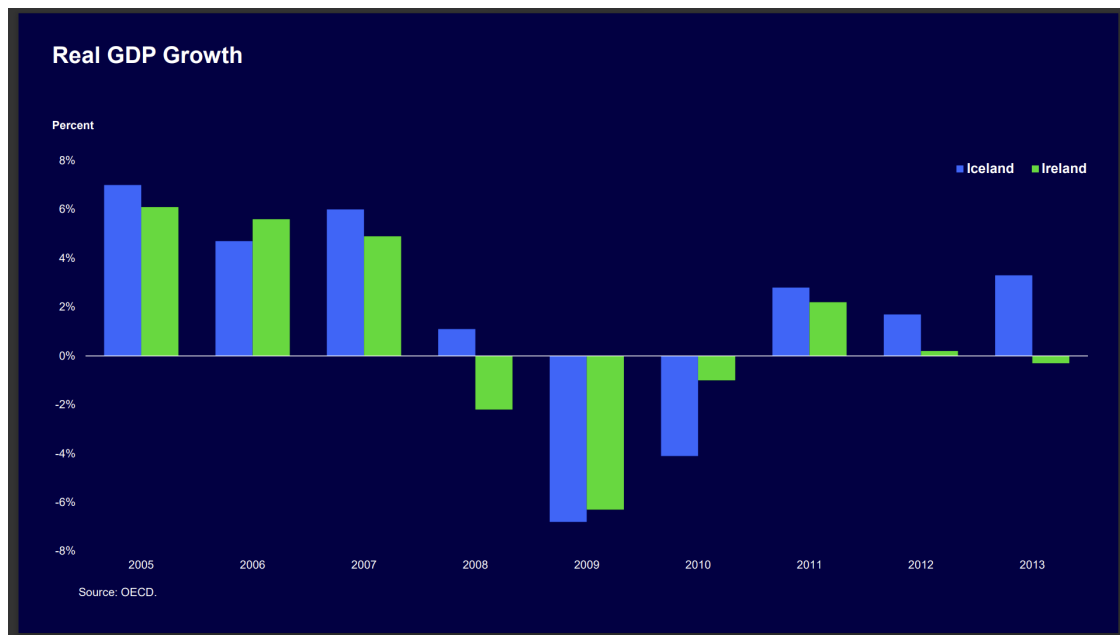


Figure 131.

12.7 Ireland and Iceland Revisited, Part II

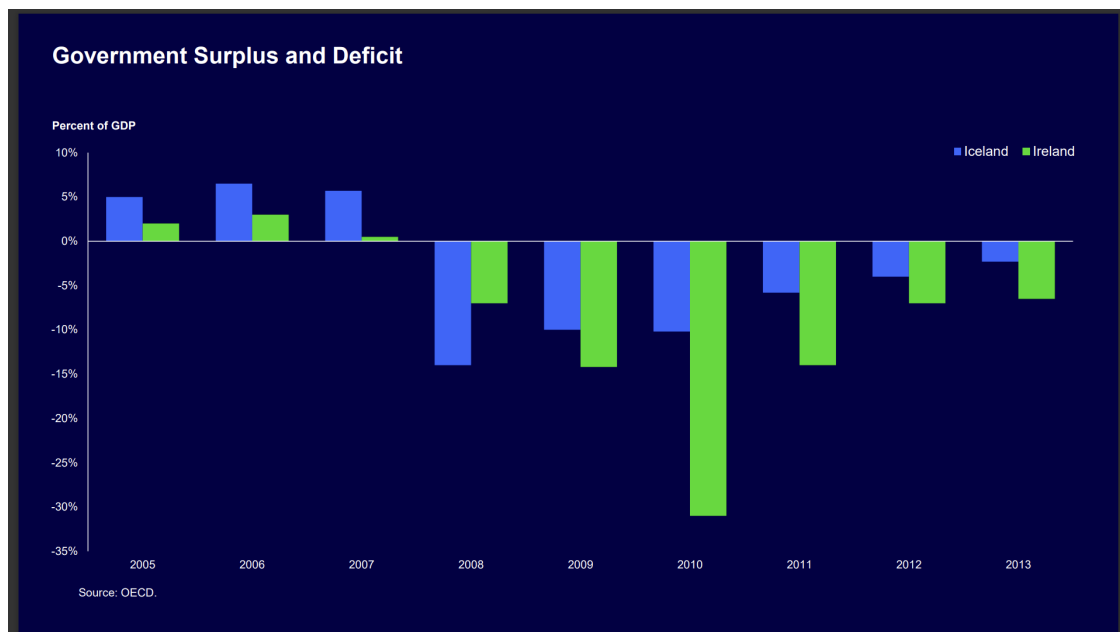


Figure 132.

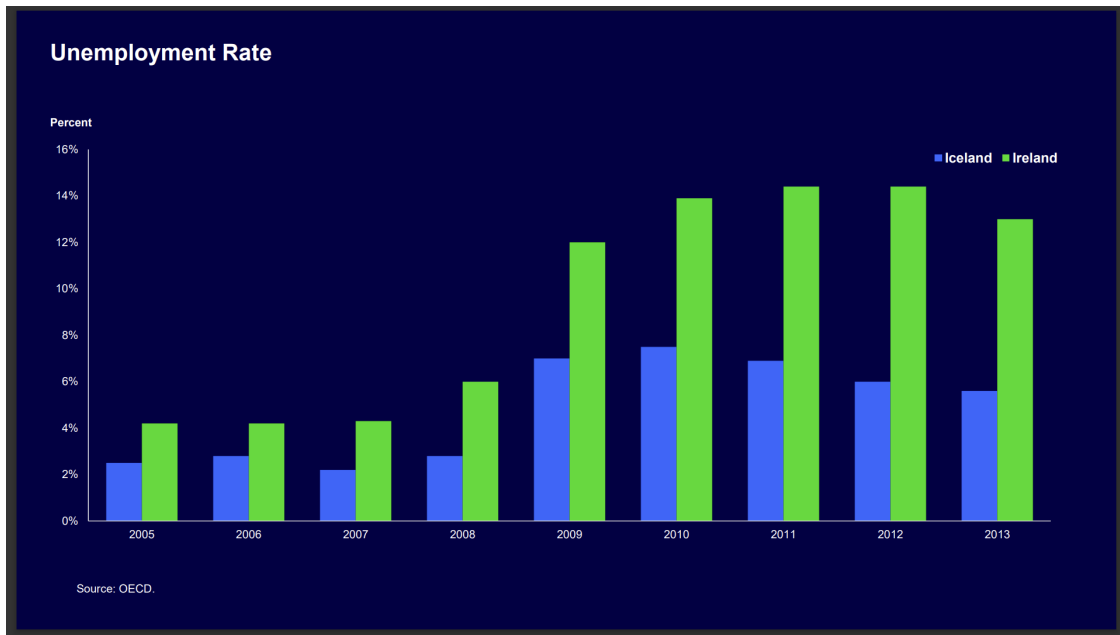


Figure 133.

12.8 Ireland and Iceland Revisited, Aftermath

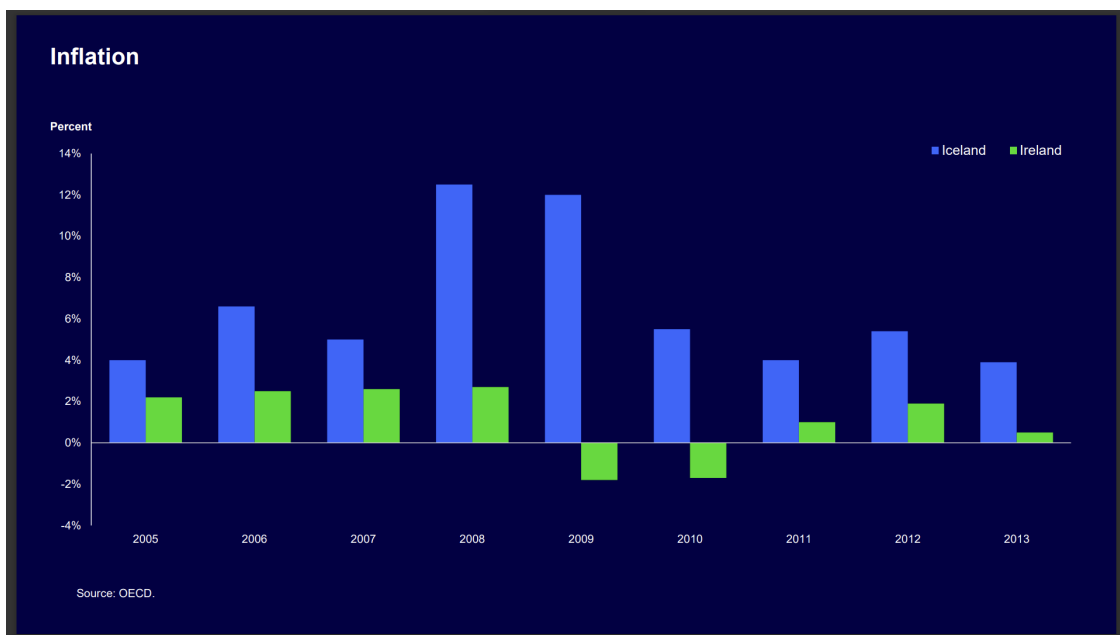


Figure 134.

12.9 Internal Devaluations

Internal Devaluations and Devaluations within Currency Unions			
Sample	Criterion for devaluation		
	3% over 1 year	5% over 3 years	7% over 5 years
26 mostly developed countries			
Internal devaluations only, * since 1964	25	26	17
Internal devaluations only, after 1990	3	6	3
All devaluations, since 1964	255	136	90
All devaluations, after 1990	114	57	44
Members of currency unions			
U.S. metro areas, since 1961	7	11	11
U.S. metro areas, after 1990	0	1	1
Euro-area countries, since 1999	1	1	1

* An internal devaluation is a fall in a country's overall price level relative to world prices that is not accompanied by a nominal depreciation of that country's currency.

Source: Bank for International Settlements, Bureau of Labor Statistics, Eurostat, as shown in Shambaugh, Jay C., 2012, The euro's three crises [with comments and discussion by Ricardo Reis and Helene Rey], Brookings Papers on Economic Activity 157-231

Figure 135.

12.10 The Euro and the Gold Standard: Lessons from the Great Depression

IDEA:

- Prior to World War I: Almost all of the wealthy countries of the world were on the Gold Standard, which is effectively a single currency for the entire world.
- During the War: All countries suspended convertibility into gold.
- After the War: Almost all countries went back on the Gold Standard, but relative prices across countries were not properly aligned. This misalignment largely set the stage for deep recessions across many countries. This commitment was the main contributor to the global propagation of the Great Depression.

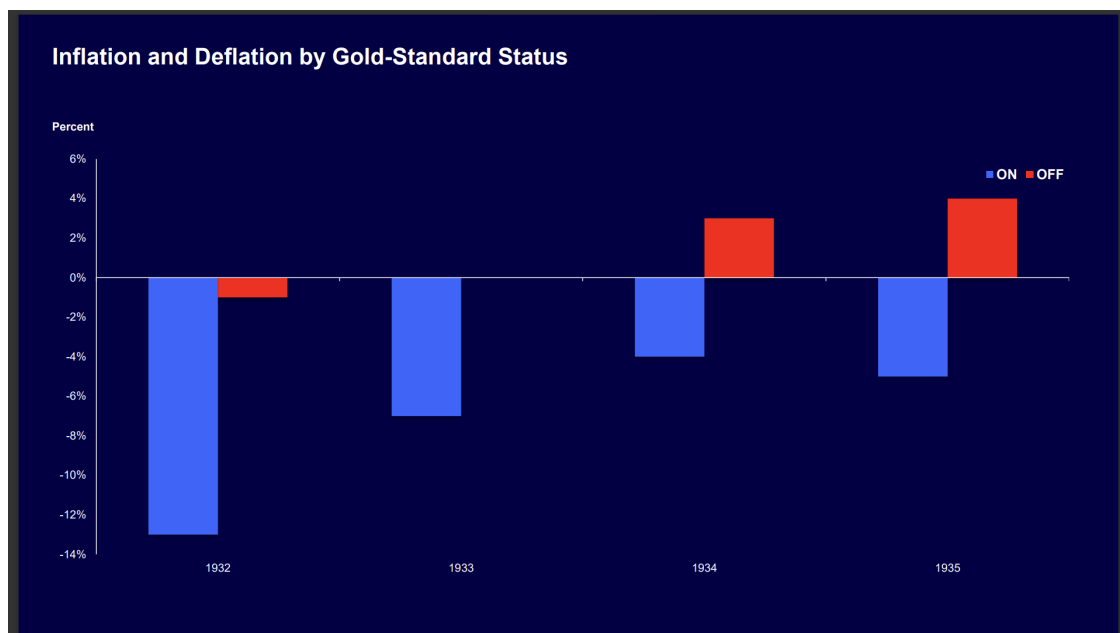


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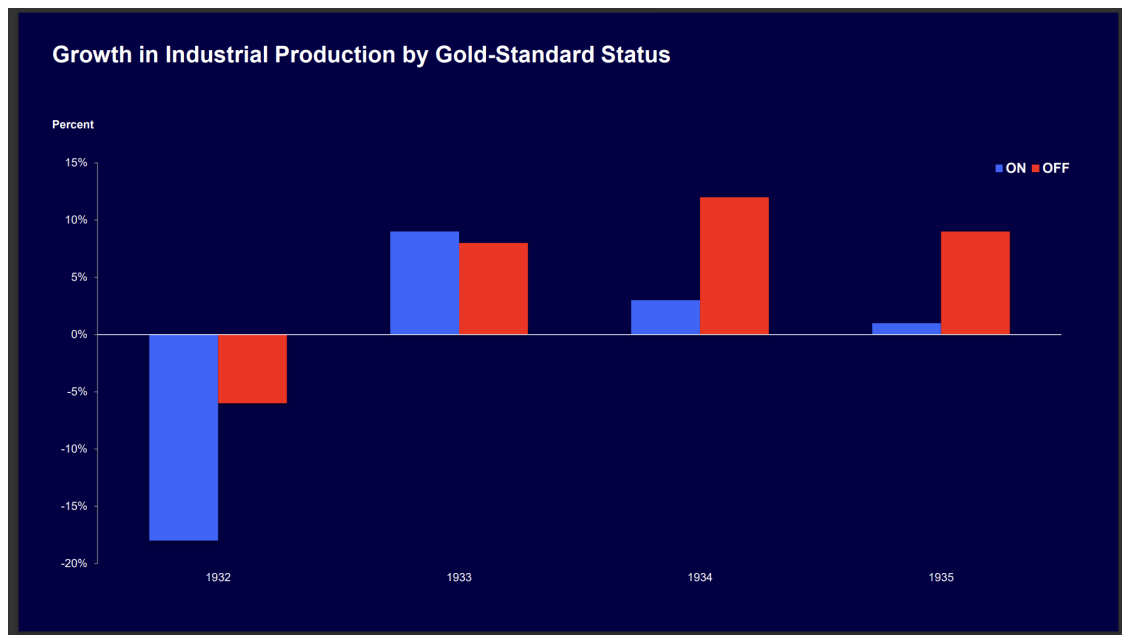


Figure 137.